

COMPACT REACTOR MODELING

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*This is to certify that the thesis entitled “**Compact Reactor modeling**”, which is being submitted by **Shouvik Chaudhuri** in partial fulfilment for the award of the degree of **Master of Nuclear Engineering** to Jadavpur University, Kolkata 700032, during the academic year 2011-2013, is the record of the student’s own work carried out by him under our supervision.*

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ABSTRACT

In the field of Nuclear Power generation, the Pressurized Water Reactors (PWRs) are known as Compact Reactors for their compact design. Amongst all the reactor types in operation around the globe, they constitute the major portion. However, in India we have just started to exploit the PWR technology as an additional scheme to our already existent three stage nuclear programme.

In this study, an analytical model of the PWR has been developed based on the prior research available regarding the topic both indigenously and otherwise. The primary loop of the PWR has been modelled from first principles for the power operation. Then by using the State Space approach, a simple model of the plant has been developed which can be exploited for control related studies. Added to that, a reactor regulating system has also been designed for the purpose of improving the system's responses to power changes and disturbances. Performance analysis of the model has been carried out under various expected scenarios in a nuclear plant to validate their consistency with the known behaviour of PWR.

Another aspect of this study includes the formation of the Block Control model of the PWR on a commercially available software and simulation package. The performance of this model has been compared to that of the previous model developed. This is done in order to verify the fact that the designed analytical model gives non-ambiguous results despite using different simulation methodologies.

Finally, the stability issue of the developed models has been addressed. Since, even the simplest nuclear reactor core model shows non-linearity, an appropriate non-linear stability criterion was necessary. This has been done by using the Lyapunov stability criterion. Some of the models previously developed have been subjected to this analysis, so as to determine their stability in the sense of Lyapunov.

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LIST OF ACRONYMS

AHWR	:	Advanced Heavy Water Reactor
BARC	:	Bhabha Atomic Research Centre
BIBO	:	Bounded Input Bounded Output
BOC	:	Beginning of Cycle
BOL	:	Beginning of Life
BWR	:	Boiling Water Reactor
CANDU	:	Canadian Deuterium Uranium Reactor
CR	:	Control Rod
CRDM	:	Control Rod Drive Mechanism
DNP	:	Delayed Neutron Precursor
DOE	:	Department of Energy
EPR	:	European Pressurized Reactor
FBR	:	Fast Breeder Reactor
FCV	:	Feed Check (or Control) Valve
FP	:	Full Power
FWP	:	Feed Water Pump
H.P	:	High Pressure
IGCAR	:	Indira Gandhi Centre for Atomic Research
KNPP	:	Koodankulam Nuclear Power Plant
LBP	:	Lumped Burnable Poison
LEU	:	Low Enriched Uranium
LMFBR	:	Liquid Metal Fast Breeder Reactor
LOCA	:	Loss of Coolant Accident
L.P	:	Low Pressure
LTi	:	Linear Time Invariant
LWR	:	Light Water Reactor
MHTS	:	Main Heat Transport System
MIMO	:	Multiple Input Multiple Output system
MOX	:	Metal Oxide
MSIV	:	Main Steam Isolation Valve
MSR	:	Moisture Separator and Re-heaters

LIST OF ACRONYMS

MWe	:	Mega Watt electrical
MWth	:	Mega Watt thermal
NAPS	:	Narora Atomic Power Station
NPP	:	Nuclear Power Plant
OTSG	:	Once Through Steam Generator
PFBR	:	Prototype Fast Breeder Reactor
PHWR	:	Pressurised Heavy Water Reactor
PWR	:	Pressurised Water Reactor
PZR	:	Pressurizer
R&D	:	Research and Development
RAPS	:	Rajasthan Atomic Power Station
RCCA	:	Rod Cluster Control Assembly
RCP	:	Reactor Coolant Pump
RR	:	Regulating Rod
RRS	:	Reactor Regulating System
SDS-1	:	Shut Down System 1
SDS-2	:	Shut Down System 2
SG	:	Steam Generator
SISO	:	Single Input Single Output
USNRC	:	United States Nuclear Regulatory Commission
VVER	:	Vodo-Vodyanoi Energetichesky Reaktor
WWER	:	Water-Water Energetic Reactor

NOMENCLATURE

A	:	System Matrix
A_f	:	Fuel to Coolant Heat Transfer Area (m ²)
B	:	Input Matrix
B_g^2	:	Geometric Buckling
C	:	Output Matrix
C	:	One group of DNP Concentration (nuclei/cm ³)
C_0	:	Equilibrium State DNP Concentration of One Group delayed neutrons
C_c	:	Specific Heat of Primary Coolant (J/kg. °C)
C_f	:	Specific Heat of fuel (J/kg. °C)
C_i	:	Concentration of the i th Precursor Group (nuclei/cm ³)
C_r	:	Normalised DNP Concentration
C_{r0}	:	Equilibrium State Normalised DNP Concentration
C_{SG}	:	Specific Heat of Water in Steam Generator (J/kg. °C)
D	:	Direct Transmission Matrix
D	:	Neutron Diffusion Coefficient (cm)
F_r	:	Neutron Population to Reactor Thermal Power Conversion Factor
f	:	Thermal Utilization Factor
$G(s)$	:	Transfer Function of a System
G_c	:	Primary coolant mass flow rate (kg/sec)
G_{feed}	:	Feed water mass flow rate (kg/sec)
$G_{feed}(ini)$	:	Initial steady state value of Feed water mass flow rate (kg/sec)
$g(x)$	:	Non-linear Part of a system, $f(x)$
h_f	:	Fuel to coolant heat transfer coefficient (W/m ² .°C)
I	:	Iodine Concentration
i	:	Index of Precursor Groups
k_∞	:	Infinite Multiplication Factor
k_{eff}	:	Effective Multiplication Factor
K_1, K_2	:	Gain Matrices
K_{rod}	:	Slope of S-Curve for Control Rod Position (constant)
L	:	Diffusion length (cms)

NOMENCLATURE

l	:	Prompt Neutron Life Time (seconds)
m_c	:	Total mass of Primary Coolant (kg)
m_f	:	Total mass of Fuel (kg)
m_{SG}	:	Primary Steam and Water Mass in Steam Generator (kg)
$n(r, t)$	:	Neutron density or Neutron Population
$\mathbf{O}$	:	Observability Matrix
$\mathbf{P}$	:	Symmetric Positive Definite Matrix
P	:	Actual Reactor Power (MWth)
P_f	:	Fast Non-Leakage Factor
P_0	:	Equilibrium State Reactor Power (MWth)
P_r	:	Normalised Reactor Power
P_{r0}	:	Equilibrium State Normalised Reactor Power
P_t	:	Thermal Non-Leakage Factor
p	:	Resonance Escape Probability
$\mathbf{Q}$	:	Symmetric Positive Definite Matrix
R_{fcv}	:	Rate of Opening or Closing of Feed Check Valve
$\mathbf{r}$	:	Reference Input Vector
$sgp(t)$	:	Steam Generator Power
sgp_0	:	Equilibrium State Steam Generator Power
T_2	:	Dead Band Time
$T_c(t)$	:	Average Coolant Temperature (°C)
T_{c0}	:	Equilibrium State Average Coolant Temperature (°C)
$T_{cout}(t)$	:	Coolant Outlet (hot leg) Temperature (°C)
T_{cout0}	:	Equilibrium State Coolant Outlet (hot leg) Temperature (°C)
$T_{cin}(t)$	:	Coolant Inlet (cold leg) Temperature (°C)
T_{cin0}	:	Equilibrium State Coolant Inlet (cold leg) Temperature (°C)
$T_f(t)$	:	Average Fuel Temperature (°C)

NOMENCLATURE

T_{f0}	:	Equilibrium State Average Fuel Temperature (°C)
T_{set}	:	Predetermined value of Average Coolant Temperature (°C)
$T_{SG}(t)$	:	Average Primary Coolant Temperature inside Steam Generator (°C)
T_{SG0}	:	Equilibrium State Average Primary Coolant Temperature in SG (°C)
$T_{SGin}(t)$	:	Steam Generator Inlet Temperature (°C)
T_{SGin0}	:	Equilibrium State Steam Generator Inlet Temperature (°C)
$T_{SGout}(t)$	:	Steam Generator Outlet Temperature (°C)
T_{SGout0}	:	Equilibrium State Steam Generator Outlet Temperature (°C)
t	:	Integrated Duration of Control Rod Movement
$\mathbf{U}$	:	State Controllability Matrix
$\mathbf{u}$	:	Input Vector
u	:	Control Signal from the Second Subunit of RRS for CR movement
u_{es}	:	Error Signal
u_{thres}	:	Threshold Value of Error Signal
$V(x, t)$	:	Positive Definite Scalar Function
$\mathbf{x}$	:	State Vector
$\dot{\mathbf{x}}$	:	Velocity Vector
X	:	Xenon Concentration
$\mathbf{y}$	:	Output Vector
α_f	:	Fuel Temperature Coefficient ($\Delta k/k/^\circ\text{C}$)
α_c	:	Coolant Temperature Coefficient ($\Delta k/k/^\circ\text{C}$)
β	:	Total Delayed Neutron Fraction
β_i	:	Delayed Neutron Fraction of the i^{th} Precursor Group
γ_I	:	Fractional yield of Iodine
γ_X	:	Fractional yield of Xenon
δ	:	Incremental Change
Δ	:	Infinitesimal Change

NOMENCLATURE

ε	:	Fast Fission Factor
ζ	:	Logarithmic Energy Decrement per Collision
ζ_1, ζ_2	:	Proportionality Constants for Control Rod Error Signal, u_{es}
η	:	Reproduction Factor
Λ	:	Effective Neutron Lifetime (seconds)
$\mathcal{L}^{-1}$	:	Inverse Laplace Operator
λ	:	Decay Constant for One Group of Delayed Neutrons (sec^{-1})
λ_i	:	Decay Constant of the i^{th} Precursor Group (sec^{-1})
λ_I	:	Decay Constant of Iodine (sec^{-1})
λ_X	:	Decay Constant of Xenon (sec^{-1})
v	:	Velocity of CR movement
$\rho(t)$	:	Reactivity (mK)
$\rho_f(t)$	:	Reactivity Feedback due to Fuel Temperature (mK)
$\rho_c(t)$	:	Reactivity Feedback due to Primary Coolant Temperature (mK)
$\rho_X(t)$	:	Reactivity Feedback due to Xenon (mK)
$\rho_{CR}(t)$	:	Reactivity Introduced by the CRs (mK)
Σ_a	:	Macroscopic absorption cross section of neutrons (barns)
Σ_f	:	Macroscopic fission cross section of neutrons (barns)
Σ_s	:	Macroscopic scattering cross section of neutrons (barns)
σ_I	:	Microscopic absorption cross section of Iodine (barns)
σ_X	:	Microscopic absorption cross section of Xenon (barns)
τ_{cld}	:	Cold Leg Delay
τ_{ct}	:	Transit Delay from Core to Hot Leg
τ_{hld}	:	Hot Leg Delay
τ_{SG}	:	Resident Delay within SG
$\phi(r, t)$	:	Neutron flux (neutrons / cm^2sec)
$\psi_1(r)$	:	Fundamental Mode Solution of the Helmontz Equation

CHAPTER 1

INTRODUCTION

The concept of modeling of dynamical systems is prevalent across various fields of engineering. In some cases, the primary aim of modeling is to aid in the design of the system. In other cases, the models are used to explain measured data from a system and to understand the observed phenomenon. In a general sense, modeling is used as an aid to describe and understand reality. Amongst the genre of models possible for systems, mathematical and software models [19] are the most popularly used ones. Mathematical models describe the relationship amongst the system variables in terms of mathematical formulations like differential or difference equations. Software models, on the other hand, are simulation programs based on the system and so, differ from the area of mathematical modeling.

In the nuclear power domain, the modeling of Reactor systems plays a vital role in the study of their operational and safety features. Mathematical models in particular have become a prime tool for researchers working in this field. However, the presence of complex non-linear equations for the reactor core renders difficulty in the task of mathematically modeling a nuclear reactor plant. So as to overcome this basic challenge two different approaches are available in the literature - one being the linearization of the nonlinear model and the second being the formulation empirical mathematical models [42].

In this study, the focus is on the development of a mathematical model for a particular kind of nuclear reactor, the Pressurised Water Reactors (PWRs) by using the first approach. These reactors have earned the name '*Compact Reactors*' due to their compact design. The reason behind such a design is that PWRs have a small Diffusion length (L) for thermal neutrons (~ 0.0275 m) & also small Neutron Lifetime ($l \sim 0.00021$ sec). The use of light water (H_2O) as a moderator in PWRs is the main contributing factor behind these features.

The PWRs are the most widely used type of nuclear reactors around the world [28]. Their popularity is based on their two most important attributes, viz. self-stability and self-regulation. Self-stability is the ability of the PWR nuclear power plant (NPP) to automatically stabilize the reactor power after a certain transition process without the function of a control system, when it experiences a perturbation in the form of reactivity variation. Similarly, the self-regulation is the ability that the PWR-NPP reactor core could regulate its output power automatically according to change of load or demand. This is better known as the load following operation.

In the Indian context, we are just starting to exploit the PWR technology as an additional scheme to our already existent nuclear programme [13]. Hence, in order to understand the relevance of this study at such a juncture, we need to have a brief overview of the Indian nuclear programme.

1.1. THE INDIAN NUCLEAR POWER PROGRAMME

The Nuclear Power Program in India relies on a three stage policy based on the rationale of optimal utilization of thorium reserves in the country. At present, the first stage programme is underway, which is mainly based on a series of Pressurized Heavy Water Reactors (PHWRs). The choice of PHWRs in the current stage of India's Nuclear Power Program is based on long-term objectives [12] to be achieved in the available uranium resources and industrial infrastructure (refer **Appendix C.1**).

However, it has been realised that there would be a long delay before direct thorium utilisation can formally begin with the three stage programme. Keeping this in mind, the nuclear programme is now focussing at reactor designs that will lead to the timely development of the thorium based technologies for the entire fuel cycle necessary for the third stage of the programme. The main highlights of this new approach has been the development of the indigenous AHWRs that allow more direct use of thorium [36] and the addition of PWRs in the form of VVERs & EPRs [14],[12] (refer **Appendix C.2**).

1.2. BACKGROUND

From the above discussions, it is clear that the PWR design is preferred in the nuclear circles due to its compact build and self-stabilization properties. Many investigations have been performed over the years to develop suitable PWR models which accurately describe their behaviour, in the context of control and safety of the plants.

Canzian *et al.* [1] discusses simulation results of a real time digital simulator for a pressurized water reactor (PWR). They focus on the primary cycle of the plant and lumped mathematical models are used for designing the simulator. They show that the results of the digital simulator are analogous to that of real PWR plants.

Fazekas *et al.* [4] presents the modelling and identification procedure for a VVER-type PWR. They describe a non-linear state space model of the reactor taking into account the temperature and Xenon poisoning effects. They use real transient measurement data from the Paks Nuclear power plant in Hungary for identification based on standard prediction error minimization. They show that the model is fairly well identifiable and the use of temperature and xenon effects significantly improves the quality of the fit between the measured and computed model outputs.

Fazekas *et al.* [5] presents a simple dynamical model of the primary circuit in a VVER-type PWR. They focus on the development of a low dimensional lumped non-linear dynamic model for the primary circuit of the plant from first engineering principles. The model includes the description of the main control loops in the system. The model is verified and validated by using measured data from three VVER-440 units of Paks Nuclear Power Plant in Hungary. They show that a good fit has been obtained.

Hamieh *et al.* [10] discusses an analytical solution of the point kinetics equations with one group of averaged delayed neutrons and temperature feedback. They derive the relations of reactivity and neutron density with neutron lifetime. Using this analytical model, a digital

simulator has been shown. Though this approach for solving the equations turn out to be tedious, from the results they conclude that the expansion of reactivity and neutron density to at least the first order of neutron lifetime should be done for having acceptable results.

Li *et al.* [18] discusses the analysis of the delayed supercritical process of a nuclear reactor considering one delayed neutron group and temperature feedback, while inserting small step reactivity. New analytical expressions of reactor power and reactivity during the delayed supercritical process are derived. The expressions are validated by comparison with other methods, which indicate the superiority of temperature prompt jump approximations.

Nahla [24] presents an analytical solution of the point kinetic equations with one group of delayed neutrons and the adiabatic feedback model. The relation between time and reactivity for reactor excursions near prompt critical is derived. Also, the neutron density and the average density of delayed neutron precursors as functions of reactivity are presented. By comparing with other analytical methods, he shows that this solution is better and accurate.

Ping *et al.* [28] presents a digital simulator of the PWR-NPP based on lumped mathematical model (point heap reactor) of the system. They focus on the development of a PWR simulator having both primary and secondary cycles. From the results of the simulation, they conclude that the use of thermal feedbacks and poisoning effect induces the self-stability and self-regulation features in the reactor and the developed simulator is able to capture these effects with precision. However, due to the mechanical links of the steam generator and the power control system the self-regulation of the NPP takes a long time.

Torabi *et al.*[42] discusses a robust power control system for PWR core using the Quantitative Feedback Theory (QFT) method. They consider a nonlinear uncertain dynamic system with different working condition for the reactor. A robust control system for the nuclear reactor has been designed by identifying two different control oriented linear models of the plant. The QFT controller is designed and simulated to control the reactor power. They show that the proposed control system has a favourable performance over a wide range of reactor operating conditions.

1.3. OBJECTIVE AND OUTLINE OF THE DISSERTATION

The discussion in sections 1.2 indicates that the modeling and simulation of a PWR-NPP based on first design principles is an important domain in the Nuclear power society. Discussion in the section 1.1 shows that PWRs are emerging as a new approach to meet the long term objectives, in the Indian context.

Based on such a background, the *objective* of the present study is to form an analytical model of the PWR which best describes the behaviour of an actual PWR plant. Emphasis is given on the use of internal feedbacks (temperature and xenon poisoning) in the model and the formulation of equations necessary to describe the primary loop of the PWR in terms of temperature variation. A commercially available software and simulation package is used to investigate the behaviour of the model under transient conditions.

The outline of this study is furnished below.

- At first, the mathematical model of the primary circuit of the PWR-NPP is formulated based on the literature available on the topic [4],[5],[10],[42]. This mathematical model is then implemented using the State space approach and a performance study of the State Space model is conducted on a commercially available software and simulation package.
- After acquiring the results from the simulation of the state space models, a Block Control model of the PWR is developed based on the same mathematical formulations. The simulation results obtained from the Block model are compared to those acquired from the State Space model and differences found, if any are enlisted. This is done to test the performance of the mathematical model under different simulation procedures.
- Lastly, the stability analyses of the State Space models are performed by using the Lyapunov direct method. This topic is addressed in brief.

1.4. STRUCTURE OF THE DISSERTATION

The dissertation is organised as follows.

Chapter 1 introduces the topic of modelling and the relevance of PWRs in the Indian context. A brief overview of the studies that exist in the field of PWR modelling is provided, from whom the objective of this study is drawn.

Chapter 2 gives a complete overview of PWR systems and their dynamics. It also discusses the control aspects of the PWR. The stability issues of Nuclear reactors are presented in brief.

Chapter 3 presents the PWR models designed during this study using the State space modelling approach. It also discusses the importance of modelling in state space. Some general rules and assumptions made during the formation of the models are discussed. A state feedback based design of the reactor regulating system (RRS) is also presented.

Chapter 4 discusses the results of the simulations that were conducted on the PWR models that are designed during this study. It also presents a Block Control model of the PWR system. The results obtained from the Block model are discussed in comparison to the results obtained from the previously developed models. The stability analyses of the PWR models presented in the previous chapter are demonstrated in brief.

Chapter 5 concludes this study with a summary of the stepwise development of the present work and the results presented by it. The future scope for this work is also indicated.

Some deductions, topics and theories relevant to this study are discussed in the Appendix A, Appendix C and Appendix D, respectively. The set of data used in the simulations are provided in Appendix B.

CHAPTER 2

LITERATURE SURVEY

Pressurized water reactors (PWRs) are a class of Light Water Reactors (LWRs). These were initially designed to serve in nuclear propulsion for nuclear submarines owing to their compact size, and hence they have been referred to as *Compact Reactors*. The research and development work was performed by Knolls Atomic Power Laboratory and Westinghouse Betties Laboratories. As a result of this initial R&D work, a commercial PWR was designed and developed for nuclear power plant applications. The first commercial PWR plant that came up was in the United States at Shipping Port, located near Pittsburgh, Pennsylvania [41], [30].

This chapter gives a complete overview of the PWR systems and theories relevant to this study. Firstly, it discusses the PWR physical system, its working principle and few constituent systems. In the next two sections, principles related to the dynamics of the reactor core and its thermal hydraulic behaviour has been focussed upon. The last two sections have been devoted to the study of control and stability aspects of the reactor.

2.1. PWR SYSTEM DESCRIPTION

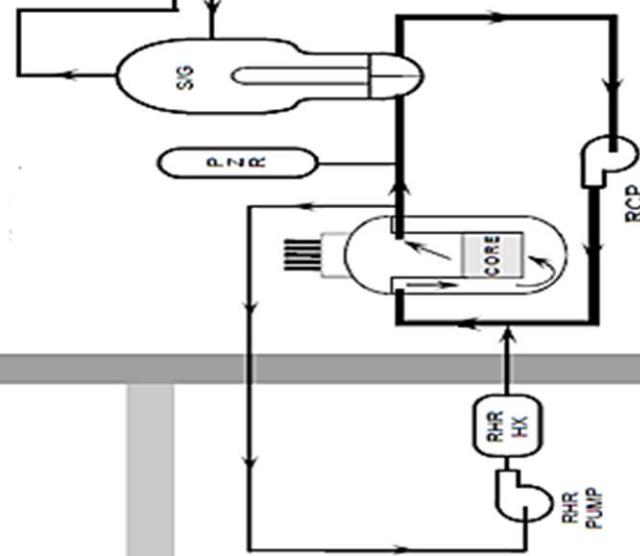
The PWR is a vertical, pressure vessel type, light water moderated and cooled class of LWRs [41], [7]. It is fuelled with enriched UO_2 pins & is an indirect cycle reactor. It uses 2 coolant systems, namely *The Reactor Coolant System (or the Primary Cooling System)* & *The Secondary Coolant System* [29]. Added to these a third cooling system is employed on the secondary side, namely *The Condenser Cooling System*, concerned with the condensate cooling. These reactors are inherently stable owing to a negative fuel and coolant temperature coefficient of reactivity (explained later in section 2.2.1.2). To this date, PWRs constitute a large majority of the world's nuclear power plants. **Fig. 2.1** shows a schematic diagram of the overall PWR system.

2.1.1. WORKING OF THE OVERALL PWR SYSTEM

In basic PWR design, the primary coolant (H_2O under pressure) picks up the nuclear heat generated in the *Core*. The primary coolant is circulated through the reactor core by one or more of high-pressure reactor coolant pumps. It proceeds to the *Steam Generator* where steam is generated and heat energy is transferred to the secondary coolant by means of heat exchangers. The primary coolant is then pumped back to the core. The pressure adequate to limit bulk boiling of the primary coolant is maintained by an electrically heated pressurizer. Steam formed due to heat absorption by the secondary coolant is transported to the steam turbine where it is used for power generation. In commercial reactors, the reactor core is contained within a large pressure vessel with a removable head & is moderated and cooled by light water.

CONTAINMENT BUILDING

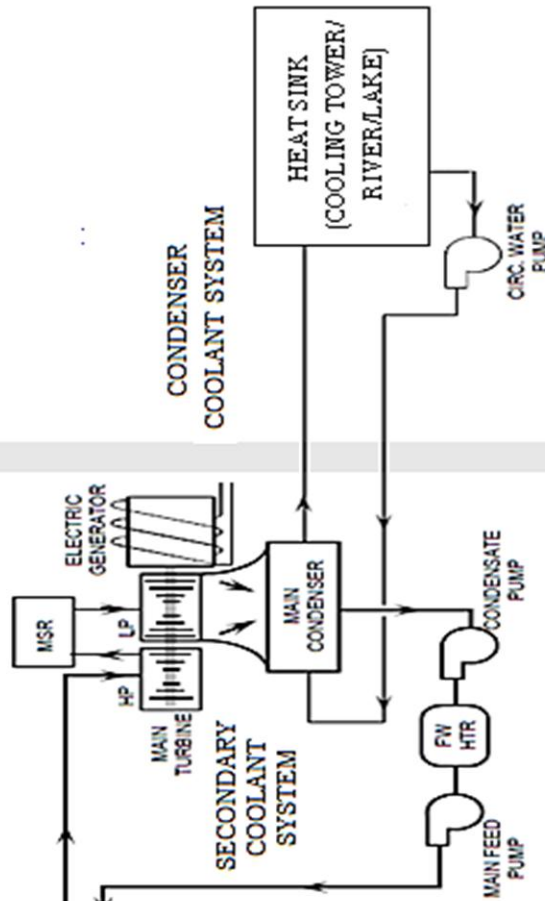
REACTOR COOLANT SYSTEM



AUXILIARY BUILDING

CONTAINMENT SUMP

CONDENSER COOLANT SYSTEM



ACRONYMS

S/G - Steam Generator
PZR - Pressurizer
RCP - Reactor Coolant Pump
HP - High Pressure Turbine
LP - Low Pressure Turbine
MSR - Moisture separator - Reheaters
FW HTR - Feed Water Heater
RHR - Residual Heat Removal

Fig 2.1: Schematic Diagram of Overall PWR system
(Adapted Courtesy: USNRC Technical Training Manual)

The reactor vessel, primary loop components, Steam Generators and auxiliaries are located within a structure, termed as the *Reactor Containment Building* (as shown in **Fig. 2.1**). This structure can contain the steam and fission products, released in the highly unlikely event of a large break in the primary piping, thus acting as the *Tertiary physical Barrier* for nuclear safety. The secondary side consisting of turbines, condenser assembly and the feed water pumps are contained within the *Turbine Building*. Although the indirect cycle has the advantage of providing a radioactively clean secondary system but it requires the use of expensive heat exchangers [41].

2.1.2. PLANT CONSTITUENT SYSTEMS

The PWR plant, as mentioned earlier has three separate cooling systems, viz. Reactor Coolant System, Secondary Coolant System and Condenser Coolant System. The Reactor Coolant System includes the reactor core contained within a reactor vessel, the Steam Generator, the Reactor Coolant pumps and the Pressurizer as shown in **Fig. 2.2**. The Secondary Coolant system on the other and has two main systems viz. the Main steam system and the Condensate feed water system. The Condenser Coolant System mainly consists of the Condenser and Circulating Water Pumps, which take suction from water supplied from the ocean, sea, lake, river etc. In the context of the present study, the Reactor core, the Steam Generator and the Condensate feed water has been described.

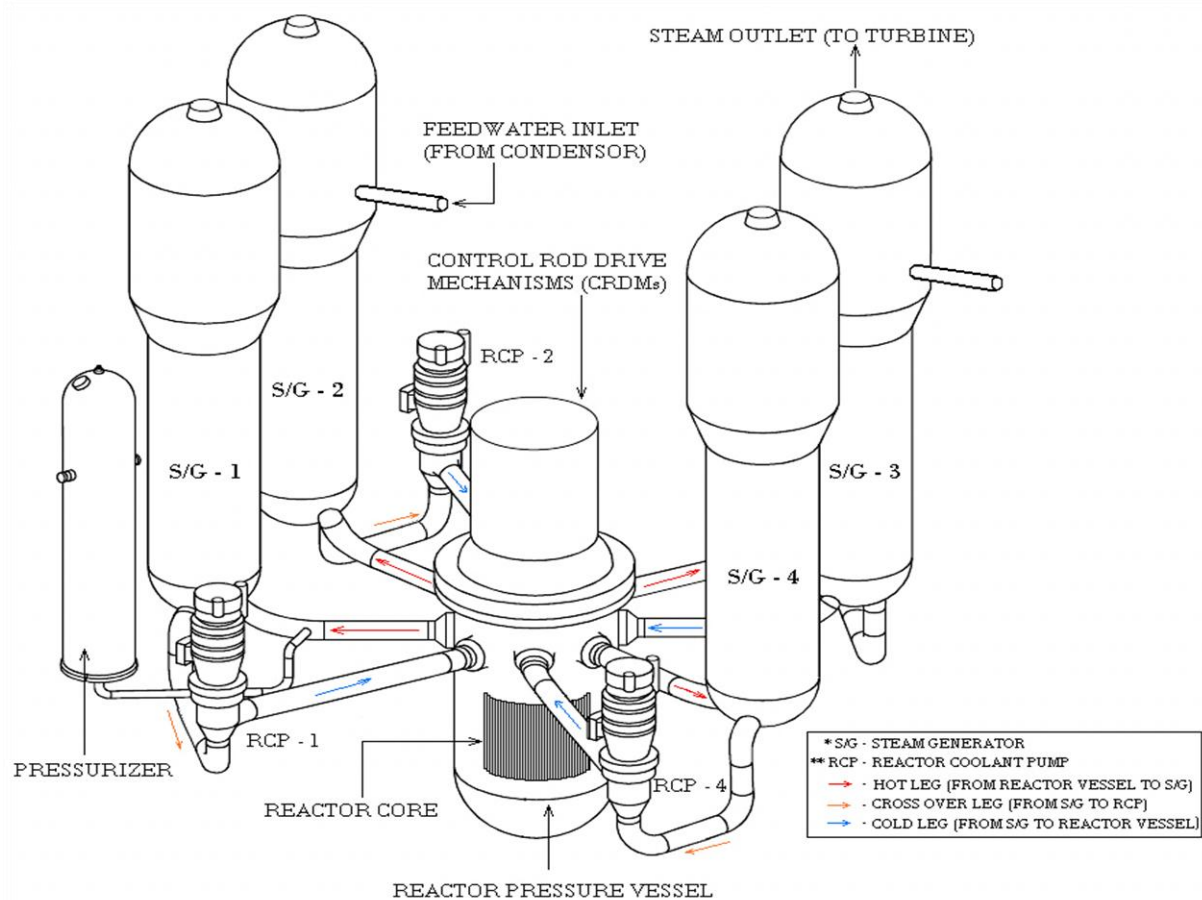


Fig 2.2: 4 - Loop Westinghouse PWR Coolant Assembly
 (Adapted Courtesy: **USNRC Technical Training Manual**)

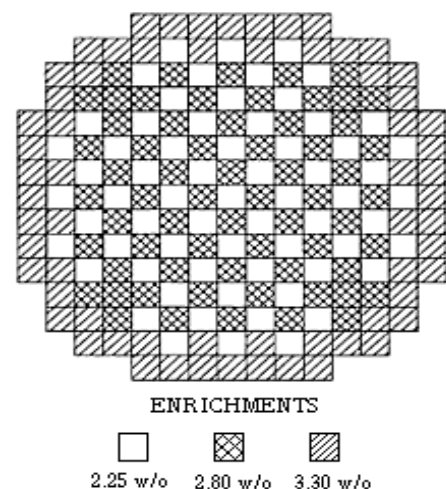
1. Reactor Core

The Reactor Core of a PWR is housed within a Reactor vessel, which is a cylindrical container having a hemispherical bottom head and a removable hemispherical top head. PWRs use a pressure vessel configuration [16], [41] as the Reactor vessel.

The fuel for PWRs is partly enriched ($\sim 3\%$ ^{235}U) UO_2 sintered pellets, with or without burnable absorber (Gadolinium, Gd_2O_3) [7], [41] & the level of enrichment vary from the outer core fuel assemblies towards the middle. The core has *fuel rods* or tubes made up of small cylindrical UO_2 pellets (1 cm. dia. & 2 cm. long [16]) usually concave on the ends, stacked upon one another & enclosed in Zircaloy cladding about 4 m long [16]. The cladding offers the *Primary physical Barrier* to prevent leakage of radioactive materials. These completed pressurized fuel rods are arranged in a square lattice structure called the *fuel assembly* [16]. These fuel assemblies are then arranged in a near cylindrical array to form the *Reactor Core*. **Fig. 2.3** shows the cross sectional view of a typical PWR core at the BOL.

The core contains about 200 assemblies of fuel and control rods. Each fuel assembly consists of a 17×17 square array [7]. Of the 289 spaces available in an assembly, almost 264 are occupied by fuel rods, the remaining contain guide tubes for control rods of the Rod Cluster Control Assemblies (RCCAs) [3], with a central tube available for instrumentation [7].

Fig 2.3: Fuel loading pattern in the core at the beginning of a PWR's life
(Adapted Courtesy: **Introduction to Nuclear Engineering**
– Lamarsh & Baratta)



2. Steam Generator

It is known that the reactor coolant (H_2O under pressure) flows from the reactor to the steam generator; this part of the coolant line is termed as *hot leg*. Within the steam generator, the high pressure, hot reactor coolant flows inside many U-shaped tubes. The secondary coolant (H_2O), or low pressure cooled feed water, coming from the condenser flows around these tubes & captures heat from the hot water inside the tubes, causing it to boil & produce steam. The lower portion of the steam generator where the boiling occurs is called *Evaporator Section* & the wet steam produced in the evaporator passes upwards into a portion known as the *Steam Drum Section* [16]. Moisture from the steam is separated by means of separators & then the steam moves to the turbine. It is important to maintain the moisture content of the steam as low as possible to prevent damage to the turbine blades.

3. Condensate - Feed Water System

A major part of the secondary coolant system is the Condensate –feed water system which starts at the outlet of the condenser, as shown in **Fig. 2.4**. The condensed steam collects in the hot well area of the main condenser. The condensate pumps increase the pressure of the water. The condensate then passes through some *low pressure (L.P) feed water heaters* [32]. The temperature of the condensate is increased to a suitable value in the heaters by using steam from the low pressure turbine. This is done because heating the feed water helps to increase the efficiency of the plant. The condensate flow then enters the suction of the *Main feed water pumps*, which increases the pressure of the water high enough to enter the steam generator. The feed water now passes through a set of *high pressure (H.P) feed water heaters*, which are heated by extracting steam from the high pressure turbine. The flow rate of the feed water is controlled via *Feed Check Valve (FCV)* as it enters the steam generators [32].

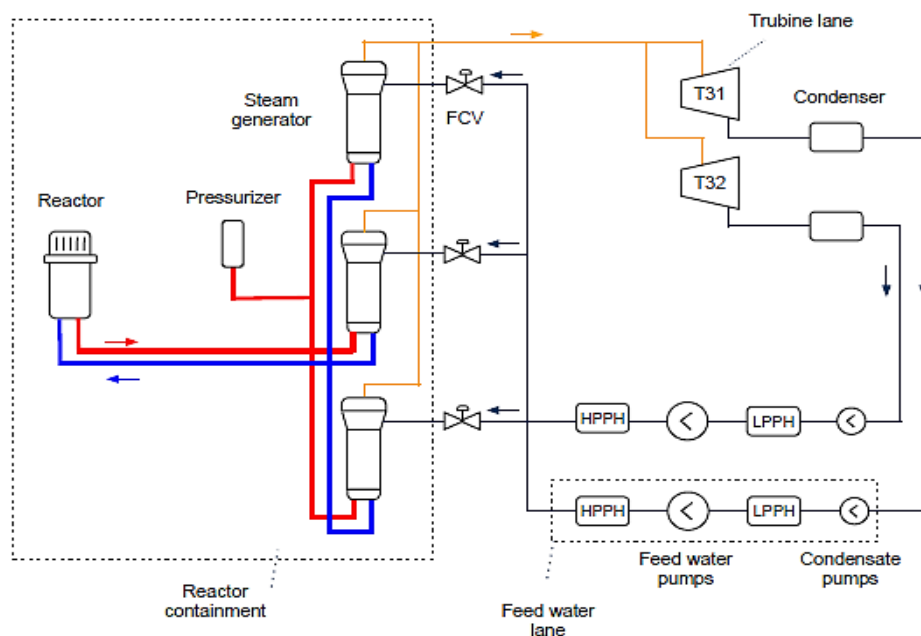


Fig 2.4: Condensate feed water system of a PWR showing L.P, H.P heaters & FCVs
(Adapted Courtesy: Chalmers University of Technology)

With the objective of capturing the performances of these constituent systems in the PWR model, a thorough mathematical study of the dynamics of the reactor is needed. The next two sections discuss this issue in detail.

2.2. **PWR CORE DYNAMICS**

The Reactor core is encapsulated within the reactor pressure vessel, as mentioned earlier. It forms the most vital part of the entire plant. The core is governed by both Neutronic and Thermal hydraulic principles. In this section however, discussions will be confined only to the Neutronic part of it and the mathematical background necessary for representing the Reactor Kinetics shall be developed.

2.2.1. NEUTRONIC PRINCIPLES

The key aspect of reactor core design lies in the description of the production, transport and absorption of neutrons made available from fission within the core. The physical design requirements of the core depend on the delicate balance between the rate of neutron production and loss. In order to understand these complex behaviours, certain parameters are provided in literature which are discussed below.

2.2.1.1. PARAMETERS USED FOR NEUTRONIC STUDIES

In a multiplying medium, i.e. in a medium containing fissile materials which could increase neutron density, such as a nuclear reactor (PWR), some basic properties are expressed in terms of parameters as explained beneath.

1. Infinite Multiplication factor (k_{∞})

The minimum condition for sustaining a chain reaction is: each nucleus that undergoes fission should produce on an average at least one neutron that causes fission of another nucleus. This condition is conveniently expressed in terms of a multiplication factor. Generally, the larger the reactor, lower the fraction of neutron leakage. So for sake of simplicity, a reactor is considered that is infinitely large, and therefore has no neutron leakage. A measure of the increase or decrease in neutron flux in an infinite reactor is the infinite multiplication factor; k_{∞} . The *infinite multiplication factor* is the ratio of the neutrons produced by fission in one generation to the number of neutrons lost through absorption in the preceding generation [7], expressed mathematically as,

$$k_{\infty} = \frac{\text{Neutron Production from fission in one generation}}{\text{Neutron Absorption in the preceeding generation}} \quad (2.1)$$

2. Effective Multiplication factor (k_{eff})

The infinite multiplication factor can fully represent only a reactor that is infinitely large, because it assumes that no neutrons leak out of the reactor. To completely describe the neutron life cycle in a real, finite reactor, it is necessary to account for neutrons that leak out. The multiplication factor that takes leakage into account is the *effective multiplication factor* (k_{eff}), which is defined as the ratio of the neutrons produced by fission in one generation to the number of neutrons lost through absorption and leakage in the preceding generation [7]. The effective multiplication factor may be expressed mathematically as shown below.

$$k_{eff} = \frac{\text{Neutron Production from fission in one generation}}{\text{Neutron (Absorption+Leakage) in the preceeding generation}} \quad (2.2)$$

3. Six Factor Formula

A group of fast neutrons produced by fission can enter into several reactions. Some of these reactions reduce the size of the neutron group while other reactions allow the group to increase in size or produce a second generation. There are six factors that are completely

independent of the size and shape of the reactor which give the inherent multiplication ability of the fuel and moderator materials taking into account the leakage. This *Six factor formula* accurately represents the effective multiplication factor as shown in the equation below.

$$k_{eff} = \varepsilon P_f p P_t f \eta \quad (2.3)$$

Here, ε = Fast Fission Factor; P_f = Fast Non-Leakage Factor; p = Resonance Escape Probability; P_t = Thermal Non-Leakage Factor; f = Thermal utilization factor; η = Reproduction Factor. Each of these six factors represents a process that adds to or subtracts from the initial neutron density produced by fission in a generation (refer **Appendix D.1**).

4. Reactivity

Reactivity represents the fractional change in neutron population per generation. Consider that there are N_0 neutrons in the preceding generation, and then there are $N_0 k_{eff}$ neutrons in the present generation. The numerical change in neutron population is $(N_0 k_{eff} - N_0)$. The gain or loss in neutron population $(N_0 k_{eff} - N_0)$, expressed as a fraction of the present generation $(N_0 k_{eff})$ is shown below. This ratio represents *reactivity*.

$$\rho = \frac{N_0 k_{eff} - N_0}{N_0 k_{eff}} \Rightarrow \rho = \frac{k_{eff} - 1}{k_{eff}} \quad (2.4)$$

Equation (2.4) represents the general form in which reactivity is expressed [7]. It can be seen that ρ may be positive, zero, or negative, depending upon the value of k_{eff} . The larger the absolute value of reactivity in the reactor core, the further the reactor is from criticality. It may be convenient to think of reactivity as a measure of a reactor's departure from criticality. By definition, the value for reactivity, a result directly from the calculation of Equation (2.4) is in units of $\Delta k/k$. Alternative units are $\% \Delta k/k$; pcm (percent millirho) and dollars [7].

5. Reactor Criticality Conditions

The condition where the neutron chain reaction is self-sustaining and the neutron population is neither increasing nor decreasing is referred to as the *critical condition* for a reactor. The effective multiplication factor (k_{eff}) is the best indicator of this condition. The three criticality conditions can be defined as listed below [7], [11].

- *Reactor is Critical:* If a reactor has $k_{eff} = 1$, the reactor is Critical. This is the preferred normal operating condition of a reactor.
- *Reactor is Super - critical:* If the neutron production is greater than the absorption and leakage, the reactor is called supercritical. In a supercritical reactor, $k_{eff} > 1$ and the neutron flux increases each generation. Super criticality is not a dangerous or abnormal condition for a reactor. The reactor is to be made supercritical in order to increase the power of the reactor.
- *Reactor is Sub - critical:* If the neutron production is less than the absorption and leakage, the reactor is called subcritical. In a subcritical reactor, $k_{eff} < 1$ and the

neutron flux increases each generation. This condition is not abnormal and is required to reduce the reactor power or to shut down the reactor.

For simplicity however, reactivity (ρ) is used to convey the criticality of a reactor, rather than k_{eff} , since a change in ρ is easier to understand than a change in k_{eff} . **Table 2.1** describes the three criticality conditions and the effect of change in k_{eff} with ρ .

Table 2.1: Variation of k_{eff} with ρ indicating Reactor Criticality

Reactor Condition	Effective Multiplication factor (k_{eff})	Reactivity (ρ)
Supercritical	>1	Positive (> 0)
Critical	$=1$	Zero ($= 0$)
Subcritical	<1	Negative (< 0)

2.2.1.2. REACTIVITY EFFECTS DUE TO CERTAIN PARAMETERS

The amount of reactivity (ρ) in a reactor core determines the behaviour of the neutron population and consequently the reactor power at any given time. However, transient changes in reactivity in an operating reactor may be witnessed owing to certain parameters like temperature, pressure, fuel depletion, or poisons. To quantify the effect that a variation in parameter will have on the reactivity of the core, *reactivity coefficients* are used. Reactivity coefficients are the amount by which the reactivity will change ($\Delta\rho$) for a given change in the parameter & are generally denoted by α_x , where x represents some reactor parameter that affects reactivity [2]. Mathematically, it can be expressed as,

$$\alpha_x = \frac{\Delta\rho}{\Delta x} \quad (2.5)$$

If the parameter x increases and positive reactivity is added, then α_x is positive. If the parameter x increases and negative reactivity is added, it's negative.

1. Effect of Temperature Changes on reactivity

Temperature variation in the system is one of the major causes for transient changes in the effective multiplication factor (or reactivity) in an operating reactor. These variations may be localised or may affect the reactor as a whole. Hence, it is of paramount importance to understand in advance whether such temperature transients cause the reactivity of the reactor to increase or decrease and how fast the changes manifest themselves.

Here, the *temperature coefficient of reactivity* is defined based on equation (2.5) as the change in reactivity ($\Delta\rho$) per unit change in temperature. Mathematically, if ΔT is the change in temperature of any reactor material, then the temperature coefficient of reactivity α_T is expressed as,

$$\alpha_T = \frac{\Delta\rho}{\Delta T} \quad (2.6)$$

According to the equation, its units are $\text{mk}/^{\circ}\text{C}$. The coefficient can be positive or negative depending on the behaviour of the component to temperature change [2], [7].

CASE 1: If the temperature coefficient of reactivity is *positive*, i.e. the reactivity of the reactor increases with transient increase in temperature. Practically this would be manifested in the form of increase in neutron density (n), and as a consequence increase in fission rate and henceforth increase in rate of heat generation. This would lead to a further increase in temperature and hence increase in reactivity. In other words, unless means are employed externally to reduce reactivity or increase the rate of heat removal, this could make such a reactor potentially unstable. PHWRs exhibit this property.

CASE 2: If the temperature coefficient of reactivity is *negative*, i.e. the reactivity of the reactor decreases with transient increase in temperature. Practically this would be manifested in the form of decrease in neutron density. As a consequence, the fission rate would decrease and henceforth there will be a decrease in rate of heat generation. This would in turn cause a decrease in temperature.

Similarly, if there were a transient decrease in temperature it would in turn cause an increase in reactivity, which would increase the rate of heat generation and hence tend to restore the original temperature. In such a reactor, a temperature transient is always self-limiting and hence desirable feature for reactor stability & operational safety. PWR exhibits the property of self-stability.

PWR, under consideration here, is a heterogeneous reactor where the fuel and moderator are separate. Hence, a distinction must be made between the fuel and the moderator temperature coefficient. The fuel and moderator coefficients are different in magnitude and possess different time constant, i.e. the time required for a temperature change to produce an appreciable change in reactivity. The time constant is short for fuel but is much longer for moderator [7], since heat produced in the fuel takes time to get transferred to the moderator.

a. Fuel Temperature coefficient of reactivity

The *fuel temperature coefficient* is the change in reactivity per degree change in fuel temperature. This coefficient is called the *prompt temperature coefficient* because increasing the reactor power, would immediately cause a change in fuel temperature. This is denoted by α_f .

In water-moderated, low uranium-235 enriched reactors like PWR, raising the temperature of the fuel will raise the resonance absorption in uranium-238 and in turn decrease the fission cross-section. This causes emission of less fission neutrons, due to the Doppler Effect [7] or more specifically *Doppler Broadening*. The increase in resonance absorption lowers the resonance escape probability (p). Hence, the fuel temperature coefficient is determined mainly by the change in the *resonance escape probability* (p) and is generally considered to be negative [11].

b. Moderator(or Coolant) Temperature coefficient of reactivity

The change in reactivity per degree change in moderator temperature is called the *moderator temperature coefficient* of reactivity and is denoted by α_c or α_m . This coefficient is also called the *delayed temperature coefficient* because it takes time for the heat from the fuel to be transferred to the moderator. The magnitude and sign (+ or -) of the moderator temperature coefficient is primarily a function of the *moderator-to-fuel* ratio (refer **Appendix D.3**). If a reactor is under moderated, it will have a negative moderator temperature coefficient. If a reactor is over moderated, it will have a positive moderator temperature coefficient [2]. A negative moderator temperature coefficient is desirable because of its self-regulating effect. For example, an increase in reactivity causes the reactor to produce more power. This raises the temperature of the core and adds negative reactivity, which slows down the power rise.

In the case of a PWR, light water acts as both the moderator and coolant. Hence, this coefficient can be used to denote both of these. Now, as the moderator temperature increases, the density of the moderator decreases. This decrease in density causes the moderation of the neutrons to reduce and fewer neutrons are thermalized. This causes the fission rate of the thermal neutrons to reduce and thus reducing fission power in turn. Hence, the moderator temperature coefficient is determined mainly by the change in the *thermal utilization factor* (f) and is generally considered negative [7].

2. Effect of Pressure Changes on reactivity

The reactivity in a reactor core can be affected by the system pressure. Here, the *pressure coefficient of reactivity* is defined based on equation (2.5) as the change in reactivity ($\Delta\rho$) per unit change in pressure. Mathematically, if ΔP is the change in pressure of any reactor material, then the pressure coefficient of reactivity α_p is expressed as,

$$\alpha_p = \frac{\Delta\rho}{\Delta P} \quad (2.7)$$

The pressure coefficient of reactivity for the reactor results from the effect of pressure on the density of the moderator [2], [7]. For this reason, it is sometimes referred to as the *moderator density reactivity coefficient*. As pressure increases, density correspondingly increases, which increases the moderator-to-fuel ratio [2] in the core. In a typical under moderated core, the increase in the moderator-to-fuel ratio will result in a positive reactivity addition. In reactors that use water as a moderator, the absolute value of the pressure reactivity coefficient is not a major factor because it is very small compared to the moderator temperature coefficients [2]. Hence, in case of PWRs its effect can be neglected.

2.2.2. REACTOR CORE KINETICS

Nuclear reactor kinetics is the study of the time dependent behaviour of the neutrons in a reactor core. The prediction of the time behaviour of the neutron population induced by various changes in multiplication factor is essential to the safe design and operation of a nuclear reactor. However, the accurate prediction is difficult to obtain. Literature on this topic is readily available [2],[16].

2.2.2.1. STUDY OF REACTOR POINT KINETICS EQUATIONS

The neutron population in a reactor can be modelled mathematically by the *Neutron Transport equation*. It happens to be the counterpart of the Boltzmann equation which is the main tool in the study of gas dynamics [7]. It expresses the distribution of neutrons in space, energy and time in a straightforward manner. The neutron transport equation is an integrodifferential equation whose analytical solution is usually impossible and approximate solutions are a time consuming affair [38]. This equation is however, fundamental to calculating the conditions for criticality of a reactor, the spatial distribution of the neutron flux at various energies & other parameters which form the basis for nuclear reactor design.

So, to get over this problem, approximate versions of the above theory are usually applied for the analysis. One of the simplest approximations is the *Neutron diffusion theory approximation*, which was drawn from a relationship to the Fick's law of diffusion for gases and hence is appropriately named [7], [38]. The *Neutron diffusion theory* can be solved by using two different methods; one is by assuming a single energy neutron group which leads to the Point kinetic equations and the other by assuming multiple energy neutron groups that lead to the multi-group neutron diffusion equations.

I. The *Point kinetics equations* assume that all the neutrons belong to a single energy group. It is basically a one-group diffusion model obtained via spatial discretization of partial differential equations of the diffusion theory [7]. It offers a simpler mathematical description of the neutron behaviour for which the spatial and angular neutronics properties are assumed to be constant in time. The negligence of the angular effects may not affect much, but spatial effects definitely play a major role in the neutronics behaviour and hence their negligence renders the point kinetics inadequate to describe the detailed behaviour of neutrons in a large reactor.

Hence, point kinetic models are appropriate for small and compact reactors like PWRs, where it is assumed that no spatial variations occur & the behaviour of the neutron population in the entire reactor is strongly coupled from point to point.

II. The neutrons in a nuclear reactor have energies which span the range of 0.01 eV to 10 MeV [7]. By considering that all the neutrons in a reactor possesses an average energy or they belong to a single energy group may lead to a non-realistic model for the nuclear reactor, especially where neutronic behaviour tends to be loosely coupled from point to point. A more sophisticated model is *multi-group diffusion theory* which can be obtained directly from transport equation [38] in a rather intuitive way. It describes the neutron population of a reactor much better since space, time and energy are not held constant & treated with sufficient accuracy. So, this theory is more applicable in case of large reactors like PHWRs and AHWRs, where the neutronic behaviour tends to be loosely coupled from point to point, which means that a change in the neutron flux or power density at one point in the core will not be felt at other points until after an appreciable time delay [35], [39].

Assumptions for Point Kinetic model of Nuclear Reactor

1. Every point in the Reactor core is equally treated.

2. Production, diffusion, absorption and leakage of neutrons take place at single energy.
3. Time variation of neutron flux at all points of core is uniform.

Formulation of Point Kinetics equations

The starting point of the point kinetic equation is of course the space-time neutron diffusion equations and the derivation process is readily available in literature [38].

The principle of neutron conservation can be expressed for a system of monoenergetic neutrons (one – speed neutrons) in a simple form. Here, the net rate of change of the neutron density with time $[\partial n(r, t)/\partial t]$ for a hypothetical volume element in the reactor medium is given by,

$$\text{Source} - \text{Absorption} - \text{Leakage} = \frac{\partial n(r, t)}{\partial t} \quad (2.8)$$

Where, all the terms on the LHS are time rates of these processes per unit volume.

The source term here can be any radioactive decay process producing neutrons and so can be represented by the symbol S , which is the rate of production of fission neutrons per m^3 per second. The rate of absorption of neutrons (single energy) can be expressed by $\Sigma_a \phi$, where, Σ_a is the macroscopic absorption cross section of one speed. The leakage term is calculated by employing Fick's Law and is found to be $D\nabla^2 \phi$ [38], [7].

The diffusion equation is expressed commonly in the form,

$$D\nabla^2 \phi(r, t) - \Sigma_a \phi(r, t) + S(r, t) = \frac{\partial n(r, t)}{\partial t} \quad (2.9)$$

Now, considering that the one-group source term $S(r, t)$ is composed of both prompt neutron source term (S_p) and delayed neutron source term (S_D) [38].

$$D\nabla^2 \phi(r, t) - \Sigma_a \phi(r, t) + S_p(r, t) + S_D(r, t) = \frac{\partial n(r, t)}{\partial t} \quad (2.10)$$

By considering that β is the fraction of delayed neutrons, so $(1-\beta)$ becomes the prompt neutron fraction. Then, substituting the source terms with their evaluated values gives,

$$D\nabla^2 \phi(r, t) - \Sigma_a \phi(r, t) + (1 - \beta)k_\infty \Sigma_a \phi(r, t) + \sum_{i=1}^6 \lambda_i \hat{C}_i(r, t) = \frac{\partial n(r, t)}{\partial t} \quad (2.11)$$

Where, $S_p(r, t) = (1 - \beta)k_\infty \Sigma_a \phi(r, t)$ & $S_D(r, t) = \sum_{i=1}^6 \lambda_i \hat{C}_i(r, t)$

Also, $i = (1, \dots, 6)$ represents the six groups of delayed neutrons considered based on their half – lives (λ_i) and $\hat{C}_i(r, t)$ be the concentration of the delayed neutron precursors. In the equation (3.21) both ϕ (and n) and $\hat{C}_i$ are functions of both space and time $[(r, t)]$ in a finite reactor. However, it is assumed that the variables are separate. This is what leads to the “Point- Reactor” model [7]. Hence, it is considered that,

$$\phi(r, t) = v n(t) \psi_1(r) \text{ \& } \hat{C}_i(r, t) = C_i(t) \psi_1(r) \quad (2.12)$$

Here, $\psi_1(r)$ is the fundamental mode solution of the Helmontz equation represented as, $\nabla^2 \psi_n + B_g^2 \psi_n = 0$.

Substituting the equation (2.12) in (2.11) and solving thereafter leads to the point kinetic equation. The first equation of point kinetics as,

$$\frac{dn(t)}{dt} = \frac{(\rho(t) - \beta)}{l} n(t) + \sum_{i=1}^6 \lambda_i C_i(t) \quad (2.13)$$

Here, $\rho(t)$ is the reactivity and l is the prompt neutron lifetime. The diffusion theory represents the net rate of formation of the delayed neutron precursors (DNP) of the i^{th} group as,

$$\frac{d\hat{C}_i(r,t)}{dt} = \beta_i k_{\infty} \Sigma_a \phi(r,t) - \lambda_i \hat{C}_i(r,t) \quad (2.14)$$

Again, solving the equation (2.14) by the separation of variables as stated in equation (3.12), leads to the second equation of the point kinetics as,

$$\frac{dC_i(t)}{dt} = \frac{\beta_i}{l} n(t) - \lambda_i C_i(t) \quad (2.15)$$

Where, $i = (1, \dots, 6)$ represents the six groups of delayed neutrons.

Explanation for Point Kinetics equations

The equations (2.13) and (2.15) represent a set of stiff non-linear differential equations and are the fundamental equations of reactor kinetics for the *point-reactor model*.

The first and second terms on the RHS of equation (2.13) gives the contribution of prompt neutron population per unit time and delayed neutron population per unit time, respectively. The first and second terms on the RHS of equation (2.15) represent the rate of change of precursor production and rate of change of precursor loss thus equalling the rate of change neutron precursor concentration.

Also, it is clear from equation (2.13) that the non-linearity in the point kinetic model is introduced as a result of the multiplication of the quantities $\rho(t)n(t)$. The behaviour of any such non-linear system is far more complex than a linear system. These concepts have been elaborately explained later while developing the reactor model.

2.2.2.2. STUDY OF FISSION PRODUCT POISONING IN REACTOR CORE

The other major study in reactor core kinetics is concerned with the formation of fission product poisons. They accumulate due to decay of fission fragments during the course of operation of a nuclear reactor.

These products act as neutron poisons and thereby add negative reactivity (ρ) by decreasing the thermal utilization factor (f) [2]. The concentration of fission product poisons in a reactor depends on the thermal – neutron flux. Hence, it appears from the dependency that some allowance for the effect of poisoning should be included in the point kinetics equations. However, the rate of change of neutron density is much more than that of the fission-product

concentration. And so, the point kinetics can be treated independently of them. The fission product poisons have an important influence on the reactivity (induces negative reactivity). So they must be taken into consideration while designing the reactor core & should be compensated for, by addition of positive reactivity in the form of control rods.

The most significant fission product poison is Xenon-135. Xenon-135 has an absorption cross-section 4000 times that of Uranium-235 and thus adds negative reactivity to the reactor [7]. Although very undesirable, it is completely unavoidable and therefore the reactor needs to be controlled by taking into account the behaviour of Xenon-135.

Xenon Dynamics

Xenon-135 has an exceptionally large (2.6×10^6 barns) neutron capture cross section [7]. It is produced directly from fission in small amounts, but is more commonly formed as a product of the tellurium-135 decay chain shown below in **Fig. 2.5**.

The fission yield (γ) for xenon-135 is about 0.3%, while γ for tellurium-135 is about 6% [7]. But, the half-life for tellurium-135 is so short compared to the other half-lives; it can be assumed that iodine-135 is produced directly from fission. Iodine-135 is not a strong neutron absorber, but decays to form the neutron poison xenon-135. 95% of all the xenon-135 produced comes from the decay of iodine-135 [7]. Therefore, the half-life of iodine-135 plays an important role in the amount of xenon-135 present. Xenon-135 is removed from the reactor also by two processes as shown in the figure above. The first is by nuclear decay of Xenon-135 into Cesium-135 and ultimately Barium-135, which is a stable product having low neutron absorption cross section. This accounts for around 10% of Xenon-135 removal [7]. The second is by neutron absorption (radioactive capture) yielding Xenon-136 which is not a significant neutron absorber. This accounts for around 90% of Xenon-135 removal from the reactor.

Formulation of Xenon Dynamics equations

The rate of change of iodine concentration is equal to the rate of production minus the rate of removal. This can be expressed in the equation below [7], [16].

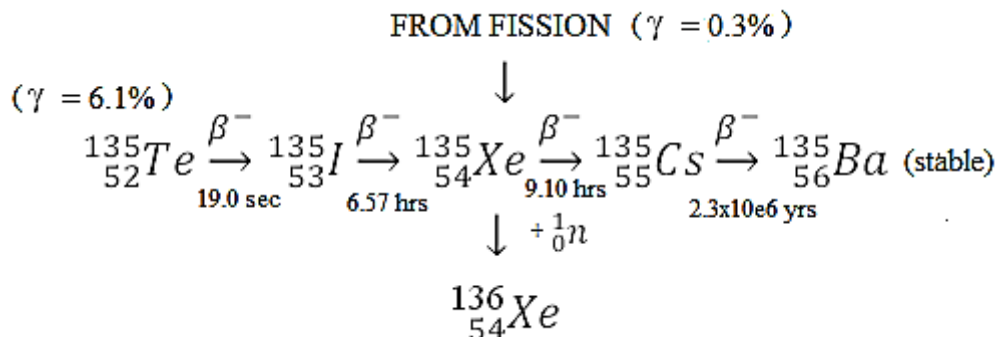


Fig 2.5: Xenon -135 Formation and Removal decay chain

Rate of iodine concentration change = Fission Yield – Decay rate– Burnup rate

$$\Rightarrow \boxed{\frac{dI}{dt} = \gamma_I \Sigma_f \phi - \lambda_I I - \sigma_I I \phi} \quad (2.16)$$

Since the quantity σ_I is very small, the burn up rate term may be ignored, and the expression for the rate of change of iodine concentration may be modified as shown below.

$$\boxed{\frac{dI}{dt} = \gamma_I \Sigma_f \phi - \lambda_I I} \quad (2.17)$$

The rate of change of the xenon concentration is equal to the rate of production minus the rate of removal. Recall that 5% of xenon comes directly from fission and 95% comes from the decay of iodine [7]. The rate of change of xenon concentration is expressed by the following equation.

*Rate of Xe – 135 concentration change = (Xe – 135 Fission Yield) +
– (Xe – 135 Decay rate) – (Xe – 135 Burn – up rate)*

$$\Rightarrow \boxed{\frac{dX}{dt} = \gamma_X \Sigma_f \phi + \lambda_I I - \lambda_X X - \sigma_X X \phi} \quad (2.18)$$

Elaboration on the Xenon Poisoning Effect

From the equations (2.17) and (2.18), the following facts have been derived.

- When the rate of production of iodine equals the rate of removal of iodine, equilibrium exists. Since the equilibrium iodine concentration is proportional to the fission reaction rate, it is also proportional to reactor power level.
- For xenon-135 to be in equilibrium, iodine-135 must also be in equilibrium. It can be seen that the equilibrium value for xenon-135 increases as power increases, because the numerator is proportional to the fission reaction rate. Thermal flux is also in the denominator; therefore, as the thermal flux increases, the $\sigma_X \phi$ term begins to dominate; the xenon-135 concentration approaches a limiting value. So, during reactor operation, xenon concentration reaches equilibrium and an equivalent reactivity is induced [2].
- It is also important to note that overriding xenon induced poisoning is important for reactors during start-up. Once the reactor is shut down, the neutron flux is reduced. However, xenon continues to build up as a decay product of iodine which was produced when the reactor was in operation. A reactor can be started immediately after shut down when the xenon concentration is small. However, it becomes difficult to override xenon after certain time duration (~ 0.5 hrs.). The reactor can then be started only after the xenon has substantially reduced by radioactive decay which takes about 3 days [2], [7].

The equations (2.17) and (2.18) represent the xenon dynamics model. Since there is no void reactivity feedback in the PWR to damp the Xenon and Iodine fission product build-up, the PWR may be subjected to Xenon oscillations.

2.3. PWR THERMAL HYDRAULICS

Thermal hydraulics is basically the study of heat transfer and fluid mechanics principles to accomplish the desired rate of heat removal from the power plant core under both operational and accidental scenarios [40].

Nuclear reactor differs only in the source of heat generation, as compared to the conventional power plants. Apart from that, they are like any other heat producing system, where heat is generated in a medium (nuclear fuel) and then transported to another medium (reactor coolant) to be utilized as useful work. Hence, the fundamental principles of thermal hydraulics get applied to the reactor systems & the PWR has to be studied in this light. Here, the focus will be on the operational conditions of the reactor only.

2.3.1. FUNDAMENTALS OF HEAT TRANSMISSION

In order to study and model the thermal hydraulic behaviour of the PWR, it is a prerequisite to know the fundamental processes that governs the heat transfer from one medium to another. There are three general mechanisms by which heat propagation occurs, viz. Conduction, Convection and Radiation. Each of these has been discussed below.

1. Conduction

The conduction mechanism of heat propagation occurs from a region of higher temperature to a region of lower temperature through a medium (solid, liquid or gaseous) or between different mediums in direct physical contact. Hence, the heat transfer occurs between two bodies primarily due to a temperature gradient [40], [7]. The rate of steady flow of heat for the one dimensional case by conduction is provided by the Fourier equation represented below.

$$Q = -kA \frac{dT}{dx} \quad (2.19)$$

Here, Q is the rate at which heat is conducted in the x – direction

A is the area of the plane normal to this direction

dT/dx is the temperature gradient

k is the thermal conductivity of the medium (in $\text{Wm}^{-1}\cdot\text{C}^{-1}$)

2. Convection

Thermal convection is an energy or heat transport mechanism which is usually concerned with the transfer of heat across a solid-fluid interface and involves macroscopic motion of the fluid. Convection is usually the dominant form of heat transfer in liquids and gases. When a circulating fluid comes in contact with a stationary surface, a thin boundary layer develops adjacent to the wall [40]. Since thermal conductivity of fluids is low, the heat flow from the moving fluid to the wall is mainly due to convection.

Convection can be categorised mainly into two types, i.e. *free convection* and *forced convection* [7]. While free convection occurs due to temperature gradient whereas, forced convection is caused by mechanical means such as coolant pumps. The fundamental equation

governing convective heat transfer (free and forced both) is the Newton's law of cooling, given as,

$$Q = hA(T_S - T_F) \quad (2.20)$$

Here, Q is the convective heat flow rate
 A is the surface area of the material exposed to the heat transfer
 T_S is the surface temperature of the material
 T_F is the temperature of the undisturbed fluid
 h is the heat transfer coefficient

3. Radiation

The transfer of heat by radiation basically occurs via radiant energy or wave motion from one body to another through the intervening space, without any physical contact between them. For hot matter (temperature greater than absolute zero), it is the emission of electromagnetic waves owing to the thermal energy it possess. Thermal energy results in kinetic energy in the random movements of atoms and molecules in matter [7]. Electromagnetic radiation or thermal radiation does not require the presence of matter to propagate and travels in the vacuum of space infinitely far if unobstructed.

In nuclear reactor core, all the three mechanisms may occur during transfer of the heat from fuel to the coolant. However, the most prominent role is played by convection and so it is assumed to be the sole contributor while developing the thermodynamic equations. Owing to this assumption, the coolant flow rate gains practical importance in nuclear reactors.

2.3.2. *STUDY OF HEAT TRANSFER IN PWR*

At an elementary level, any nuclear reactor plant, in this case the PWR basically consists of a reactor, coolant loops (both primary & secondary), the steam generator (heat exchanger) and the steam system. In order to study the PWR basic loop as a servomechanism or control loop, it is a prerequisite to understand the elementary PWR plant thermodynamic structure, apart from the point kinetic model describing the neutron kinetics of the reactor. From the section 2.1 describing PWR system, it can be inferred that there are primarily three places where heat exchange takes place [41].

- First is within the reactor core, where heat transfer occurs at the boundary of the fuel and coolant.
- Second point of heat exchange is at the steam generator between the primary and secondary coolant.
- The final point of heat exchange is at the condenser on the secondary side.

At all these places the heat transfer occurs based on the temperature difference the two interacting medium via convection and radiation. This temperature difference acts as the driving force for the thermal transport rate process.

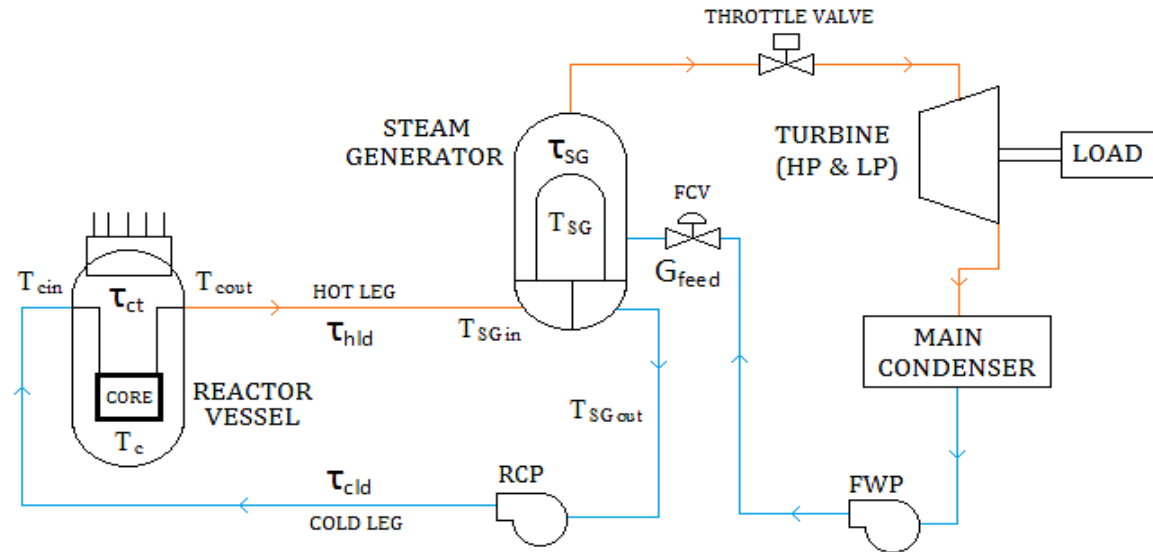


Fig 2.6: Schematic Diagram of PWR plant thermodynamic loop

It will be assumed that there exists an ideal driving force to ensure 100% heat transfer at all the three heat exchange points [21]. Here, the thermodynamic structure will be developed progressively from the differential equations for the thermodynamics of the reactor to the transport delays in the primary loop, the transfer of energy in the steam generator and the use of this energy in the steam system. However, the thermodynamics of only the primary loop will be considered. The situation here will be simplified by the fact that the moderator and coolant (heat transfer medium) in a PWR are the same, i.e. light water (H_2O), which helps to eliminate the double heat transfer mathematics. **Fig. 2.6** shows the schematic diagram of an elementary PWR thermodynamic loop.

2.3.2.1. THERMAL HYDRAULICS OF THE PWR CORE

In order to have an accurate model for describing the variations in reactor thermal power, detailed thermal hydraulic analysis of the core is needed. In the core, the fission of each U-235 nuclei liberates around 210 MeV of energy [7]. This fission energy gets liberated in two ways. One is the instantaneous release of energy in various forms primarily as the kinetic energy of fission products (~ 167 MeV), the energy of fission neutrons (~ 5 MeV) etc., summing up to a total of ~ 187 MeV. The other way is the delayed liberation of energy primarily from the decay of fission products (~ 23 MeV) [7]. This total energy manifests itself in a rise of the temperature of the fuel. So, if no energy is withdrawn from the fuel, the temperature will continue to rise indefinitely as long as the fission occurs. Hence, to keep the fuel at a steady temperature this heat needs to be transported by a suitable medium, i.e. the coolant (H_2O in this case). So in reactors, the temperature difference between the fuel and the surrounding coolant causes heat to convectively flow from the fuel to the coolant, which forms the basis for extraction of heat from fission process.

Assumptions for Core thermal hydraulics

1. As the PWR is compact and small in design, it will be assumed that the fuel in the core is lumped fuel.
2. The total energy of fission is produced instantaneously (no delayed energy) and it gets stored as a whole in the fuel, i.e. there is no loss of energy/heat from the fuel boundaries, since the major energy sources, release their energy within the core itself. Also, it is assumed that no heat enters the core from the outside.
3. The fuel elements possess constant density and constant specific heat [33].
4. The reactor coolant flow rate is assumed to be constant, so the heat transfer coefficient (h) remains constant (as h is function of flow rate) [33].

Formulation of Core thermal hydraulic equations

With the above assumptions in place, the differential equations for the thermodynamics of the reactor could be developed considering the heat balance of fuel and coolant. Based on our first assumption, a one dimensional heat flow from the fuel to the coolant will be considered. Here, the heat generated in a given region plus the heat entering the region from outside equals the heat required in raising the temperature of the fuel and that of the coolant plus the heat escaping from the boundaries, all effects are in per unit time. Although from our second assumption, the heat entering the core from outside and also the heat escaping from the boundaries is neglected. Also, since no loss of heat occurs, the entire fission heat is absorbed by the fuel. Hence, it is convenient to formulate that,

Rate of heat added to the fuel by fission =

$$\text{Rate of heat absorbed by the fuel} - \text{Rate of heat rejected out of the fuel} \quad (2.21)$$

From basic thermodynamics, it is known that the change in the heat content of a medium (in this case fuel) is equal to the product of mass (m_f), specific heat (C_f) and the temperature change of the medium (ΔT_f). Mathematically,

$$Q = m_f C_f \Delta T_f = m_f C_f \frac{dT_f}{dt} \quad (2.22)$$

And, for the convective heat transfer of heat to the coolant, the Newton's law of cooling is utilized as,

$$Q = h_f A_f (T_f - T_c) \quad (2.23)$$

Here, T_f is the fuel temperature and T_c is considered as the average coolant temperature within the core, which is the average of the inlet coolant temperature (cold leg temperature, T_{cin}) & the outlet coolant temperature (hot leg temperature, T_{cout}), as shown in the **Fig. 2.6**. This consideration comes from the practical field, the logic being that when a reactor is operating at a power level high enough for the negative temperature coefficients to be significant, the temperature within the reactor will never be uniform and an isothermal temperature coefficient will not be applicable [33]. Such a situation calls for the use of an average temperature for the coolant. Mathematically,

$$T_c = (T_{cin} + T_{cout})/2 \quad (2.24)$$

Also, the amount of heat absorbed by the fuel is the same as the amount of heat generated at a particular operating power defined by n (neutron power). Hence, $F_r n$ must define the overall reactor specific power, where, F_r is the conversion factor and ranges between 0 and 1 [27], [28], [42].

Then, substituting these values from (2.22) & (2.23) in (2.21), the Fuel temperature (T_f) equation for PWR can be obtained as given below [10], [24], [39].

$$m_f C_f \frac{dT_f}{dt} = F_r n - h_f A_f (T_f - T_c)$$

$$\Rightarrow \boxed{\frac{dT_f}{dt} = \frac{F_r n}{m_f C_f} - \frac{h_f A_f}{m_f C_f} (T_f - T_c)} \quad (2.25)$$

Similarly, the one-dimensional heat transfer balance can be set up for the heat flowing out of the fuel and being carried away by the coolant plus the heat lost in the form of neutron population. This heat is represented by the last term on the RHS of the equation (2.25). Hence, mathematically it becomes [10], [24], [39],

$$\begin{aligned} &\text{Rate of heat added to the coolant from the fuel} = \\ &\text{Rate of heat rejected out of the fuel} - \text{Rate of heat carried away by the coolant} \\ &+ \text{Heat lost during the transfer due to neutron loss} \end{aligned} \quad (2.26)$$

Based on similar laws, as for equation (2.25), the equation (2.26) can be represented as,

$$m_c C_c \frac{dT_c}{dt} = h_f A_f (T_f - T_c) - G_c C_c (T_{cout} - T_{cin}) + (1 - F_r) n$$

$$\Rightarrow \boxed{\frac{dT_c}{dt} = \frac{h_f A_f}{m_c C_c} (T_f - T_c) - \frac{G_c C_c}{m_c C_c} (T_{cout} - T_{cin}) + \frac{(1 - F_r)}{m_c C_c} n} \quad (2.27)$$

This equation is known as the Primary coolant temperature (T_c) equation for PWR. Here, G_c represents the reactor coolant flow rate which is assumed to be constant in assumption (4). Hence, equations (2.25) and (2.27) together describe the thermal hydraulics of an elementary reactor core for the PWR [27], [28], [42].

2.3.2.2. THERMAL HYDRAULICS OF PRIMARY COOLANT LOOP OF PWR

Now, the heat gained by the coolant from the reactor core is transported to the steam generator & back to the reactor via coolant pipes, namely the hot leg connecting the reactor to the steam generator and the cold leg connecting the steam generator back to the reactor. The effects of the coolant pumps (hence the cross over leg) have been neglected here, since they don't contribute much to the thermal energy flow and also because the primary coolant flow is assumed to be constant in assumption (4) in the previous section. Here, only a single coolant loop will be considered instead of three/four in practical cases just for simplicity of formation of the elementary model of PWR. However, it is interesting to note that on both ends of each of these two sections there happens to be time lag in the temperature updation, which is formally termed as *Transport delays* [33].

Assumptions for Primary coolant loop thermal hydraulics

1. The heat losses during the heat transportation from Reactor to the Steam Generator and back are negligible. The coolant has constant specific heat and density and doesn't vary with temperature [21].
2. The heat transfer occurs at a point in the reactor to the coolant.

Formulation of Primary coolant loop thermal hydraulic equations

With the above assumptions in place, the differential equations representing the transport delays in the primary coolant loop are developed.

1. Coolant movement from Core to the Hot leg

The coolant flows in a reactor from the bottom to the top, i.e. the coolant enters the core at the inlet plenum (at the bottom), then extracts heat from the core and leaves the core from the outlet plenum (at the top). Now, as the fuel generates heat, the coolant surrounding it gets heated and so the average coolant temperature changes as given by equation (2.27). This change in average temperature of coolant should result in a change in its outlet temperature (enters the hot leg), represented by T_{cout} . However, this effect will become effective in the T_{cout} after a certain time delay, which in this case is termed as, *transit delay* [21] denoted by τ_{ct} . Based on this physical insight, the differential equation for the coolant outlet temperature can be obtained as,

$$\boxed{\frac{dT_{cout}}{dt} = \frac{(2T_c - T_{cin} - T_{cout})}{\tau_{ct}}} \quad (2.28)$$

Under steady state conditions, the LHS of equation (2.28) is zero and hence, $2T_c - T_{cin} = T_{cout}$. Although under transient conditions, $2T_c - T_{cin} \neq T_{cout}$, since there would be transit delay involved [21]. So, in this perspective, the equation (2.28) is valid.

2. Coolant movement through Hot leg to the Steam Generator inlet

As the coolant exits the core, it travels through the hot leg and enters the steam generator. Based on the 1st assumption, the inlet steam generator temperature denoted by T_{SGin} should have been equal to the outlet coolant temperature (T_{cout}). However, this is not the case and the two temperatures will not be equal at the same instant as there will be a transport delay for the hot coolant to flow from the core to the steam generator. This delay is termed as the *Hot Leg Delay* [33] and is denoted by τ_{hld} . Mathematically the steam generator inlet temperature variation can be formulated as below [33].

$$T_{cout}(t) = T_{SGin}(t + \tau_{hld}) \quad (2.29)$$

Expanding equation (3.40) in a Taylor series gives,

$$T_{cout}(t) = T_{SGin}(t) + \tau_{hld} \frac{d}{dt} T_{SGin}(t) + \frac{\tau_{hld}^2}{2!} \frac{d^2}{dt^2} T_{SGin}(t) \quad (2.30)$$

For slow changes in the temperature, the higher order terms beyond the second can be neglected. Hence, a first order differential equation is formed to indicate the time delay as,

$$T_{cout}(t) = T_{SGin}(t) + \tau_{hld} \frac{d}{dt} T_{SGin}(t) \quad (2.31)$$

$$\Rightarrow \boxed{\frac{d}{dt} T_{SGin} = \frac{T_{cout} - T_{SGin}}{\tau_{hld}}} \quad (2.32)$$

3. Coolant movement through Cold leg back to the Reactor inlet

Again, as the coolant exits the steam generator, it travels through the cold leg and enters the reactor core. Taking into account similar arguments as given for the previous section, the outlet steam generator temperature denoted by T_{SGout} will not be equal to the inlet coolant temperature (T_{cin}) at the same instant as there will be a transport delay for the cooled down coolant to flow from the steam generator to the reactor core inlet. This delay is termed as the *Cold Leg Delay* and is denoted by τ_{cld} . Mathematically the reactor inlet temperature variation can be formulated as below [33].

$$T_{SGout}(t) = T_{cin}(t + \tau_{cld}) \quad (2.33)$$

Again, expanding equation (2.33) in a Taylor series and ignoring the higher order terms beyond the second for slow changes in the temperature,

$$T_{SGout}(t) = T_{cin}(t) + \tau_{cld} \frac{d}{dt} T_{cin}(t) \quad (2.34)$$

$$\Rightarrow \boxed{\frac{d}{dt} T_{cin} = \frac{T_{SGout} - T_{cin}}{\tau_{cld}}} \quad (2.35)$$

Considering T_{SG} as the average coolant temperature inside the steam generator which would be the average of the inlet steam generator temperature (T_{SGin}) & the outlet steam generator temperature (T_{SGout}) as shown in the figure **Fig. 3.2**. Mathematically, [33]

$$T_{SG} = (T_{SGin} + T_{SGout})/2 \quad (2.36)$$

$$\Rightarrow T_{SGout} = 2T_{SG} - T_{SGin} \quad (2.37)$$

Substituting equation (2.37) in equation (2.35), an equation can be formed which deals with the variation of the coolant inlet temperature [33].

$$\boxed{\frac{d}{dt} T_{cin} = \frac{2T_{SG} - T_{SGin} - T_{cin}}{\tau_{cld}}} \quad (2.38)$$

2.3.2.3. THERMAL HYDRAULICS OF THE STEAM GENERATOR OF PWR

The next step in the process is the transfer of heat within the steam generator. Here, the temperature difference between the hot primary coolant and the surrounding secondary coolant within the vertical U-tube type steam generator causes heat to convectively flow from the primary to the secondary coolant.

Assumptions for Steam Generator thermal hydraulics

1. The heat transfer between the primary coolant side and the steam side of the SG is represented by means of a single area and a single heat – transfer coefficient [33].
2. It is assumed that 100% heat is converted into steam in the steam generator , i.e the SG Power is the same as that of the reactor [21].

Formulation of Steam Generator thermal hydraulic equations

Considering these assumptions, the thermal properties of the primary coolant side of the steam generator will be determined. Let the amount of heat extracted from the steam generator depend on the difference between the inlet and outlet temperature of the primary coolant within the steam generator. Then, mathematically,

$$\begin{aligned}(T_{SGin} - T_{SGout}) &= \{T_{SGin} - (2T_{SG} - T_{SGin})\} \\ \Rightarrow (T_{SGin} - T_{SGout}) &= 2(T_{SG} - T_{SGin})\end{aligned}\quad (2.39)$$

Now, the amount of heat extracted also depends on the amount of time the primary coolant stays within the steam generator. This time delay is regarded as the *Resident delay* within the SG, denoted by τ_{SG} and forms the basis of primary side equation of the steam generator [21].

$$\begin{aligned}\frac{dT_{SG}}{dt} &= \frac{(T_{SGin} - T_{SGout})}{\tau_{SG}} - \frac{F_r \cdot sgp}{m_{SG} C_{SG}} \\ \Rightarrow \quad &\boxed{\frac{dT_{SG}}{dt} = \frac{2(T_{SG} - T_{SGin})}{\tau_{SG}} - \frac{F_r \cdot sgp}{m_{SG} C_{SG}}}\end{aligned}\quad (2.40)$$

In the above equation, the first term on the RHS represents the source of heat and the second term represents the amount of heat extracted from the steam generator. The term *sgp* used in the second term denotes the power generated by the steam generator [21].

Under steady state conditions, the total amount of heat extracted from the steam generator denoted by the second term of equation (2.40), can be assumed to be the same as that defined by the feed flow denoted here by G_{feed} . Since the feed flow changes depend on the demand power at the secondary side (since PWR is a load follower), so it can be said that the power extracted from the SG equals that of the demand power [21]. However, under transient mode, there will be a certain time delay between the changing feed flow and the feed following SG power (*sgp*). Mathematically, the SG power can be represented as [21],

$$\boxed{\frac{d}{dt}sgp = \frac{G_{feed} - sgp}{3}}\quad (2.41)$$

This study about the PWR system dynamics can be utilized to form the desired model of the PWR. However, in order to simulate the behaviour of the model as close as possible to that of an actual PWR plant, it is necessary to study the PWR from the control perspective. This issue has been addressed in the following section.

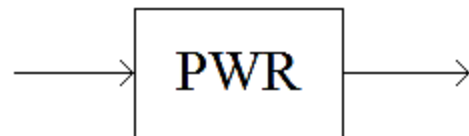
2.4. CONTROL ASPECTS OF PWR

From the control point of view, PWR can be represented as a system, where the Reactor itself forms an open loop system, whereas the closed loop reactor system is formed as a result of temperature coefficients and poisoning feedback, along with the external control mechanisms.

2.4.1. ELEMENTARY REACTOR – AN OPEN LOOP SYSTEM

To a control system designer, a Reactor is like any other element in the control loop and can be represented simply by a black box model, whose significant parameters are the neutron density (n), the neutron lifetime (l) and the reactivity (ρ). Its kinetic behaviour or open loop responses [33] can be studied easily by considering various control approaches like transfer function approach or state space approach. It will be seen that the reactor itself is a non-linear system described by the non-linear point kinetic equations and the quantity (β/l) plays a significant role in determining its time behaviour, thus distinguishing a thermal reactor from a fast reactor [7]. The neutron lifetime is generally the parameter upon which the largest roots of the equation depend. In case of a PWR, a significantly large root is placed on the left side of the s-plane which is a characteristic of this reactor & depends upon the value of l . From the control perspective, this root plays a major role in stabilizing the reactor responses.

**Fig 2.7: Black Box model of PWR
– An open loop system**



2.4.2. REACTOR FEEDBACK LOOPS – A CLOSED LOOP SYSTEM

An elementary reactor by itself doesn't completely portray the entire array of processes that occur within the reactor and the plant. These processes can be considered as feedback loops which upon addition to the open loop reactor modify its responses. These feedbacks can be distinctly divided into two categories namely, the internal or local feedbacks and the external feedbacks.

1. Internal or Local Feedbacks

The internal feedbacks in a reactor are caused by temperature coefficients and Xenon poisoning, mainly which cause a change in the plant reactivity [7], [33].

In a PWR as mentioned earlier, there are two temperature coefficients namely, the fuel and moderator temperature coefficients which are both negative. Added to that, the xenon poisoning effect also adds negative reactivity in small amounts. Hence, these three systems within the reactor core together form the internal feedback loop for the PWR and as mentioned earlier owing to their negative coefficients stabilize the reactor without using external control sources. Hence, PWRs have their own internal control mechanism and are inherently stable [28], [21].

As explained later, addition of external reactivity to the open loop reactor would cause the power to climb indefinitely. However, with the internal feedback loop in place the same input would make the reactor power stable after a certain time, around a desired point.

2. External Feedbacks

The external feedbacks in a reactor are the loops associated with the external plant. It is tuned to the circulation of the primary coolant through the reactor. External feedback loops comprises of many control systems described in the next section. The major loop is that of the reactor operational control system which acts through the use of CRs.

These two feedbacks together control and modify the responses of the PWR black box. Since feedbacks are used as an inevitable operation in the analysis of nuclear reactors, the problem of system stability naturally follows which shall be dealt with later in this chapter. **Fig. 2.8** shows a schematic view of a PWR with its feedback control loops [28], [33].

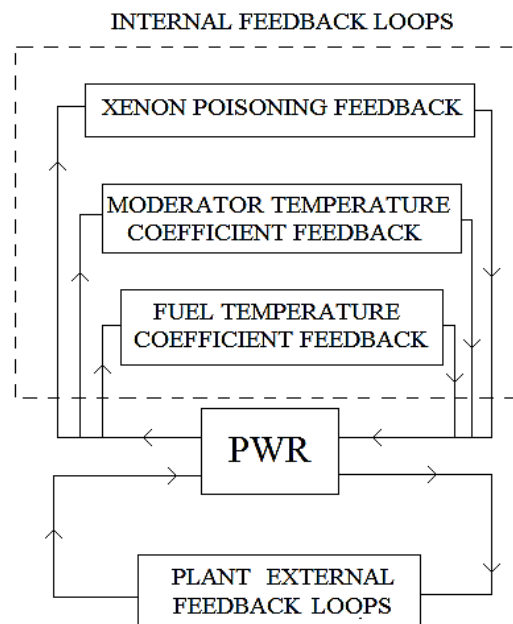


Fig 2.8: PWR Feedback loops – A Closed loop system

2.4.3. REACTOR CONTROL SYSTEMS

In order to produce heat within a reactor in a controlled environment without letting it run away (exponential increase in power output), NPPs require reliable external feedback control loops as shown in **Fig. 2.8**.

A reactor operation can be envisaged to have three operating regimes, namely, *Start-Up regime*, *Heating –Up regime* and *Power operational regime* [21]. The start-up regime is concerned with the initial power, the Heating up regime concentrates on the power during the time between the reactor start up and power operational regimes. The Power Operational regime is assumed to exist when the average coolant temperature has reached a certain specified value and demand power is within 20-100% of full power [21]. The Power

regulation in these three regimes of a nuclear reactor is achieved through actuation of different external control systems like the Reactor Operational control system , the Fuel management system and the Xenon control system [11], [21]. In the context of this study, the Reactor operational system is relevant because it acts as the main control system during power operation of a reactor and so, it has been discussed beneath.

2.4.3.1. REACTOR OPERATIONAL CONTROL SYSTEM

The Reactor operational control system is of major significance in the reactor control studies. The general control actions or control loops of this system [11] as illustrated in the **Fig. 2.9**.

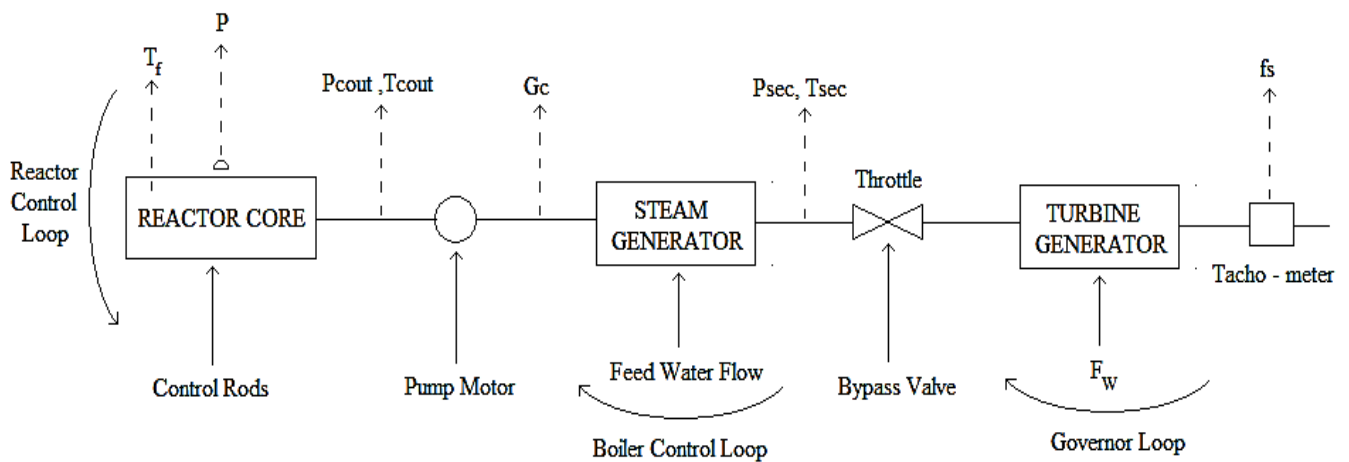


Fig 2.9: Schematic Diagram of PWR control loops

With reference to the **Fig. 2.9**, the following signals are available for automatic control.

- The neutron or flux measurements give the reactor power, P .
- The pressure, P_{cout} , and also the temperature, T_{cout} , of the primary coolant exiting the reactor and entering the steam generator.
- The primary circuit flow rate G_c .
- The turbine and generator shaft speed f_s . This is normally expressed as the system frequency.
- The pressure, P_{sec} , and also the temperature, T_{sec} , of the steam generator secondary loop exit, and the entrances to the high and intermediate turbines.
- The secondary water level in the steam generator via the Feed water flow through the Feed check valves (FCVs).

These measured signals are applied to perform the control functions for the entire plant as mentioned beneath.

- Control signals are applied to control the control rods which are responsible for controlling the reactor core reactivity and power. This action is taken by the *Rod control system or Reactor Regulating system*.

- Control signals can also be used to control the primary coolant pump speed. This controls the primary coolant temperature. Since the coolant and moderator are the same substance, it indirectly changes the reactivity. Also, the primary coolant pressure needs to be maintained which is done by the *Pressuriser pressure and level control system*.
- Control signals are also applied to control the secondary steam flow rate from the turbine throttle opening. This also controls the feed water pump as the steam flow to the steam generator is quite important to control with changes in the throttle. The main systems under this category are the *Turbine control system*; *Steam generator water level control system* which includes *Main feed regulating valve control system*, *Feed pump speed control system* etc.

2.4.3.2. PWR – A LOAD FOLLOWER

Owing to their internal feedback systems and the presence of reliable external control loops as mentioned, PWRs show the ability to change its power according to the load on the secondary side. In fact, PWRs have the ability of changing load in the range of 15/30 to 100% of rated power at a rate of around 1 to 3%/minute. However, 5 to 10%/minute is possible in a limited range. Compared to this, the PHWRs can adjust their power only between 60 and 100% of its full power since, the natural uranium with its large Xenon poisoning, limits power adjustments in the lower power ranges [16],[11].

Every time the demand power changes, a sequence of steps are followed by the plant which guides the reactor power to get synchronized with the demand power level. Say, if an increase in the demand power were to occur, the following sequence of events would enforce the load following of the PWR.

- **STEP 1:** To meet the increase in demand, the control valves (throttle) are opened to allow more steam into the turbine.
- **STEP 2:** As a result of step 1, the water level in the steam generator drops. To restore the water level in the steam generator the feed water flow will increase via the Feed Check valves (FCVs).
- **STEP 3:** Reactor coolant temperature and pressure will decrease due to the mismatch between the reactor power level and steam generator load. Moderating fluid (light water) density decreases as a result and in turn the neutron multiplication rate increases which is the initial increase in the reactor power.
- **STEP 4:** Owing to the lower coolant temperature, the control system will withdraw the control rods to raise the reactor power.
- **STEP 5:** The coolant pressure control will activate the pressuriser heaters as a response to the pressure change.
- **STEP 6:** Eventually pressuriser water level will be restored and a new steady state condition is reached, and therefore equilibrium between the generator electrical power and the reactor power level is achieved.

A similar array of steps, only in the opposite sense would be applicable when a decrease in the demand power occurs [11].

2.4.3.3. REACTOR REGULATING SYSTEM (RRS)

Any regulating system is a closed loop feedback system that brings the system, to which it is applied, to an equilibrium state if deviated from steady state by perturbation of any external or internal parameter. A Reactor regulating system (RRS) is an automated control system which forms an integral part of any reactor system and plays a significant role in output power regulation. Although, other operational control systems are required for the safe reactor operation, only the RRS operation shall be considered here. The important functions of the RRS are mentioned below [21].

- i. Automatic control of reactor power in start-up, heating up and power regime of operations.
- ii. Automatic control of average coolant temperature in heating and power regime of operations.
- iii. Control signal generation for movement of Control rods (CRs) in pre-defined sequence in automatic mode.

So, it is noted that RRS is designed to work effectively in all the operational regimes of the reactor. However, transients in a reactor need to be kept from growing when there are substantial reactivity changes. Two sorts of transients need to be accounted for: Sudden change in demand power when operating in the power regime and Reactivity fluctuations, apart from the start up or shut down of a reactor or an accident.

1. Constraints on the RRS

In order to increase or decrease the reactor power, the load follower steps given above are utilized. However, this operation requires the movement of the control rods which are initiated by the RRS such that the effective multiplication factor (k_{eff}) deviates from unity. The control mechanisms are so adjusted that the factor returns to unity (reactor becomes critical again) subsequently, as the power reaches the desired power level. For changing the power level, three conditions must be satisfied by the RRS [33]:-

- i. The power output level of the reactor must follow the demand power level within a given error.
- ii. The effective multiplication factor (k_{eff}) must ultimately be 1.
- iii. The control loop should be unconditionally stable, i.e. until the reactor is in operation no conceivable sustained oscillations are permitted.

2. Structure & Operation of the RRS

The RRS is an automated control system which usually consists of a control loop around the CRs (absorber rods for PWR) & can be operated either as a *Proportional Regulating system* or a *Discontinuous Regulating system* [33].

- I. A Proportional Regulating system is one in which the position of the CRs are changed in proportion and in phase opposition with any error created by a power demand change or by an internal system transient. The advantage that this system gives is its improved accuracy.

II. A Discontinuous Regulating system is one in which no control is attained, unless an error, which is some fixed percent away from a demanded set of conditions, is set up in the control loop. When sufficient deviation occurs from the demanded conditions, the CRs position is changed at a fixed velocity. These systems can be designed to hold the reactor power level within a predetermined tolerance level. Since these systems have a dead- zone, they become less susceptible to noise. However, they cannot match the accuracy of the proportional system.

Given below in **Fig. 2.10** is the schematic diagram for a Discontinuous RRS system.

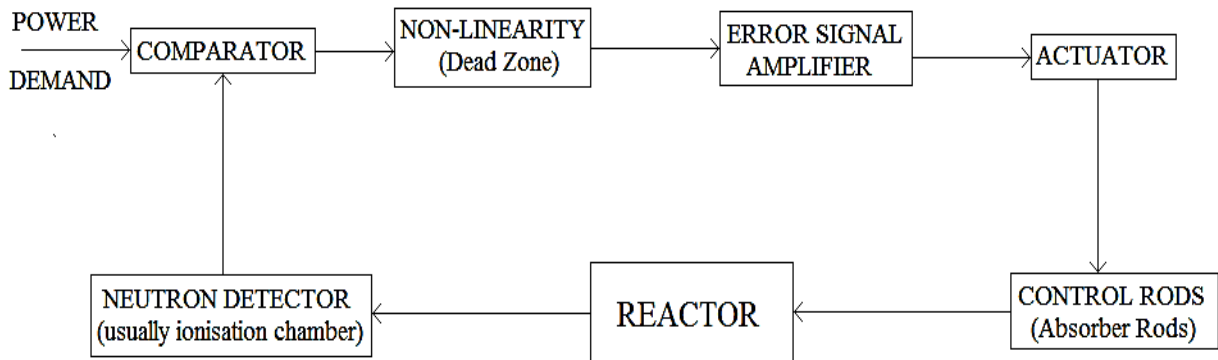


Fig 2.10: Schematic Diagram of PWR Discontinuous RRS
(Adapted Courtesy: **Control of Nuclear Reactors and power plants – M.A.Schultz**)

Let us briefly describe the sequence of operation of the above system. Firstly, the output of the reactor, in terms of the power output or neutron level is measured by a Neutron detector, usually an ionisation chamber. This output is then compared with the desired power demand in the comparator and any error between the actual and the demand power is monitored. This error is then sent to the error amplifier. In a discontinuous system, as the one shown, there are non-linearities in the system like dead zone. Hence, this signal has to pass through the non-linearity, causing a delay in the action to be taken by the error signal amplifier. However, proportional systems won't have the non-linearity block. Upon receiving the error signal, the error amplifier which controls the actuator, moves the control rods in such a way that the error is eliminated [33].

The Control rods as mentioned earlier, are the most important reactivity management devices (neutron absorbers), which in turn changes the reactor power. It can also be determined that the change in the reactivity caused by the movement of the CRs is a function of the rod-position. This relation between rod position and reactivity is represented by an S- curve, which shows that as a CR is inserted for a small distance, the reactivity change is small. Beyond a certain point, the change in reactivity starts to vary linearly with the extent of rod insertion. Finally, when the rod is almost completely inserted, further insertion has little effect on reactivity. For the power regime operation however, the linear zone of reactivity variation with rod insertion is a valid assumption [21].

3. Control Methods of the RRS

As mentioned in the functions earlier, apart from the power level control the RRS can also be equipped to control the average coolant temperature of a reactor. Hence, this leads to two distinct control strategies for the RRS, given below.

- I. **POWER CONTROL** – The power control method acts by evaluating the difference between the set value of demand power and the measured value of the actual reactor power. If an error exists, the control system will actuate the CRs to move at a certain speed until the difference is resolved or is less than a dead band as in a discontinuous system shown in **Fig. 4.4**. [43].
- II. **TEMPERATURE CONTROL** – Similarly, the temperature error can be determined from the set value of the average coolant temperature, the measured average coolant temperature and the equivalent temperature caused by the power mismatch. This error signal will again actuate the CRs to move at a certain speed until the difference is resolved or is less than a dead band. Hence, in place of a neutron detector in **Fig. 4.4**, this RRS would have a temperature measuring device in place [43].

These control principles and systems are basic to the performance of any reactor. These control systems work in conjunction with the dynamics of the reactor plant. However, stability analysis of the entire system becomes necessary due to the non-linearity involved in the system. This has been addressed in the next section.

2.5. ***STABILITY ANALYSIS OF REACTOR SYSTEMS***

Nuclear reactors behave in a non-linear fashion which can be justified by the strong coupling between reactivity and power as expressed by the point kinetics equations. The use of feedback loops with such a system enhances this effect. Since this non-linearity plays a major role in practical situations, it is necessary to analyse its effect on the system stability.

There are various techniques available for carrying out stability analysis of non-linear systems. In this study, the *Lyapunov's direct method of stability analysis* was used to accomplish this task [37], [15]. There were several reasons behind its choice, some of which are mentioned beneath.

- This analysis uses the system equations without resorting to approximations that are generally used for complex systems.
- There is no theoretical limit on the number of nonlinearities or on the order of the differential equation to which Lyapunov's Direct Method can be applied.
- The relationship between system stability and the system parameters is directly extracted from the Lyapunov's stability conditions.

Any non-linear system can be expressed as a combination of linear and non-linear parts. In the Lyapunov direct method, if it can be proved that the linear part of the system is asymptotically stable, and then a finite region of stability can be specified for the entire non-linear system. This technique shall be employed to analyse the stability of the PWR models. The Lyapunov stability theory has been provided in **Appendix D.2**.

2.6. CHAPTER SUMMARY

In this chapter, a brief review of the literature available on various aspects of Pressurized Water Reactors (PWRs) has been presented. Before getting started with the modelling of PWRs, an understanding of the PWR systems and dynamics is established. In the context of this study, firstly a discussion on the relevant PWR-NPP systems has been provided. These systems include the Reactor Core, the Steam Generator and the Condenser-Feed water system. Succeeding this discussion, the mathematical framework necessary for modelling these aforesaid systems has been stated. The assumptions and deduction logic necessary to form the equations for Reactor core dynamics and thermal hydraulics of these systems have been thoroughly explained.

Although the previous section renders the mathematical background necessary to model the PWR system, it becomes necessary to understand the control aspects of the reactor in order to capture the actual behaviour of the PWR-NPP. So, a study of the PWR has been presented from the control perspective. In this section, the major emphasis is on the Reactor feedback loops. It is clear from this discussion that PWRs can stabilize themselves without the use of any external control system, mainly due to their internal feedback mechanism (self-stability). Also, the Reactor Regulating system which forms a part of the external feedback loop has been focussed upon, so as to fulfil the objective of properly designing such a system for the reactor model used here. Lastly, the issue of reactor stability has been addressed in brief. Since, the behaviour of nuclear reactors is non-linear in nature, the direct method Lyapunov is chosen to analyse the stability of the reactor models. The reasons behind this choice are clearly indicated.

CHAPTER 3

PWR SYSTEM MODELING

In this chapter, modeling of the PWR carried out during the course of the dissertation has been described in detail. At the onset, the modeling approach has been put forth & along with that, the general assumptions are specified and duly justified. The next section contains the model building procedure described in a progressive manner from the base model to the entire system. Lastly, the design of an RRS for the developed models has been discussed.

3.1. *MODELING PREREQUISITES*

A model is a precise mathematical representation of a system's dynamics which allows the designer to make predictions regarding its behaviour. In order to mathematically model any kind of system, three prerequisites need to be fulfilled. The first and foremost requirement is a set of equations which describes the behaviour of the entire system. Next, the necessity of an appropriate modeling approach & operational mode arises. And lastly, some general rules and assumptions need to be considered regarding the system under investigation.

In this case, the set of equations necessary for describing the PWR system has already been acquired. The other two requirements have been discussed in this section.

3.1.1. *STATE SPACE MODELING APPROACH*

In this dissertation, the PWR was focussed upon from the control point of view. From that perspective, certain points should be kept in mind regarding the Reactor system. Firstly, it is a dynamic system where many variables keep changing with small perturbations. Simply put, even the Point Kinetic model, which is a simplified representation of the reactor core, is nonlinear in nature. Also, in order to model the PWR, feedback loops should be used. Hence, a modeling approach suitable to these requirements was required.

Although, the conventional control theory presents a rather intuitive and conceptually simple framework, due to their inherent shortcomings they could not be used. Instead, a representative of the modern control theory, the state variable modeling known as the *State – Space technique* of analysis and design were employed to develop an analytical model of the integrated Reactor plant, i.e. the core, the plant and the RRS.

The salient points which lead to the preference of the State Space Analysis over conventional control theory (based on transfer function or pulse transfer function and graphical approaches such as Root locus, Bode plots and Nyquist plots) are pointed below [8],[26].

- Conventional transfer function models cannot be applied to non-linear systems except in few simple cases. It can be applied only to LTI systems with SISO (a specific input) framework. As mentioned earlier, the reactor is inherently non-linear (point kinetics) and can be thought of in terms of MIMO (a class of inputs) framework as and when required.

Also, the reactor control system (specifically RRS in this case) contains elements which exhibit nonlinearity in their nature. So, this trait becomes a shortcoming. On the other hand, the state space techniques, can be used for linear and nonlinear systems alike, with either time invariant or time varying property and either SISO or MIMO property.

- The transfer function approach considers the initial conditions of the system to be zero, i.e. the system is initially at rest. But this may not be the case in many practical systems. This property has also been taken care of in the state space approach.
- The conventional theory considers only input and output signals based on the transfer function of the system. So, only output is available as feedback in such an approach, unless an observer is designed to estimate the states.

However, the state space modeling provides state feedbacks while designing the closed loop models of the system, which tunes exactly with our requirement for the PWR system as shown by the feedback loops in the previous chapter.

3.1.1.1. CONCEPTIONS OF STATE SPACE

The state space approach describes the system equations in terms of n first order difference equations or differential equations by selecting suitable state variables. These are represented in a combined form as a first order vector matrix difference or differential equations [26]. The state variables of any dynamic system are the smallest set of variables (say n) that determine the state of the dynamic system or the complete behaviour of the dynamic system, once the input and initial states of the system are specified. Also, these state variables may not even be physically measurable or observable quantities. This is another advantage of the state space modeling approach.

Representation of the State Space Equations

State space analysis is concerned with three kinds of variables, namely the state variables (denoted by $\mathbf{x}$), the input variables (denoted by $\mathbf{u}$) and the output variables (denoted by $\mathbf{y}$). A general representation of the state space equations for an LTI system in the continuous time case is given as,

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (3.1)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad (3.2)$$

Here, $\mathbf{x}(t)$ = State vector ($n \times 1$); $\dot{\mathbf{x}}(t)$ = Velocity vector ($n \times 1$); $\mathbf{u}(t)$ = Input vector [control vector, ($m \times 1$)] & $\mathbf{y}(t)$ = Output vector ($p \times 1$). Also, $\mathbf{A}$ = State matrix ($n \times n$); $\mathbf{B}$ = Input matrix ($n \times m$); $\mathbf{C}$ = Output matrix ($p \times n$) and $\mathbf{D}$ = Direct transmission matrix ($p \times m$).

Here, n is the order of the system. Equation (3.1) is known as the *State equation* and (3.2) as the *Output equation*. This conception and representation will become much clear when the system is modelled in state space [26].

3.1.1.2. OBSERVABILITY & CONTROLLABILITY PRINCIPLES

The discussion on state space cannot be completed without mentioning the two most fundamental principles of control systems: Observability and Controllability.

Observability

Observability is concerned with the problem of determining the state of a dynamic system from the observations of the output and the control vectors in a finite amount of time. In other words, a system is completely controllable if every initial state of the system $x(0) = x_0$, can be determined from the observation of the output $y(t)$, over a finite interval of time [8],[26].

Mathematically, the conditions for observability of a general system as described by equations (3.1) and (3.2), can be found with the help of the observability test matrix, $\mathbf{O}$ ($n \times np$), given as,

$$\mathbf{O} = [C: CA: CA^2: \dots : CA^{n-1}]^T \quad (3.3)$$

The condition for the system to be observable is that $\det. (\mathbf{O}) \neq 0$ and the matrix $\mathbf{O}$ has a rank n , where n is the order of the system.

Controllability

Controllability is concerned with the problem of whether it is possible to steer the system from a given initial state to an arbitrary state using a control vector. It is basically of two types, State controllability and Output controllability. In this study, the major concern will be on the state controllability criterion for the modelled PWR systems.

A system is said to be completely state controllable if it is possible by means of an unbounded control vector $u(t)$ to transfer the system from any arbitrary initial state say $x(0) = x_0$, to any desired state $x(t_f)$ in a specified interval of time. If any of the state variables do not respond to the control signal, then this state cannot be controlled and so the system becomes uncontrollable [8],[26].

Mathematically, the conditions for state controllability of a general system as described by equations (3.1) and (3.2), can be forecasted with the help of the controllability test matrix, $\mathbf{U}$ ($n \times nm$) given as,

$$\mathbf{U} = [B: AB: A^2B: \dots : A^{n-1}B] \quad (3.4)$$

The condition for the system to be state controllable is that the matrix $\mathbf{U}$ has a rank n , where n is the order of the system. However, if the matrix $\mathbf{U}$ is a square matrix ($n \times n$), then the condition for controllability is that $\det. (\mathbf{U}) \neq 0$, i.e. the matrix is non-singular.

3.1.2. GENERAL RULES AND ASSUMPTIONS

Before, beginning to model the PWR system with state space approach, few general rules and assumptions need to be set, mainly based on the explanations of the previous chapters. These set of assumptions are given below for each part of the system.

1. Reactor Operating Schemes –

- a) The reactor was considered to be operating in the *Power Operational regime*. The load changing region to be considered while collecting data was selected at 20% to 100% of reactor full-power. This was considered as the normal load changing range of a PWR, which took place in minutes to even seconds, as described by most of the sources [16],[11].
- b) *Start-up and shut down operations* were not included as these do not form part of normal reactor load following operations. Also, it was assumed that the reactor had been running for a sufficiently long time after a start-up in order to avoid the conditioning period (Heating – up regime). This doesn't form part of the normal reactor load operation.

2. Reactor Neutronics –

- a) The size of the reactor was considered (PWR) to be small (100 MWth). So, as explained in section 2.2.2.1, the *Point Kinetic* model was assumed for describing the neutron behaviour (or reactor power) over time. Another advantage of using point kinetics was that it is easily expressible in state space form.
- b) Since the reactor size was considered to be small, the neutron flux was automatically selected as low.
- c) *Temperature coefficients of reactivity* due to fuel and coolant were considered to be constant w.r.t the operating temperature or the power operation of the reactor. It should be pointed out here that actually the magnitude of the temperature coefficient is a function of the temperature. Also, the moderator (or coolant) temperature coefficient remains constant as long as the water density remains high. In reactors, where boiling occurs the water density tends to fall sharply causing the negative temperature coefficient to be more negative. However, it is not the case with PWRs, where the reactor coolant is pressurised and doesn't boil.

3. Reactor Thermal systems –

- a) Only the Primary loop of the PWR system was considered for modeling. Since, a distinct relation exists between the temperatures and pressures of the primary and secondary sides of the plant, so once the constants of the primary side of the plant are known, the secondary constants are automatically specified.
- b) The assumptions considered in section 2.3.2.1 for the thermal hydraulics of the *PWR core* were all valid for our model. They included assumptions regarding lumped fuel; no heat loss from fuel boundaries; constant fuel density, specific heat and heat transfer coefficient along the entire power region of operation. The prominent heat transfer mechanism was assumed to be convection.
- c) The assumptions considered in section 2.3.2.2 for the thermal hydraulics of the *Primary coolant loop of the PWR* were also valid for our model. They included negligible heat loss during heat transportation from reactor to steam generator and constant specific heat and density of the coolant. Here, the effect of the pressurizer was not considered during the modeling.

- d) Next, the assumptions made in the section 2.3.2.3 for the thermal hydraulics of the *Steam Generator of the PWR* were applicable to our case. They included single area heat transfer between the primary and secondary & steam generator power to be the same as the reactor power.
- e) It was also assumed that the driving force behind the thermal heat transport processes were ideal and ensured 100% heat transfer at all the three exchange points as mentioned in section 2.3.2. ($F_r = 1.0$).

4. Reactor Control systems –

- a) Here, it was assumed that the Fuel management system consisting of burnable poison such as *Boron* was ideally balanced with excess reactivity at the BOL of the reactor and fuel burn up subsequently. Also, the addition of boron was a slow acting effect which didn't cause any variation in the normal load changing operation of the reactor which took place within minutes or even seconds.
- b) Amongst the other *Reactor power control systems*, only the RRS mentioned in section 2.4.3.2 was considered. The Shutdown systems (SDS) or the Reactor protection system (RPS) were not considered since only the power operation was of major concern.
- c) The *Reactor regulating system* considered here was a discontinuous regulating system (having dead bands) so as to minimize the frequent use of control rods. Only the automatic control algorithm for the RRS was considered here.
- d) The RRS was considered to have a uniform behaviour over its entire range of operations unlike the multirate systems with varying sampling rates.
- e) Since, the reactor was assumed to be operating in the power operational regime; the reactivity variation due to the *movement of the control rods* was represented linearly in this study as mentioned in section 2.4.3.2
- f) As mentioned in the section 2.4.3.1, the load following operation requires that the *feed water flow* via the Feed Check valves (FCVs) [condensate- feed water system as in section 2.1.2] to the steam generator to vary as per the movement of the throttle valves to allow required amount of steam to the turbine which is actually initiated by the change in the demand power. Since, here only the primary loop of the PWR was considered, it was assumed that the rate of change of the feed flow was dictated directly by the demand power change and was put in to use by the rate of movement of the feed check valves.

3.1.3. OPERATIONAL PROGRAM

The pattern that the temperatures, pressures and flows throughout the plant follow or assume as a function of the output power is called the *program* [33].

Here, the same basic plant of the reactor as shown in the **Fig. 2.6** was assumed. Now, there happens to be two modes of control program that can be used for controlling the Reactor power. First one is the constant T_c (Average coolant temperature within the core) program and another being the constant T_s (steam temperature) program. Another program considered here is the constant T_{cout} (primary coolant outlet or hot leg temperature) program [33].

1. *Constant T_c program* – In this program the Average coolant temperature denoted by T_c is kept constant at all power levels. For reactors having negative temperature coefficients like the PWR, this is the natural program of the reactor and requires least amount of control.

It is assumed that the output power required by the load increases. This extra energy extracted from the system requires the inlet temperature of the primary coolant (T_{cin}) to drop. This drop in temperature causes a drop in the average coolant temperature and since the reactor has negative temperature coefficient, it possess more reactivity now. If initially the reactor was critical, it now becomes supercritical temporarily. So, the output temperature of the coolant (T_{cout}) rises and more energy is available from the reactor. Finally, in the steady state the reactor returns to its critical condition with T_c constant as initially. Hence, without any control mechanism, the system stabilizes itself about a given temperature and supplies the demand placed upon the reactor. The only disadvantage of this system is that the secondary plant temperature and pressures need to be adjusted accordingly, resulting in poor turbine performance.

2. *Constant T_s program* – In this program, the secondary side steam temperature denoted by T_s , is kept constant throughout the operation. This program is preferred by the secondary steam system of the reactor plant.

An increase in the power demand of the secondary plant requires additional thermal energy transfer from the primary circuit to the secondary circuit in the steam generators. Now with the secondary circuit temperature kept constant the primary circuit is required to adjust its temperatures. With this mode it is assumed that the pressuriser control system is capable of maintaining the pressure of the coolant volume changes. The constant steam temperature mode provides very good turbine performance.

3. *Constant T_{cout} program* – Another program, as mentioned in the literature [33], is the constant coolant outlet or hot leg temperature program denoted by T_{cout} .

Again it is assumed that the reactor is operating at a critical level and possesses a negative temperature coefficient. If the power demand increases, the inlet coolant temperature drops and more reactivity is inserted into the reactor because of the negative coefficient. The outlet temperature (T_{cout}) tends to rise, but now so as to hold it constant a control mechanism needs to be implemented in the form of variation of the Reactor coolant pump speed. However, with this technique the average coolant temperature (T_c) decreases with the increase in demand, meaning that the reactor with its negative temperature coefficient would now attempt to operate in a supercritical condition. Consequently another control loop usually automatic external control rod loop must be added to extract the excess reactivity due to the change of the T_c .

In this study, according to assumptions (1) and (3) of the Reactor thermal system, only the primary loop of the reactor was modelled without considering the effect of the pressurizer. Hence, the constant steam temperature program (T_s) was not used. The effects of the constant T_c program and T_{cout} program were investigated.

3.2. PWR MODEL

The modeling of the PWR system was done in a progressive manner starting from the elementary reactor to the closed loop reactor with thermal and poison feedbacks. To accomplish this task, the modeling was divided into 3 modules. Later an additional module was added which comprised of the RRS control algorithm or technique that could be incorporated. The first module took care of the basic reactor designed in state space with the Point Kinetic equations. In the second module, the thermal feedback part and the implementation of the rest of the primary coolant loop was considered. The third module incorporated the Xenon poisoning feedback to the plant developed in the second module.

3.2.1. MODULE 1 – POINT KINETIC PWR (PKPWR)

The elementary PWR as mentioned in section 2.4.1 is an open loop system in terms of classical control. So, before investigating the behaviour of the entire system, it was necessary to understand the reactor's open loop responses or the kinetic characteristics of the reactor. Since, it was an elementary reactor, the effect of negative temperature coefficients and other thermal factors were assumed to be absent and so, didn't affect the responses of the open loop reactor. Although it is true that spatial effects inside a reactor can cause changes in reactivity and so affect the output, but based on our assumption (a) of the Reactor neutronics it can be said that these effects did not play any significant role since the reactor considered was small.

3.2.1.1. SINGLE GROUP POINT KINETIC MODEL

Firstly, according to assumption (a) of the Reactor neutronics, the Point kinetics model was considered, given by equations (2.13) and (2.15) as explained in section 2.2.2.1. Here, it was first assumed that a single group of delayed neutrons were present instead of six groups. So, a decay constant λ equal to the properly weighted average of six actual groups was assumed such that [7],

$$\boxed{\frac{\beta}{\lambda} = \sum_{i=1}^6 \frac{\beta_i}{\lambda_i}} \quad (3.5)$$

Here, the total delayed neutron fraction was considered as,

$$\beta = \sum_{i=1}^6 \beta_i \quad (3.6)$$

Where, $i = 1, \dots, 6$ represented the six groups of delayed neutrons.

Now with these assumptions the equations (2.13) and (2.15) were rearranged to give the single group point kinetics equations as,

$$\frac{dn(t)}{dt} = \frac{(\rho(t) - \beta)}{l} n(t) + \lambda C(t) \quad (3.7)$$

$$\frac{dC(t)}{dt} = \frac{\beta}{l} n(t) - \lambda C(t) \quad (3.8)$$

Now, the equation (3.7) showed the change in neutron density (n) over time. However, neutron density (n) is directly proportional to the reactor power, say P . So, the equations (3.7) and (3.8) were rewritten in the following form,

$$\boxed{\begin{aligned} \frac{dP(t)}{dt} &= \frac{(\rho(t) - \beta)}{l} P(t) + \lambda C(t) \\ \frac{dC(t)}{dt} &= \frac{\beta}{l} P(t) - \lambda C(t) \end{aligned}} \quad \begin{aligned} (3.9) \\ (3.10) \end{aligned}$$

3.2.1.2. NORMALISATION & LINEARISATION OF POINT KINETIC MODEL

The next step in the PKPWR modeling was to normalise and then linearize the Point kinetic equations defined by equations (3.9) and (3.10).

Normalisation

It was assumed that the reactor was operating at steady state and the value of operating power was P_0 . The steady state value of DNP concentration was obtained from equation (3.10) as,

$$\frac{dC(t)}{dt} = 0 \quad \Rightarrow \quad C_0 = \frac{\beta}{\lambda l} P_0 \quad (3.11)$$

Now, the equations (3.9) and (3.10) were normalised w.r.t their steady state values P_0 and C_0 . Firstly the equation (3.9) was taken up.

$$\frac{d}{dt} \left(\frac{P}{P_0} \right) = \frac{(\rho(t) - \beta)}{l} \left(\frac{P}{P_0} \right) + \lambda \left(\frac{C}{P_0} \right) \quad (3.12)$$

The following relation was assumed,

$$\boxed{\left(\frac{P}{P_0} \right) = P_r \text{ and } \left(\frac{C}{C_0} \right) = C_r} \quad (3.13)$$

Here, P_r was the normalised power and C_r was the normalised delayed neutron precursor (DNP) concentration [27]. Substituting the value of P_0 from equation (3.11) in (3.12) and then using the equation (3.13) rearranged the equation (5.12) as,

$$\boxed{\frac{d}{dt} P_r = \frac{(\rho(t) - \beta)}{l} P_r + \frac{\beta}{l} C_r} \quad (3.14)$$

Similarly, equation (3.10) was rearranged to obtain the normalised form as,

$$\boxed{\frac{d}{dt} C_r = \lambda P_r - \lambda C_r} \quad (3.15)$$

Hence, equations (3.14) and (3.15) represented the Normalised Point Kinetics model for this study.

Linearization

The point kinetics equations (even normalised point kinetics) form a non-linear system as mentioned earlier. The behaviour of any non- linear system is far more complex than a linear system, having multiple equilibrium points to which it may converge depending upon the initial conditions. This is what actually allows a nuclear reactor, to operate at different power levels in equilibrium state. However, to implement the State Space approach, the point kinetics was linearized around any one of the equilibrium states assuming perturbation model. Hence, linearization involved solving the point kinetics equations for small deviations from steady state condition.

Firstly, according to the perturbation model, small deviations of the following quantities were considered from their equilibrium or steady state condition as [21],

$$\begin{aligned} P_r(t) &= P_{r0} + \delta P_r \\ C_r(t) &= C_{r0} + \delta C_r \\ \rho(t) &= \rho_0 + \delta \rho \end{aligned} \quad (3.16)$$

Here, P_{r0} , C_{r0} and ρ_0 are the steady state or equilibrium values of power, DNP concentration and reactivity respectively. Amongst these, it was considered that $\rho_0 = 0$, since at the steady state the reactor was operating at *Critical* condition as mentioned in **Table 2.1**.

By substituting the equation (3.16) in (3.14) and (3.15), the perturbed point kinetics model was formed. Now, under steady state conditions, the derivatives of the steady state values became zero. While solving the equations, terms involving the quantity $\delta P_r \delta \rho$ came up. This was actually the value of non-linearity in the model and since its value was small, it was neglected. This lead to the linearized point kinetic model of the PWR (refer **Appendix A.1**) as,

$$\begin{aligned} \frac{d}{dt} \delta P_r &= \left(\frac{-\beta}{l} \right) \delta P_r + \left(\frac{\beta}{l} \right) \delta C_r + \left(\frac{P_{r0}}{l} \right) \delta \rho \\ \frac{d}{dt} \delta C_r &= \lambda \delta P_r - \lambda \delta C_r \end{aligned} \quad (3.17)$$

3.2.1.3. STATE SPACE PKPWR MODEL

Utilizing the Linearized point kinetic equation given in (3.17), the state space model of the PKPWR was formed similar to the equations (3.1) and (3.2). The point kinetic system here had 2 differential equations. Hence, the state variables were considered to be P_r and C_r . So, firstly the state equation was formed for which the state vector $\mathbf{x}$, the input vector $\mathbf{u}$ and the velocity vector $\dot{\mathbf{x}}$ were defined as,

$$\begin{aligned} \mathbf{x} &= [P_r \ C_r]^T \\ \mathbf{u} &= \rho \\ \dot{\mathbf{x}} &= [\dot{P}_r \ \dot{C}_r]^T \end{aligned} \quad (3.18)$$

The PKPWR model was a single input system as denoted by the input vector.

With the formation of the vectors in equation (3.18) and comparing (3.17) with (3.1), the State matrix (**A**) and Input matrix (**B**) were formed as below.

$$\mathbf{A} = \begin{bmatrix} \frac{-\beta}{l} & \frac{\beta}{l} \\ \lambda & -\lambda \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} \frac{P_{r0}}{l} \\ 0 \end{bmatrix} \quad (3.19)$$

Now, similarly the output equation was formed with Output vector (**y**) defined as,

$$\mathbf{y} = [P_r] \quad (3.20)$$

Then, the Output matrix (**C**) and Direct transmission matrix (**D**) were assumed to be,

$$\mathbf{C} = [1 \quad 0] \quad \text{and} \quad \mathbf{D} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (3.21)$$

With the C and D matrices as defined above, it was concluded that the output vector (**y**) was only the reactor output power (normalised). So, the equations were arranged in the state space form as given in equation (3.1) and (3.2) beneath.

$$\dot{\mathbf{x}} = \begin{bmatrix} \frac{-\beta}{l} & \frac{\beta}{l} \\ \lambda & -\lambda \end{bmatrix} \mathbf{x} + \begin{bmatrix} \frac{P_{r0}}{l} \\ 0 \end{bmatrix} \mathbf{u}$$

$$\mathbf{y} = [1 \quad 0] \mathbf{x} \quad (3.22)$$

Now, to take into account the initial conditions, the initial normalised power was assumed to be P_{r0} . So, from the equation (3.15), at steady state it was found that, $C_{r0} = P_{r0}$, where, C_{r0} was the steady state normalised DNP concentration.

Behaviour of the eigen values of the PKPWR model

Depending on the PKPWR system matrix, the Eigen values of the system were described by the equation,

$$(sI - A) = 0 \quad (3.23)$$

Where I was the identity matrix having the dimension of A matrix. By substituting A matrix from equation (3.19) in equation (3.23) and then solving the equation, a quadratic equation was arrived upon whose solution was,

$$s = 0; -(\lambda + \frac{\beta}{l}) \quad (3.24)$$

Hence, it became clear that one Eigen value lied at zero and the other at $-(\lambda + (\beta/l))$. This is the characteristic of any reactor. However, since λ doesn't change significantly, the lower value of the neutron lifetime (l) governs the position of the Eigen value. In case of PWRs, the value of neutron lifetime is very small (~ 0.00021 sec) as mentioned earlier, so this Eigen value should be placed on the extreme left of the s-plane, which shall be evident from the simulation results for the PKPWR system. The Eigen value which lies at the origin acts as an integrator & makes the system marginally stable.

The PKPWR system thus formed was simulated with various inputs and the Eigen values of the system were obtained, which are provided later in chapter 4.

3.2.1.4. EXTENDED PKPWR (Ex-PKPWR) MODEL

Considering the same procedure, the PKPWR model was extended for all six delayed neutron groups. The equations (2.13) and (2.15) were expanded to give the six group point kinetics equations as,

$$\begin{aligned}\frac{dP(t)}{dt} &= \frac{(\rho(t) - \beta)}{l} P(t) + \lambda_1 C_1(t) + \lambda_2 C_2(t) + \dots + \lambda_6 C_6(t) \\ \frac{d}{dt} C_1(t) &= \frac{\beta_1}{l} P(t) - \lambda_1 C_1(t) \\ &\vdots \\ &\vdots \\ \frac{d}{dt} C_6(t) &= \frac{\beta_6}{l} P(t) - \lambda_6 C_6(t)\end{aligned}\tag{3.25}$$

These equations were normalised and linearized similar to the single group case. Here, the following relations were considered for their normalisation.

$$\boxed{\begin{aligned}\left(\frac{P}{P_0}\right) &= P_r; \left(\frac{C_1}{C_{1,0}}\right) = C_{1r}; \left(\frac{C_2}{C_{2,0}}\right) = C_{2r}; \left(\frac{C_3}{C_{3,0}}\right) = C_{3r}; \left(\frac{C_4}{C_{4,0}}\right) = C_{4r}; \\ \left(\frac{C_5}{C_{5,0}}\right) &= C_{5r} \text{ and } \left(\frac{C_6}{C_{6,0}}\right) = C_{6r}\end{aligned}}\tag{3.26}$$

Ultimately, the equations were linearized based on the perturbation model similar to the PKPWR and were presented as given below.

$$\begin{aligned}\frac{d}{dt} \delta P_r &= \left(\frac{-\beta}{l}\right) \delta P_r + \left(\frac{\beta_1}{l}\right) \delta C_{1r} + \left(\frac{\beta_2}{l}\right) \delta C_{2r} + \left(\frac{\beta_3}{l}\right) \delta C_{3r} + \left(\frac{\beta_4}{l}\right) \delta C_{4r} + \left(\frac{\beta_5}{l}\right) \delta C_{5r} \\ &\quad + \left(\frac{\beta_6}{l}\right) \delta C_{6r} + \left(\frac{P_{r0}}{l}\right) \delta \rho \\ \frac{d}{dt} \delta C_{1r} &= \lambda_1 \delta P_r - \lambda_1 \delta C_{1r} \\ &\vdots \\ &\vdots \\ \frac{d}{dt} \delta C_{6r} &= \lambda_6 \delta P_r - \lambda_6 \delta C_{6r}\end{aligned}\tag{3.27}$$

Now, with these equations, the state space model of this system was formed similar to the PKPWR model. Similar to the PKPWR case, the state and output equations of ex- PKPWR were formed for which the state vector ($\mathbf{x}$), the input vector ($\mathbf{u}$), the velocity vector ($\dot{\mathbf{x}}$) and the output vector ($\mathbf{y}$) were defined as given below.

$$\begin{aligned}
\mathbf{x} &= [P_r \ C_{1r} \ C_{2r} \ C_{3r} \ C_{4r} \ C_{5r} \ C_{6r}]^T \\
\mathbf{u} &= \rho \\
\dot{\mathbf{x}} &= [\dot{P}_r \ \dot{C}_{1r} \ \dot{C}_{2r} \ \dot{C}_{3r} \ \dot{C}_{4r} \ \dot{C}_{5r} \ \dot{C}_{6r}]^T
\end{aligned} \tag{3.28}$$

With these vectors in place the State space matrices were formed for this model. It was considered that the Output Matrix, $\mathbf{C} = \mathbf{I}_7$, i.e. identity matrix of the same dimensions as State matrix ($\mathbf{A}$) and the Direct transmission matrix, $\mathbf{D} = [\mathbf{0}]_{7 \times 1}$, i.e. null matrix of the same dimensions as input matrix ($\mathbf{B}$). Then, the equations were arranged in the state space form as given by the equation (3.22).

$$\text{State Matrix, } \mathbf{A} = \begin{bmatrix} -\left(\frac{\beta}{l}\right) & \frac{\beta_1}{l} & \frac{\beta_2}{l} & \frac{\beta_3}{l} & \frac{\beta_4}{l} & \frac{\beta_5}{l} & \frac{\beta_6}{l} \\ \lambda_1 & -\lambda_1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_2 & 0 & -\lambda_2 & 0 & 0 & 0 & 0 \\ \lambda_3 & 0 & 0 & -\lambda_3 & 0 & 0 & 0 \\ \lambda_4 & 0 & 0 & 0 & -\lambda_4 & 0 & 0 \\ \lambda_5 & 0 & 0 & 0 & 0 & -\lambda_5 & 0 \\ \lambda_6 & 0 & 0 & 0 & 0 & 0 & -\lambda_6 \end{bmatrix} \tag{3.29}$$

$$\text{Input Matrix, } \mathbf{B} = \begin{bmatrix} \frac{P_{r0}}{l} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T \tag{3.30}$$

In this model also, the initial conditions were assumed such that the initial normalised power was specified as P_{r0} . So, from the equation of variation of precursor concentration (normalised) at steady state, it was found that, $C_{ir0} = P_{r0}$, where, C_{ir0} was the steady state normalised DNP concentrations for the delayed groups ($i = 1, \dots, 6$). Also, the location of the Eigen values (7 in this case) were similar to that of the PKPWR case, with one being at the origin and the others dependent on the value of the neutron lifetime (l) given as,

$$s = 0; -\left(\lambda_i + \frac{\beta_i}{l}\right) \tag{3.31}$$

3.2.2. MODULE 2 – THERMO-NEUTRONIC PWR (TNPWR)

In the previous section the model of the PKPWR, the elementary reactor model was discussed. Now as mentioned earlier in section 2.4.2, the internal feedbacks were applied to the PKPWR model. In this module however, only the thermal feedbacks were considered, i.e. the fuel and coolant temperature feedback loops. Here, the state space model for the entire primary loop of the PWR was built by considering thermal feedbacks to the PKPWR.

3.2.2.1. FORMULATION OF THE TNPWR MODEL EQUATIONS

As mentioned in section 2.2.1.2, PWR being a heterogeneous reactor possess two temperature coefficients of reactivity, namely the fuel temperature coefficient of reactivity (prompt coefficient) and the moderator temperature coefficient (delayed coefficient). Since, in a PWR, moderator and coefficient are the same, the coolant temperature coefficient was used without

loss of generality. Also, it was mentioned that for the PWR both these coefficients are negative, thus providing inherent stability to the reactor.

Now, by considering equation (2.6),

$$\Delta\rho = \alpha_T * \Delta T \quad (3.32)$$

Where, α_T = Temperature coefficient of reactivity; ΔT = change in the temperature & $\Delta\rho$ = corresponding change in reactivity.

Based on equation (3.32), the fuel and coolant temperature reactivity induction equations were formulate as,

$$\Delta\rho_f = \alpha_f * (T_f - T_{f0}) \quad (3.33)$$

$$\Delta\rho_c = \alpha_c * (T_c - T_{c0}) \quad (3.34)$$

In the above two equations, α_f, α_c are the fuel and coolant temperature coefficients of reactivity respectively. And, T_f, T_{f0}, T_c, T_{c0} are the present fuel temperature, the steady state fuel temperature, the present coolant temperature and the steady state coolant temperature respectively.

Now, the normalised point kinetic equations given by (3.14) and (3.15) were considered and the equations (3.33) and (3.34) were incorporated within its framework while linearizing the system.

Firstly, it was considered that, $\rho(t) = \rho_f(t) + \rho_c(t)$ (3.35)

By substituting this equation in (3.14), it was found that,

$$\frac{d}{dt}P_r = \frac{(\rho_f(t) + \rho_c(t) - \beta)}{l}P_r + \frac{\beta}{l}C_r \quad (3.36)$$

In order to linearize the equation (3.36), the perturbation model was considered. The small deviations of the following quantities from the equilibrium or steady state condition were assumed to be,

$$\begin{aligned} P_r(t) &= P_{r0} + \delta P_r \\ C_r(t) &= C_{r0} + \delta C_r \\ \rho_f(t) &= \rho_{f0} + \delta\rho_f \\ \rho_c(t) &= \rho_{c0} + \delta\rho_c \\ T_f(t) &= T_{f0} + \delta T_f \\ T_c(t) &= T_{c0} + \delta T_c \end{aligned} \quad (3.37)$$

Also, according to assumption (c) of reactor neutronics, it was noted that α_f and α_c were constants. With these ideas in mind, equations (3.33) and (3.34) were written as [39],

$$\begin{aligned} \delta\rho_f &= \alpha_f * (T_f - T_{f0}) \\ \Rightarrow \delta\rho_f &= \alpha_f * \delta T_f \end{aligned} \quad (3.38)$$

Similarly,

$$\begin{aligned} \delta\rho_c &= \alpha_c * (T_c - T_{c0}) \\ \Rightarrow \delta\rho_c &= \alpha_c * \delta T_c \end{aligned} \quad (3.39)$$

Here, it was again considered that, $\rho_0 = 0$, since at the steady state the reactor was operating at *Critical* condition. So, $\rho_{f0} = 0$ and $\rho_{c0} = 0$. With these conditions, while linearizing the equations, it was found that, (refer **Appendix A.1**)

$$\frac{d}{dt} \delta P_r = \left(\frac{-\beta}{l} \right) \delta P_r + \left(\frac{\beta}{l} \right) \delta C_r + \left(\frac{P_{r0}}{l} \right) \delta \rho_f + \left(\frac{P_{r0}}{l} \right) \delta \rho_c \quad (3.40)$$

By substituting equations (3.38) and (3.39) in the equation (3.40), it was written as,

$$\boxed{\frac{d}{dt} \delta P_r = \left(\frac{-\beta}{l} \right) \delta P_r + \left(\frac{\beta}{l} \right) \delta C_r + \left(\frac{P_{r0}}{l} \right) \alpha_f \delta T_f + \left(\frac{P_{r0}}{l} \right) \alpha_c \delta T_c} \quad (3.41)$$

This equation represented the power variation equation for the TNPWR model. However, the linearized DNP concentration equation (3.17) remained unaltered. After formulating the linearized neutronics model with thermal terms, the thermal hydraulic equations describing the primary loop of the reactor as given in section 2.3.2.1, 2.3.2.2 and 2.3.2.3 were taken up. Firstly, the equations were normalised for Power (P), with steady state value P_0 . Then they were linearized as per the perturbation model given below [5], [6].

$$\begin{aligned} T_f(t) &= T_{f0} + \delta T_f \\ T_c(t) &= T_{c0} + \delta T_c \\ T_{cin}(t) &= T_{cin0} + \delta T_{cin} \\ T_{cout}(t) &= T_{cout0} + \delta T_{cout} \\ T_{SGin}(t) &= T_{SGin0} + \delta T_{SGin} \\ T_{SG}(t) &= T_{SG0} + \delta T_{SG} \\ sgp(t) &= sgp_0 + \delta sgp \end{aligned} \quad (3.42)$$

These conditions were used on the equations (2.25), (2.27), (2.28), (2.32), (2.38), (2.40) and (2.41), to formulate the linearized model for the primary loop of PWR as,

$$\begin{aligned} \frac{d}{dt} \delta T_f &= \frac{F_r P_0}{m_f C_f} \delta P_r - \frac{h_f A_f}{m_f C_f} (\delta T_f - \delta T_c) \\ \Rightarrow \quad \boxed{\frac{d}{dt} \delta T_f &= k_1 \delta P_r - \frac{1}{\tau_f} \delta T_f + \frac{1}{\tau_f} \delta T_c} \end{aligned} \quad (3.43)$$

Here $k_1 = F_r P_0 / m_f C_f$ and $\tau_f = m_f C_f / h_f A_f$ (unit in seconds). Similarly,

$$\begin{aligned} \frac{d}{dt} \delta T_c &= \frac{(1 - F_r) P_0}{m_c C_c} \delta P_r + \frac{h_f A_f}{m_c C_c} (\delta T_f - \delta T_c) - \frac{G_c C_c}{m_c C_c} (\delta T_{cout} - \delta T_{cin}) \\ \Rightarrow \quad \boxed{\frac{d}{dt} \delta T_c &= k_4 \delta P_r + k_2 \delta T_f - k_2 \delta T_c - k_3 \delta T_{cout} + k_3 \delta T_{cin}} \end{aligned} \quad (3.44)$$

Here, $k_4 = (1 - F_r)P_0/m_c C_c$; $k_2 = h_f A_f / m_c C_c$ and $k_3 = G_c C_c / m_c C_c$.

Also, from equation (2.24), $\delta T_{cout} = 2\delta T_c - \delta T_{cin}$ (3.45)

By using (3.45) in (3.44), another form of the equation was formed as,

$$\boxed{\frac{d}{dt}\delta T_c = k_4\delta P_r + k_2\delta T_f - (k_2 + 2k_3)\delta T_c + 2k_3\delta T_{cin}} \quad (3.46)$$

Either of the two equations (3.44) or (3.46) could have been used for the state space model. Equation (3.46) minimized the number of state variables. The next few equations were formed as given below.

$$\boxed{\frac{d}{dt}\delta T_{cout} = \frac{2}{\tau_{ct}}\delta T_c - \frac{1}{\tau_{ct}}\delta T_{cout} - \frac{1}{\tau_{ct}}\delta T_{cin}} \quad (3.47)$$

$$\boxed{\frac{d}{dt}\delta T_{cin} = -\frac{1}{\tau_{cld}}\delta T_{cin} - \frac{1}{\tau_{cld}}\delta T_{SGin} + \frac{2}{\tau_{cld}}\delta T_{SG}} \quad (3.48)$$

$$\boxed{\frac{d}{dt}\delta T_{SGin} = \frac{1}{\tau_{hld}}\delta T_{cout} - \frac{1}{\tau_{hld}}\delta T_{SGin}} \quad (3.49)$$

Equation for the thermal hydraulics of the steam generator was also normalised, since it had the terms of steam generator power (sgp) as follows,

$$\frac{d}{dt}T_{SG} = \frac{2(T_{SG} - T_{SGin})}{\tau_{SG}} - \frac{F_r P_0}{m_{SG} C_{SG}} sgp \quad (3.50)$$

Using this equation also minimised the number of state variables. By linearizing equation (3.50) and rearranging, it was rewritten as,

$$\boxed{\frac{d}{dt}\delta T_{SG} = \frac{2}{\tau_{SG}}\delta T_{SG} - \frac{2}{\tau_{SG}}\delta T_{SGin} - k_5\delta sgp} \quad (3.51)$$

Where, $k_5 = F_r P_0 / m_{SG} C_{SG}$.

The last equation for our model described the relation between the steam generator power and the feed flow as given by equation (3.51).

$$\boxed{\frac{d}{dt}\delta sgp = -\frac{1}{3}\delta sgp + \frac{1}{3}G_{feed}} \quad (3.52)$$

Hence, the equations (3.41), (3.17), (3.43), (3.44) or (3.46), (3.47), (3.48), (3.49), (3.51) and (3.52) were used to form the state space TNPWR model.

3.2.2.2. STATE SPACE TNPWR MODEL

Utilizing the Linearized equations mentioned above, the state space model of the TNPWR was formed, imitating the PKPWR model. The TNPWR system had 9 differential equations. So, 9 state variables were considered as defined by the state vector ($\mathbf{x}$). So, to start with the state equation was formed for which the state vector $\mathbf{x}$, the input vector $\mathbf{u}$ and the velocity vector $\dot{\mathbf{x}}$ were defined as,

$$\begin{aligned}\mathbf{x} &= [P_r \ C_r \ T_f \ T_c \ T_{cout} \ T_{cin} \ T_{SGin} \ T_{SG} \ sgp]^T \\ \mathbf{u} &= G_{feed} \\ \dot{\mathbf{x}} &= [\dot{P}_r \ \dot{C}_r \ \dot{T}_f \ \dot{T}_c \ \dot{T}_{cout} \ \dot{T}_{cin} \ \dot{T}_{SGin} \ \dot{T}_{SG} \ \dot{sgp}]^T\end{aligned}\quad (3.53)$$

Here, $P_r \rightarrow$ normalised power & $C_r \rightarrow$ normalised DNP concentration.

Upon comparison of these vectors to that of the PKPWR, two points were noticed. Firstly, the state vector and velocity vector contained all the thermal variables of the PWR. Next, and most importantly, the input vector was the feed flow rate in this case. As mentioned in the assumption (f) of the Reactor control systems, the feed flow rate follows the demand power. Since the PWR is inherently stable, the load following operation was shown with the single input, G_{feed} . However, another method would be to create a MIMO system with two inputs, one being the reactivity (ρ), same as in PKPWR and the next being the feed flow rate. The input matrix would seem like,

$$\mathbf{u} = [\rho \ G_{feed}]^T \quad (3.54)$$

Such a formation would substantially change the power equation (3.41) with the reactivity variables $\rho_f(t)$ and $\rho_c(t)$ becoming inputs to the system. Also, this technique would require the input matrix ($\mathbf{B}$) to be time varying since the power would get updated for every iteration. Hence, considering the TNPWR equations, as they were formed in this case, all these complexities were reduced and it provided a compact model with the internal feedbacks contained within the system matrix ($\mathbf{A}$). So, with the formation of the vectors in equation (3.53) and comparing all the 9 differential equations with (3.1), the State matrix ($\mathbf{A}$) and Input matrix ($\mathbf{B}$) were formed as given in equations (3.56) and (3.57). Similarly, the output equation was formed with Output vector ($\mathbf{y}$) defined as,

$$\mathbf{y} = [P_r \ C_r \ T_f \ T_c \ T_{cout} \ T_{cin} \ T_{SGin} \ T_{SG} \ sgp]^T \quad (3.55)$$

So, from equation (3.55), it could be interpreted that each element of the output vector is mapped one to one with the state vector ($\mathbf{x}$) variables as in equation (3.53) and the values of these variables get stored in it. Hence, the output matrix was considered to be an identity matrix with dimensions similar to that of the state matrix.

Hence, $\mathbf{A}$ became a 9x9 square matrix and $\mathbf{B}$ a 9x1 column matrix. It was considered that the Output Matrix, $\mathbf{C} = \mathbf{I}_9$, i.e. identity matrix with same dimensions as State matrix ($\mathbf{A}$) and the Direct transmission matrix, $\mathbf{D} = [\mathbf{0}]_{9 \times 1}$, i.e. null matrix with the same dimensions as $\mathbf{B}$.

$$\text{Input Matrix, } \mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \left(\frac{1}{3}\right) \end{bmatrix}^T \quad (3.56)$$

State Matrix, $A =$

$$\begin{bmatrix} -\left(\frac{\beta}{l}\right) & \frac{\beta}{l} & \left(\frac{P_{r0}}{l}\right)\alpha_f & \left(\frac{P_{r0}}{l}\right)\alpha_c & 0 & 0 & 0 & 0 & 0 \\ \lambda & -\lambda & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ k_1 & 0 & -\left(\frac{1}{\tau_f}\right) & \left(\frac{1}{\tau_f}\right) & 0 & 0 & 0 & 0 & 0 \\ k_4 & 0 & k_2 & -(k_2 + 2k_3) & 0 & (2k_3) & 0 & 0 & 0 \\ 0 & 0 & 0 & \left(\frac{2}{\tau_{ct}}\right) & -\left(\frac{1}{\tau_{ct}}\right) & -\left(\frac{1}{\tau_{ct}}\right) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\left(\frac{1}{\tau_{cld}}\right) & -\left(\frac{1}{\tau_{cld}}\right) & \left(\frac{2}{\tau_{cld}}\right) & 0 \\ 0 & 0 & 0 & 0 & \left(\frac{1}{\tau_{hld}}\right) & 0 & -\left(\frac{1}{\tau_{hld}}\right) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \left(\frac{2}{\tau_{SG}}\right) & -\left(\frac{2}{\tau_{SG}}\right) & -k_5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\left(\frac{1}{3}\right) \end{bmatrix} \quad (3.57)$$

Then, the equations were arranged in the state space form as given by equation (3.1) & (3.2).

Steady state conditions of the TNPWR model

It is already known that the state space modeling allows us to take into account the initial steady state conditions of the system. In the TNPWR model, this was done by calculating the initial parameters of the model by using the developed thermodynamic equations for a chosen operating power. Also, it was assumed that the average coolant temperature (T_c) or the hot leg temperature (T_{cout}) was known for that particular operating power.

It was considered that the initial normalised power was P_{r0} and the average coolant temperature was T_{c0} . Now, from the equation (3.15), at steady state, $C_{r0} = P_{r0}$, where, C_{r0} was the steady state normalised DNP concentration. From the other thermodynamic equations at steady state condition, the following equations were formulated.

$$Z = (T_{f0} - T_{c0}) = k_1 * P_{r0} * \tau_f \quad (3.58)$$

$$T_{f0} = T_{c0} + Z \quad (3.59)$$

$$T_{cin0} = T_{c0} - ((k_2 * Z)/(2 * k_3)) - ((k_4 * P_{r0})/(2 * k_3)) \quad (3.60)$$

$$T_{cout0} = (2 * T_{c0}) - T_{cin0} \quad (3.61)$$

$$T_{SGin0} = T_{cout0} \quad (3.62)$$

$$sgp_0 = G_{feed}(ini) \quad (3.63)$$

$$T_{SG0} = T_{SGin0} - ((k_5 * sgp_0 * \tau_{SG})/2) \quad (3.64)$$

According to assumption (e) of Reactor thermal systems, $F_r = 1.0$, so $k_4 = 0$. The A matrix and the steady state values were changed accordingly to account for this parameter. Similar, steady state values could be calculated, when the hot leg temperature is known for the operating power instead of T_{c0} . Then, in order to calculate the other parameters, the T_{cout0} value should be used.

External input to the TNPWR model

For this kind of a system, the behaviour of the plant depends on the external input, in this case the input vector ($\mathbf{u}$). It was assumed that an external input could be a deliberate change in demand power by the operator or an insertion of unwanted noise. Now, as mentioned in the assumption (f) of the reactor control systems, the feed flow (G_{feed}) is directly dictated by the change in demand power from the operating power. So, as mentioned in section 2.4.3.1, this would initiate a set of events ultimately leading the reactor power to obey the demand power, mainly owing to the addition of positive reactivity due to negative temperature coefficients of fuel and coolant for an increase in demand power and addition of negative reactivity for a decrease in demand power. Hence, the external input to the model in the form of the G_{feed} variable may lead to ideal PWR plant behaviour. While changing the feed flow to cater for the demand power, it was also assumed that the opening of the FCV was linear and there were no internal delays within the FCV. The rate at which FCV opened or closed was denoted by R_{FCV} . Mathematically, G_{feed} and R_{FCV} were expressed by the relation,

$$G_{feed} = G_{feed}(ini) + R_{FCV} * dt \quad (3.65)$$

Here, it was considered that,

- At steady state, the equilibrium power was the same as the demand power, so, the feed flow, $G_{feed}(ini) = P_{r0}$.
- The rate of change of the FCV was directly governed by the error power between the demand power and the steady state or actual power. So, it was expressed as a ratio iteration as given below.

$$R_{FCV} = ((demand\ power - actual\ power)/actual\ power) \quad (3.66)$$

So, while simulating the TNPWR system, the initial equilibrium power (P_{r0}) and the demand power (normalised) were taken as the input values and the rest were automatically calculated for the system. The TNPWR system thus formed was simulated with various inputs and the Eigen values of the system were obtained, which are provided later in chapter 4. Also, the issue of stability analysis using Lyapunov theory was carried out.

3.2.2.3. REDUCED ORDER TNPWR (RO-TNPWR) MODEL

The TNPWR model formed above, was also formulated with a reduced dimension simply by eliminating the state variable, T_{cout} . This was made possible by substituting the T_{cout} in equation (3.49) with the following equation as given by (3.45).

$$T_{cout} = (2 * T_c) - T_{cin} \quad \Rightarrow \quad \delta T_{cout} = (2\delta T_c) - \delta T_{cin} \quad (3.67)$$

By substituting equation (3.67) in equation (3.49), the following equation was obtained.

$$\boxed{\frac{d}{dt}\delta T_{SGin} = \frac{2}{\tau_{hld}}\delta T_c - \frac{1}{\tau_{hld}}\delta T_{cin} - \frac{1}{\tau_{hld}}\delta T_{SGin}} \quad (3.68)$$

When the set of equations (3.41), (3.17), (3.43), (3.46), (3.48), (3.68), (3.51) and (3.52) was considered, it was seen that the variable T_{cout} had been eliminated. Still, these equations represents the primary loop of the PWR and were used to form the RO-TNPWR model with just 8 state variables as obtained from the 8 differential equations mentioned above. Similar to the TNPWR case, the state & output equations of RO-TNPWR were formed for which the state vector ($\mathbf{x}$), the input vector ($\mathbf{u}$), the velocity vector ($\dot{\mathbf{x}}$) and the output vector ($\mathbf{y}$) were defined as,

$$\begin{aligned} \mathbf{x} &= [P_r \ C_r \ T_f \ T_c \ T_{cin} \ T_{SGin} \ T_{SG} \ sgp]^T \\ \mathbf{u} &= G_{feed} \\ \dot{\mathbf{x}} &= [\dot{P}_r \ \dot{C}_r \ \dot{T}_f \ \dot{T}_c \ \dot{T}_{cin} \ \dot{T}_{SGin} \ \dot{T}_{SG} \ \dot{sgp}]^T \\ \mathbf{y} &= [P_r \ C_r \ T_f \ T_c \ T_{cin} \ T_{SGin} \ T_{SG} \ sgp]^T \end{aligned} \quad (3.69)$$

With these vectors in place, the State space matrices for this model were formed. It was seen that the State matrix ($\mathbf{A}$) was a 8x8 square matrix (T_{cout} row deleted) and B was a 8x1 column matrix. It was considered that the Output Matrix, $\mathbf{C} = \mathbf{I}_8$, i.e. identity matrix with same dimensions as State matrix ($\mathbf{A}$) and the Direct transmission matrix, $\mathbf{D} = [\mathbf{0}]_{8 \times 1}$, i.e. null matrix with same dimensions as input matrix ($\mathbf{B}$). Then, the equations were arranged in the state space form as given by equations (3.1) and (3.2).

Steady state conditions and External Input of the RO-TNPWR model

While specifying the steady state conditions for RO-TNPWR model, it was found that only the T_{cout0} variable got eliminated by substitution of (3.61) in (3.62) as,

$$T_{SGin0} = (2 * T_{c0}) - T_{cin0} \quad (3.70)$$

Other than this change in the steady state value, all the other equations, i.e. (3.58), (3.59), (3.60), (3.63) and (3.64) remained unaltered. Also the initial normalised power (P_{r0}) and the average coolant temperature (T_{c0}) were known. From the equation (3.15), $C_{r0} = P_{r0}$, the steady state normalised DNP concentration. This specified all the steady state values for the 8 state variables. In this model, the external input as specified by the input vector, remained the same as for TNPWR, i.e. G_{feed} . So, the conditions for the input remained unaltered as expressed by the equations (3.65) and (3.66).

3.2.3. MODULE 3 – TNPWR WITH XENON FEEDBACK (TNXPWR)

In this module, the Xenon poisoning feedback was added with the TNPWR model developed in the previous module, thus completing the internal feedback loop as explained in section 2.4.2. As it is a slow acting process, it doesn't have much of an effect on power variations, during the changes in demand power which are considered as relatively fast changes.

Although it is an integral part of the reactors' internal feedback mechanism and so, was incorporated with the TNPWR model.

3.2.3.1. INCORPORATION OF XENON EFFECT INTO THE TNPWR MODEL

Xenon being a neutron poison adds negative reactivity to the system. In literature [7], [4] this reactivity variation in a reactor, due to xenon is expressed in the form of,

$$\Delta\rho_X = -\left(\frac{\sigma_X}{\Sigma_a}\right)X \quad (3.71)$$

Where, σ_X is the microscopic thermal neutron absorption cross section of xenon. And, X is the Xenon concentration at that time.

Now, going by the formulation of equation (3.35), the reactivity within the reactor was expressed as a summation of three terms each pertaining to an internal feedback phenomenon as given below.

$$\rho(t) = \rho_f(t) + \rho_c(t) + \rho_X(t) \quad (3.72)$$

So, the reactivity due to xenon ($\rho_X(t)$) got added to the already existent thermal feedback. Then, by arranging the power equation similar to equation (3.36) and later, using the perturbation model to linearize it, lead to an equation similar to the equation (3.40) only with an added term for the xenon induced reactivity (refer **Appendix A.1**).

$$\frac{d}{dt}\delta P_r = \left(\frac{-\beta}{l}\right)\delta P_r + \left(\frac{\beta}{l}\right)\delta C_r + \left(\frac{P_{r0}}{l}\right)\delta\rho_f + \left(\frac{P_{r0}}{l}\right)\delta\rho_c + \left(\frac{P_{r0}}{l}\right)\delta\rho_X \quad (3.73)$$

By substituting the equations (3.38), (3.39) and (3.71) into (3.73), the overall power equation with full internal feedback was obtained as below.

$$\frac{d}{dt}\delta P_r = \left(\frac{-\beta}{l}\right)\delta P_r + \left(\frac{\beta}{l}\right)\delta C_r + \left(\frac{P_{r0}}{l}\right)\alpha_f\delta T_f + \left(\frac{P_{r0}}{l}\right)\alpha_c\delta T_c - \left(\frac{P_{r0}}{l}\right)\left(\frac{\sigma_X}{\Sigma_a}\right)\delta X \quad (3.74)$$

However, the set of equations [(3.43) – (3.52)] which described the thermal behaviour of the plant in TNPWR remained unaltered. Two additional equations were added to the already existent nine equation model of TNPWR, one for the Xenon concentration and the other for the Iodine concentration as mentioned in section 2.2.2.2. By rearranging the equations (2.17) and (2.18) based on a similar argument presented for equation (3.9) and (3.10) gave,

$$\frac{dI}{dt} = \gamma_I\Sigma_f P - \lambda_I I \quad (3.75)$$

$$\frac{dX}{dt} = \gamma_X\Sigma_f P + \lambda_I I - \lambda_X X - \sigma_X X P \quad (3.76)$$

Then these two equations were normalised for only Power (P) w.r.t. the steady state value P_0 . Normalisations of these equations lead to the formulation of the equations given beneath.

$$\frac{dI}{dt} = (\gamma_I \Sigma_f P_0) P_r - \lambda_I I \quad (3.77)$$

$$\frac{dX}{dt} = (\gamma_X \Sigma_f P_0) P_r + \lambda_I I - \lambda_X X - \sigma_X P_0 P_r X \quad (3.78)$$

To linearize the equations (3.77) and (3.78) based on the perturbation model, it was assumed,

$$\begin{aligned} P_r(t) &= P_{r0} + \delta P_r \\ I(t) &= I_0 + \delta I \\ X(t) &= X_0 + \delta X \end{aligned} \quad (3.79)$$

Linearization of these equations leads to the following forms (refer to **Appendix A.2**)

$$\frac{d}{dt} \delta I = k_I \delta P_r - \lambda_I \delta I \quad (3.80)$$

$$\frac{d}{dt} \delta X = (k_X - \sigma_X P_0 X_0) \delta P_r + \lambda_I \delta I - (\lambda_X + \sigma_X P_0 P_{r0}) \delta X \quad (3.81)$$

Here, $k_I = \gamma_I \Sigma_f P_0$ and $k_X = \gamma_X \Sigma_f P_0$. To form state space matrix of TNXPWR, these equations were added with the TNPWR state space formulation.

3.2.3.2. STATE SPACE TNXPWR MODEL

The TNXPWR system had 11 differential equations. In addition to the 9 state variables already mentioned while developing the TNPWR, 2 more variables pertaining to xenon (X) and iodine (I) concentration were added to the state vector ($\mathbf{x}$). So, to form the state and output equations, the state vector ($\mathbf{x}$), the input vector ($\mathbf{u}$), the velocity vector ($\dot{\mathbf{x}}$) and the output vector ($\mathbf{y}$) were defined as,

$$\begin{aligned} \mathbf{x} &= [P_r \ C_r \ I \ X \ T_f \ T_c \ T_{cout} \ T_{cin} \ T_{SGin} \ T_{SG} \ sgp]^T \\ \mathbf{u} &= G_{feed} \\ \dot{\mathbf{x}} &= [\dot{P}_r \ \dot{C}_r \ \dot{I} \ \dot{X} \ \dot{T}_f \ \dot{T}_c \ \dot{T}_{cout} \ \dot{T}_{cin} \ \dot{T}_{SGin} \ \dot{T}_{SG} \ \dot{sgp}]^T \\ \mathbf{y} &= [P_r \ C_r \ I \ X \ T_f \ T_c \ T_{cout} \ T_{cin} \ T_{SGin} \ T_{SG} \ sgp]^T \end{aligned} \quad (3.82)$$

Similar to the above two modules, the state space matrices were formed as given below. Here, the State matrix ($\mathbf{A}$) was a 11x11 square matrix and Input matrix ($\mathbf{B}$) was a 11x1 column matrix. It was considered that the Output Matrix, $\mathbf{C} = \mathbf{I}_{11}$, i.e. identity matrix of the same dimensions as ($\mathbf{A}$) & the Direct transmission matrix, $\mathbf{D} = [\mathbf{0}]_{11 \times 1}$, i.e. null matrix of the same dimensions as ($\mathbf{B}$).

$$\text{Input Matrix, } \mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \left(\frac{1}{3}\right) \end{bmatrix}^T \quad (3.83)$$

State Matrix, $\mathbf{A} =$

$$\begin{bmatrix} -\left(\frac{\beta}{l}\right) & \frac{\beta}{l} & 0 & -\left(\frac{P_{r0}}{l}\right)\left(\frac{\sigma_X}{\Sigma_a}\right) & \left(\frac{P_{r0}}{l}\right)\alpha_f & \left(\frac{P_{r0}}{l}\right)\alpha_c & 0 & 0 & 0 & 0 & 0 \\ \lambda & -\lambda & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ k_I & 0 & -\lambda_I & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ (k_X - \sigma_X P_0 X_0) & 0 & \lambda_I & -(\lambda_X + \sigma_X P_0 P_{r0}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ k_1 & 0 & 0 & 0 & -\left(\frac{1}{\tau_f}\right) & \left(\frac{1}{\tau_f}\right) & 0 & 0 & 0 & 0 & 0 \\ k_4 & 0 & 0 & 0 & k_2 & -(k_2 + 2k_3) & 0 & (2k_3) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \left(\frac{2}{\tau_{ct}}\right) & -\left(\frac{1}{\tau_{ct}}\right) & -\left(\frac{1}{\tau_{ct}}\right) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\left(\frac{1}{\tau_{cld}}\right) & -\left(\frac{1}{\tau_{cld}}\right) & \left(\frac{2}{\tau_{cld}}\right) & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \left(\frac{1}{\tau_{hld}}\right) & 0 & -\left(\frac{1}{\tau_{hld}}\right) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \left(\frac{2}{\tau_{SG}}\right) & -\left(\frac{2}{\tau_{SG}}\right) & -k_5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\left(\frac{1}{3}\right) \end{bmatrix} \quad (3.84)$$

With the state space matrices specified, the equations were again arranged in the state space form as given by the equations (3.1) and (3.2).

Steady state conditions and External Input of the TNXPWR model

As in case of the TNPWR model, the steady state conditions were found by calculating the initial parameters of the model, using the developed thermodynamic equations for a chosen operating power. Also, that the average coolant temperature (T_c) or the hot leg temperature (T_{cout}) was assumed to be known for that particular operating power. For the TNXPWR model, all those steady state conditions as given by equations (3.58) – (3.64) remained the same. Two additional parameters were added for the xenon and iodine initial concentrations.

$$I_0 = (k_I * P_{r0}) / \lambda_I \quad (3.85)$$

$$X_0 = ((k_X * P_{r0}) + (\lambda_I * I_0)) / (\lambda_X + (\sigma_X * P_0 * P_{r0})) \quad (3.86)$$

Here, P_{r0} was the initial normalised power (specified initially) and P_0 was the steady state power used for normalising the equations (specified initially; can be considered as rated power of the reactor). In this model also, the external input as specified by the input vector, remained the same as for TNPWR, i.e. G_{feed} . So, the conditions for the input remained unaltered as expressed by the equations (3.65) and (3.66).

3.3. DESIGN OF REACTOR REGULATING SYSTEM

As mentioned earlier in section 2.4.3.2, the Reactor regulating system (RRS) is an automated control system which forms an integral part of any reactor system and plays a significant role in output power regulation. Its major task is to closely match the power output level of the reactor with a demanded power level within a given error and to bring down the reactivity variation ultimately to zero for the demanded power level, thus making the reactor critical. Also, as mentioned in assumption (c) of Reactor control systems, the RRS in this study was assumed to be a Discontinuous system. Although, the control algorithm for the RRS takes into account all the reactor operational regimes, in this study only the power operational regime for the RRS was concentrated upon.

3.3.1. RRS IMPLEMENTATION SCHEME

The RRS implementation scheme used in this study closely resembles the actual scheme, implemented for RRS in PWRs. It is known that the power operational regime of the reactor can be categorized mainly into two modes namely, the *Steady state mode* & the *Transient mode*. The scheme used here, took into account the RRS action for both these modes separately. It was assumed that the RRS was divided into two subunits. The first subunit was used to implement the control logic for regulation of reactor power and temperature and to generate signals which activated the second subunit. The second subunit was mainly concerned with the generation of control signals for the upward or downward movement of the CRs [21].

3.3.1.1. RRS SUBUNIT (A)

As mentioned earlier in section 2.4.3.2, two different strategies exist for the implementation of the control algorithm of the RRS. One was concerned with the Power level and the other with the Average coolant temperature. For the subunit (a), the control signal used here was a combination of both of these strategies having a power level variation signal and an average coolant temperature variation signal.

1. Transient Mode Scheme

The transient mode of power operation for the PWR is the state when the rate of change of feed flow is more than the specified limit. It can occur under conditions of increase or decrease in demand power & during power setbacks. It was assumed that during a transient state, this subunit immediately starts to generate error signals proportional to summation of the Average coolant temperature variation and the difference between actual power & demand power. It was considered that the difference between the actual power and demand power was expressed as,

$$\begin{aligned}\Delta P &= \text{Actual power} - \text{demand power} \\ \Rightarrow \Delta P_r &= P_r - G_{feed}\end{aligned}\tag{3.87}$$

This was formulated considering that the feed flow represented the change in demand power, in this study.

Also, the deviation of the average coolant temperature from the set value was considered as,

$$\Delta T_c = T_c - T_{set} \quad (3.88)$$

The set value for the coolant temperature could be set as per requirement. In this study, it was the same as the initial value of the average coolant temperature which was specified earlier. Hence, the error signal, say u_{es} , which was proportional to both ΔP and ΔT_c was expressed as,

$$u_{es} = (\zeta_1 * \Delta P_r) + (\zeta_2 * \Delta T_c) \quad (3.89)$$

Here, ζ_1 and ζ_2 are two proportionality constants [21].

In the transient mode, no threshold values were provided for the u_{es} . It started to act as soon as any error signal was generated. Also, there was no delay in the execution of the control rod actuators as given in the RRS schematic in **Fig. 2.10**. The error signal initiated the movement of the control rods [given by subunit (b)], which tried to negate the increase or decrease in the reactivity of the system. The system remained in the transient mode, till time T_1 after the position of the FCV matched with the desired demand power. Hence, the RRS continued to implement this *transient mode scheme* for an additional time T_1 , even after the FCV was steady at the demand power. This ensured that the output power settled down quickly.

2. Steady State Scheme

In the steady state, the demand power doesn't change with time. So, the output power matches the demand power all the time. However, if there is any noise in the system, there can be occasional mismatch between them. Apart from the reactor's internal feedback system which made the power to move back to the initial state, an external scheme was implemented, similar to the one used in transient mode. The only difference between the transient mode scheme and the steady state scheme was that the error signal was manifested in the form of control signal after a certain delay of time say T_2 in case of steady state mode, whereas there was no such delay in the transient mode. Till this threshold value, the control system remained inactive and allowed the reactor to settle down on its own depending on its internal feedbacks. The control logic for the steady state error signal generation was [21],

- a. If $\Delta T_c \leq \pm T_c$, then $u_{es} = 0$.
- b. If $\Delta T_c \geq \pm T_c$ and $t \leq T_2$, then $u_{es} = 0$.
- c. If $\Delta T_c \geq \pm T_c$ and $t \geq T_2$, then $u_{es} = (\zeta_1 * \Delta P_r) + (\zeta_2 * \Delta T_c)$. (3.90)

Here, $\pm T_c$ (in°C) was the threshold value for the average coolant temperature, which triggered the control signal in the first place. After the dead band was crossed, the CRs moved according to the logic of the subunit (b) of the RRS and the reactivity was changed so as to minimize the error.

3.3.1.2. RRS SUBUNIT (B)

The second subunit of the RRS as mentioned was only concerned with generating the control signals for the movement of CRs either in the upward direction or in the downward direction.

Once the error signal from the subunit (a) crossed a predetermined threshold value, it triggered the movement of the CRs. The control logic for the second subunit was [21],

- a. If $u_{es} < u_{thresh}$, then $u = 1$. (CRs move upward)
- b. If $-u_{thresh} \leq u_{es} \leq u_{thresh}$, then $u = 0$.
- c. If $u_{es} > u_{thresh}$, then $u = -1$. (CRs move downward) (3.91)

Here, u_{thresh} was the threshold value of the error signal that triggered the CRs to move. And, u represented the upwards or downwards movement of the CRs. An upward movement introduced positive reactivity in the core and a downward movement introduced negative reactivity. The choice of the control schemes for this RRS made it a nonlinear control system.

3.3.2. INCORPORATION OF RRS IN SYSTEM MODELS

With the RRS implementation scheme known, the RRS was designed in such a way so that it could be incorporated with any of the PWR models developed in this study. Here, the implementation has been shown with the TNXPWR model only.

3.3.2.1. SYSTEM FORMULATION WITH RRS

Firstly, going by the formulation of equation (3.72), the reactivity within the reactor was expressed as a summation of four terms. The first three represented an internal feedback phenomenon and the last one was for the external feedback (ρ_{CR}) in the form of CRs. Mathematically, it was expressed as,

$$\rho(t) = \{\rho_f(t) + \rho_c(t) + \rho_x(t)\} + \rho_{CR}(t) \quad (3.92)$$

However, the addition of the term $\rho_x(t)$, was based entirely on the choice of the model. Then, by arranging the power equation similar to equation (3.36) and later, using the perturbation model to linearize it, lead to an equation similar to (3.73) with an additional term for the external reactivity insertion via the CRs.

$$\frac{d}{dt} \delta P_r = \left(\frac{-\beta}{l} \right) \delta P_r + \left(\frac{\beta}{l} \right) \delta C_r + \left(\frac{P_{r0}}{l} \right) \delta \rho_f + \left(\frac{P_{r0}}{l} \right) \delta \rho_c + \left(\frac{P_{r0}}{l} \right) \delta \rho_x + \left(\frac{P_{r0}}{l} \right) \delta \rho_{CR}$$

(3.93)

Since the reactor was assumed to operate in the power regime, the reactivity variation was considered to be in the linear zone of the S-curve for rod position. So, the slope of the curve, say K_{rod} was assumed to be constant. Also, the CR movement was governed by the rod velocity, v .

The reactivity variation for CRs was expressed as [21], $\delta \rho_{CR} = K_{rod} v u t$ (3.94)

Here, u was the control signal from the second subunit of RRS which signified the direction of the rod movement as per equations (3.91) and t was the integrated duration of rod movement. However, apart from the power equation, the set of equations [(3.17), (3.43) – (3.52)] describing the DNP concentration and thermal behaviour of the plant, respectively as in TNPWR remained unaltered.

3.3.2.2. STATE SPACE SYSTEM MODEL WITH RRS AS INPUT (RRS-PWR)

In the system model just developed, the only change came in the form of the external reactivity addition due to the CRs. So as to keep the system with the internal feedbacks intact, this change was introduced as an external input to the system. Thus, to form the state and output equations, the state vector ($\mathbf{x}$), the input vector ($\mathbf{u}$), the velocity vector ($\dot{\mathbf{x}}$) and the output vector ($\mathbf{y}$) were defined as,

$$\begin{aligned}\mathbf{x} &= [P_r \ C_r \ I \ X \ T_f \ T_c \ T_{cout} \ T_{cin} \ T_{SGin} \ T_{SG} \ sgp]^T \\ \mathbf{u} &= [\rho_{CR} \ G_{feed}]^T \\ \dot{\mathbf{x}} &= [\dot{P}_r \ \dot{C}_r \ \dot{I} \ \dot{X} \ \dot{T}_f \ \dot{T}_c \ \dot{T}_{cout} \ \dot{T}_{cin} \ \dot{T}_{SGin} \ \dot{T}_{SG} \ \dot{sgp}]^T \\ \mathbf{y} &= [P_r \ C_r \ I \ X \ T_f \ T_c \ T_{cout} \ T_{cin} \ T_{SGin} \ T_{SG} \ sgp]^T\end{aligned}\quad (3.95)$$

From the formulation of the vectors, it was observed that the State matrix ($\mathbf{A}$) and the Output matrix ($\mathbf{C}$), remained unaltered. If the TNXPWR model is considered, the state matrix gets represented by (3.84) and similarly for the TNPWR model it gets represented by the equation (3.57). The input matrix ($\mathbf{B}$), was changed to incorporate the new addition to our equation and it was represented as,

$$\text{Input Matrix, } \mathbf{B} = \begin{bmatrix} \left(\frac{P_{ro}}{l}\right) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \left(\frac{1}{3}\right) \end{bmatrix}^T \quad (3.96)$$

Also, the steady state conditions and input conditions for the feed flow didn't change, pertaining to the particular model that was chosen. For this particular model, the PWR system in this study became a MIMO system with two inputs, one being the Reactivity due to CRs (ρ_{CR}) and the other being the already existent feed flow (G_{feed}).

3.3.2.3. STATE FEEDBACK MODEL OF THE RRS-PWR

In order to analyse, the impact of the RRS algorithms as mentioned on the systems models that were developed (TNPWR or TNXPWR), a state feedback model of the system was designed [3], [26], [21]. The usefulness of the state feedback model lied in the fact that the components of the error signal were state variables, i.e. the neutron power and the coolant average temperature. Also, it was helpful in analysing the behaviour of the Eigen values when the transient mode scheme was implemented for the power operation of the model.

General form of state feedback

As the name suggests, the state feedback is concerned with manipulating the system input signal via feedback of the state variables to it. The general form of state feedback can be expressed as,

$$\mathbf{u} = -\mathbf{K}\mathbf{x} + \mathbf{r} \quad (3.97)$$

Here, $\mathbf{u}$ is the forcing function, $\mathbf{x}$ is the state vector, $\mathbf{r}$ is the reference input vector and $\mathbf{K}$ is the gain matrix. In our study, however, we modelled the forcing function as,

$$\mathbf{u} = K_1 \mathbf{x} + K_2 \mathbf{r} \quad (3.98)$$

Here, K_1 and K_2 are the gain matrices. Now, to understand the affect such a forcing function on the state space LTI system the forcing function in equation (3.1) was replaced with the one represented in equation (3.98).

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}(K_1 \mathbf{x} + K_2 \mathbf{r}) \\ \Rightarrow \dot{\mathbf{x}} &= (\mathbf{A} + \mathbf{B}K_1)\mathbf{x} + \mathbf{B}K_2 \mathbf{r} \\ \Rightarrow \dot{\mathbf{x}} &= \mathbf{A}_c \mathbf{x} + \mathbf{B}K_2 \mathbf{r} \end{aligned} \quad (3.99)$$

Where, $\mathbf{A}_c = (\mathbf{A} + \mathbf{B}K_1)$.

Since the System matrix changed from $\mathbf{A}$ to $\mathbf{A}_c$, the Eigen values of the system also get shifted. So, the system stability has to be re-evaluated based on the new state matrix ($\mathbf{A}_c$)

Our main objective here was to design the forcing function given in equation (3.95) in the form of equation (3.99) and in the process, define the gain matrices (K_1 and K_2). After designing the system, the system stability for this RRS based PWR model could be analysed.

State feedback implementation on RRS-PWR model

In order to form the forcing function, the reference input vector ($\mathbf{r}$) and the two gain matrices, K_1 and K_2 were to be determined.

1. First, the reference vector was defined. The reference vector for the state feedback of the RRS-PWR model had external reactivity insertion in the form of CRs and demanded feed flow as its variables, similar to input vector equation in (3.95). Mathematically, it was expressed as,

$$\mathbf{r} = [\rho_{CR} \ G_{feed}]^T \quad (3.100)$$

There were two situations under which the reference vector changes. The first one was a change in the demand power by the operator. This caused a change in the second element of the reference vector, thus causing a mismatch between the output and the demand power. This ultimately forced the input signal ($\mathbf{u}$) to change. But, during steady state the output power remained the same as the demand power, so the reference vector and in turn the forcing function ($\mathbf{u}$) didn't change. Another way to change $\mathbf{r}$, was to add external reactivity via the CRs. However, both these changes were dependent on the gain matrix K_2 .

2. Next, the gain matrices were developed and in doing so, the control logic given in equation (3.89) was incorporated, which dictated the transient mode operation of the reactor. From equation (3.94),

$$\rho_{CR} = K_{rod} vut \quad (3.101)$$

The RHS of the above equation was multiplied and divided by u_{es} , thus rewriting it as,

$$\rho_{CR} = K_{rod} vut * \left(\frac{u_{es}}{u_{es}} \right) \quad (3.102)$$

This procedure ensured that the error feedbacks due to the power and average coolant temperature as given by equations (3.87), (3.88) and (3.89) were included in the gain

matrices. However, the equation (3.102) was valid only for cases when $u = +1$ or -1 as given by the algorithm (3.91) for the subunit (b) of the RRS.

Under steady state conditions $u_{es} = 0$, this made $\rho_{CR} = 0$. So, the state space system formed in equation (3.95) had only one input, the feed flow (G_{feed}) and became equivalent to the original TNXPWR or TNPWR which had only the internal feedbacks. Also, during the dead band $[-u_{thresh} \leq u_{es} \leq u_{thresh}]$ as given by the equation (3.91), the control signal $u = 0$. This impend any rod movement and left the reactor to its internal feedbacks. So, under these two conditions the equation (3.102) was not valid and the system became equivalent to the original system without RRS.

By replacing the numerator in equation (3.102) by (3.89) gave,

$$\rho_{CR} = K_{rod}vut * (\{\zeta_1 * \Delta P_r\} + \{\zeta_2 * \Delta T_c\})/u_{es} \quad (3.103)$$

$$\Rightarrow \boxed{\rho_{CR} = \frac{K_{rod}vut}{u_{es}} \zeta_1 \delta P_r + \frac{K_{rod}vut}{u_{es}} \zeta_2 \delta T_c} \quad (3.104)$$

It was assumed that,

$$K_{11} = \frac{K_{rod}vut}{u_{es}} \zeta_1 \quad \text{and} \quad K_{12} = \frac{K_{rod}vut}{u_{es}} \zeta_2 \quad (3.105)$$

Then, the equation (3.104) was rewritten as,

$$\rho_{CR} = K_{11} \delta P_r + K_{12} \delta T_c \quad (3.106)$$

Where, K_{11} and K_{12} form the elements of the gain matrix K_1 . Hence, it was concluded that the control rod signal was proportional to two state variables and gain matrices were designed accordingly.

Now with these formulations in place and with the objective of satisfying the equation (3.99) or (3.93), the two gain matrices were found to be,

$$K_1 = \begin{bmatrix} K_{11} & 0 & 0 & K_{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad K_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad (3.107)$$

Then, the forcing function (u) given by equation (3.102) was reformed as,

$$u = \begin{bmatrix} K_{11} & 0 & 0 & K_{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} \rho_{CR} \\ G_{feed} \end{bmatrix} \quad (3.108)$$

Here, x was the state vector as mentioned in equation (3.95). Hence, the forcing function (u) remained a 2 element column matrix, where the first row was for the reactivity due to CR insertion and the second row was for the changes in the demand power (i.e. the feed flow, G_{feed}).

3.4. CHAPTER SUMMARY

This chapter presents the PWR models designed during this study. It starts with justification for the choice of State Space approach to model the PWR system. Prior to modelling the PWR systems based on the mathematical background available from the previous chapter, a general set of rules and assumptions are provided. These work as guidelines for the model.

The formation of the PWR models is done in a progressive manner, starting from the bare reactor model to the closed loop reactor with thermal and poison feedbacks. The task of modeling is divided in 3 modules. In the first module, the *Bare Reactor model* is developed based on the point kinetics equations for a single group of delayed neutrons. The point kinetics is first normalised and then linearised, before arranging it into the state space framework. A SISO model for this system is formed with reactivity (ρ) as the input and the state vectors and variables are defined. This model is extended by taking into account the equations for all the six groups of delayed neutrons and by following a similar procedure as mentioned above.

In the second module, the *thermal feedbacks* are added to the Bare Reactor model. The equations describing the thermal hydraulic behaviour of the plant as discussed in the previous chapter are used to form a complete model of the PWR primary loop. Again, a normalisation-linearization procedure is applied on the model equations to come up with a linear model of the system. This system is arranged in the SISO state space framework with the Feed Flow (G_{feed}) being the only input to the system. Also, the steady state conditions are specified for such a system. In the third module, *poisoning feedback* due to Xenon is added to the model available from the second module. A similar procedure as mentioned in the second module is followed so as to form state space model of this system.

Further, a Reactor Regulating system is designed based on a similar control logic used in practical PWR plants. To start with, the control logic is clearly indicated for various scenarios that are usually encountered during reactor operation. A state feedback system is formed to incorporate the RRS with the State space models of the systems designed in the three modules.

CHAPTER 4

SIMULATION AND RESULTS

The state space models formed during this study were simulated on a commercially available software package. These models were simulated in order of their development for the power operational range of the PWR. The system under investigation was tested for observability and state controllability and the Eigen values were also recorded.

In this chapter, the first section comprises of the performance analysis of the state space models (PKPWR, TNPWR and TNXPWR). The next section shows a Block Control model of the PWR system, built for comparison purposes with the mathematical models formed. And at last, the third section comprises of the stability analysis of these aforesaid mathematical models.

4.1. PERFORMANCE ANALYSIS OF PWR MODELS

The analytical models of the PWR were simulated so as to measure their performance and validate the same with the performances available from literature and practical sources.

4.1.1. PKPWR ANALYSIS

This model described the behaviour of the normalised reactor power (P_r) and the normalised DNP concentration (C_r) only. It was considered that the state space model of the system provides only the power as output as given by equation (3.22) and the only input to the system being the reactivity (ρ). The values of various parameters for this simulation are provided in **Appendix B**. The analysis can be divided into two parts, one concerning the eigen value behaviour, controllability and observability and the next being the behaviour of the state variables especially power (P_r).

4.1.1.1. SYSTEM BEHAVIOUR

- A. The system so formed was found to be state controllable and observable. The Eigen values of this system were found to be,

$$eigen(A) = -31.0053 ; 0.0000$$

This behaviour of the Eigen spectrum exactly matched that explained by the equation (3.24) under the section 3.2.1.3. , thus validating that the system is marginally stable. However, it should be pointed out that the first Eigen value was not exactly zero but a negative number of order e-017 and hence, was considered to be zero.

- B. During this simulation, the system was considered to be operating at steady state power level of 60 % F.P ($P_{r0} = 0.6$). The reactivity input was administered to the system at 20 seconds. The simulation was performed for duration of 50 seconds with a step size of 0.01 second. Also, the initial conditions were specified as given under the section 3.2.1.3.

C. Two significant cases have been put up here for elaboration of the system behaviour.

- (i) Reactivity input of 1mK ($\rho = 0.001$) at steady state.
- (ii) Reactivity input of -1mK ($\rho = -0.001$) at steady state.

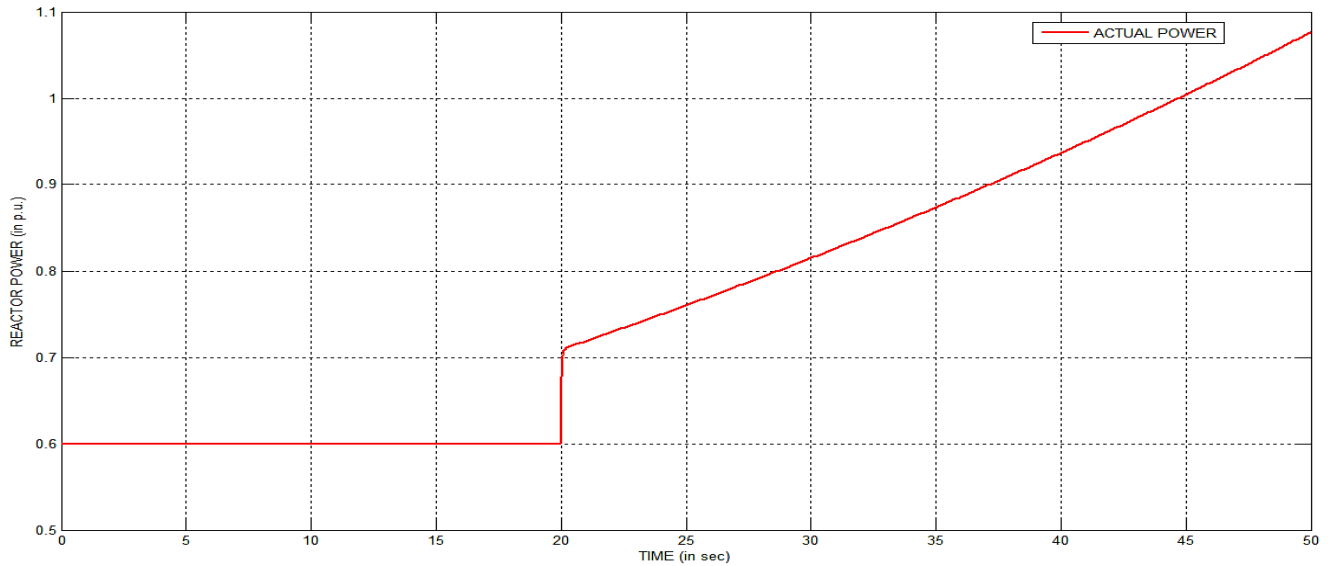


Fig 4.1: Plot of Reactor power of PKPWR for 1mK reactivity insertion

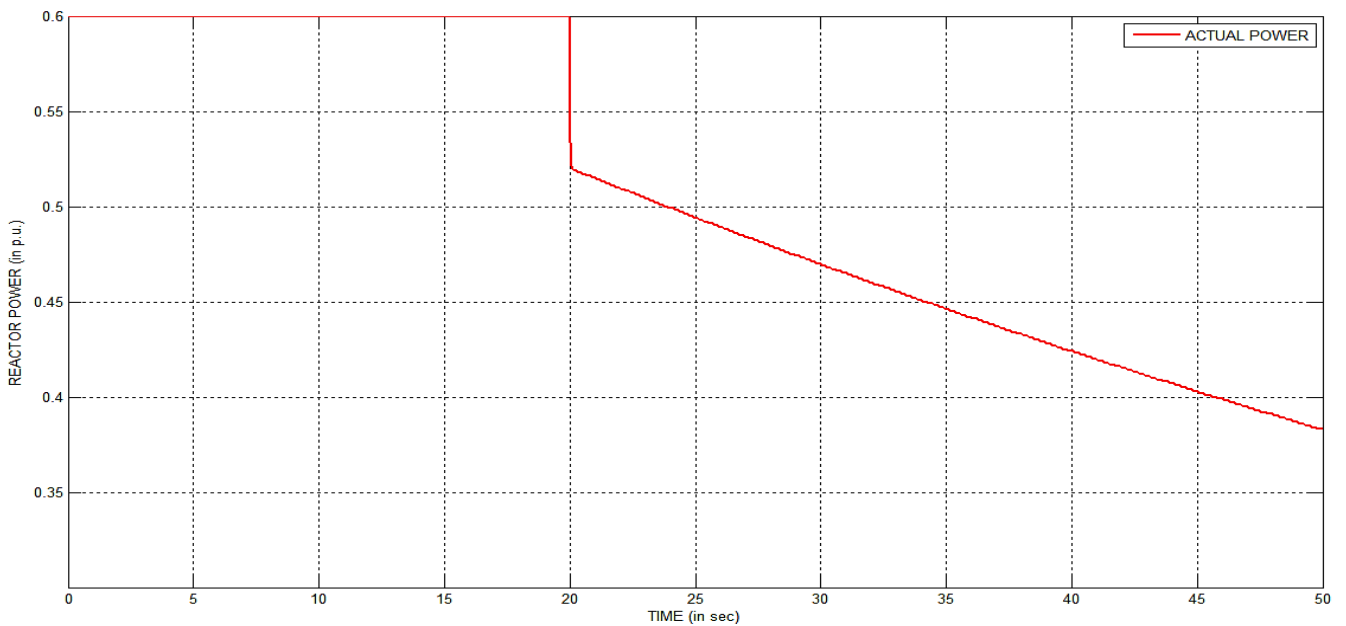


Fig 4.2: Plot of Reactor power of PKPWR for -1mK reactivity insertion

Explanation of the obtained results

It was observed that for both the cases there was a sudden jump or drop in power at the instant of insertion of reactivity and from then onwards the power went on increasing or decreasing without undergoing damping (as shown in Fig. 4.1 and 4.2). This sudden jump or drop is termed as *Prompt jump* or *Prompt drop* and can be explained by approximate solution of the unnormalised Point kinetics equation for single group of delayed neutrons given below.

$$P = P_0 \left[\left\{ \frac{\beta}{\beta - \rho} e^{\frac{\lambda \rho t}{(\beta - \rho)}} \right\} - \left\{ \frac{\rho}{\beta - \rho} e^{-\frac{(\beta - \rho)t}{l}} \right\} \right] \quad (4.1)$$

For case (i), it can be seen that the first term of the equation (4.1), increases while the second term decreases with time after addition of reactivity (1 mK). So, the power after the jump is mainly determined by the first term. It is here that the effect of the delayed neutrons comes into play. They prevent the neutron density from rising up as rapidly as caused by the prompt jump, the reason being that due to delay in release of these neutrons, the average effective neutron lifetime (Λ) increases.

For case (ii), the coefficients of both the terms are positive and the exponents are negative in the equation (4.1). The second term, however, again dies down rapidly owing to the magnitude of the exponent and the first term dominates the power decline. The presence of the delayed neutrons in this case also helps by slowing down the rate of power decline. However, in both cases, due to the lack of any control mechanism the power goes on either rising or falling indefinitely.

4.1.1.2. Ex-PKPWR SYSTEM BEHAVIOUR

The Ex-PKPWR model showed similar results to the single group case upon simulation.

- A. This system was also found to be state controllable and observable. The difference with simple PKPWR case being that it happens to have 7 Eigen values pertaining to each of the seven state variables, which are as follows.

$$eigen(A) = -31.3417, -2.8795, -0.9858, -0.1940, -0.0681, 0.0000, -0.0143$$

It immediately becomes clear from these values that the Eigen spectrum behaviour exactly matches that explained by the equation (3.31) and the system is marginally stable.

- B. Assuming the same initial conditions and steady state values as given under section 3.2.1.4, the two given cases investigated for PKPWR were also considered for this model. Upon comparison with the plots obtained for the previous model, it was seen that the nature of the behaviour hasn't changed. However, the magnitude of the response did change in the 6 group case for the same allotted time interval of 50 seconds and step size of 0.01 second as shown in the table below.

Table 4.1: Comparison of results for PKPWR and Ex- PKPWR models

MODEL NAME	1mK reactivity insertion		-1mK reactivity insertion	
	Steady Power	End Power	Steady Power	End Power
PKPWR	60% F.P.	107.6% F.P.	60% F.P.	38.3% F.P.
Ex- PKPWR	60% F.P.	146.7% F.P.	60% F.P.	33.2% F.P.

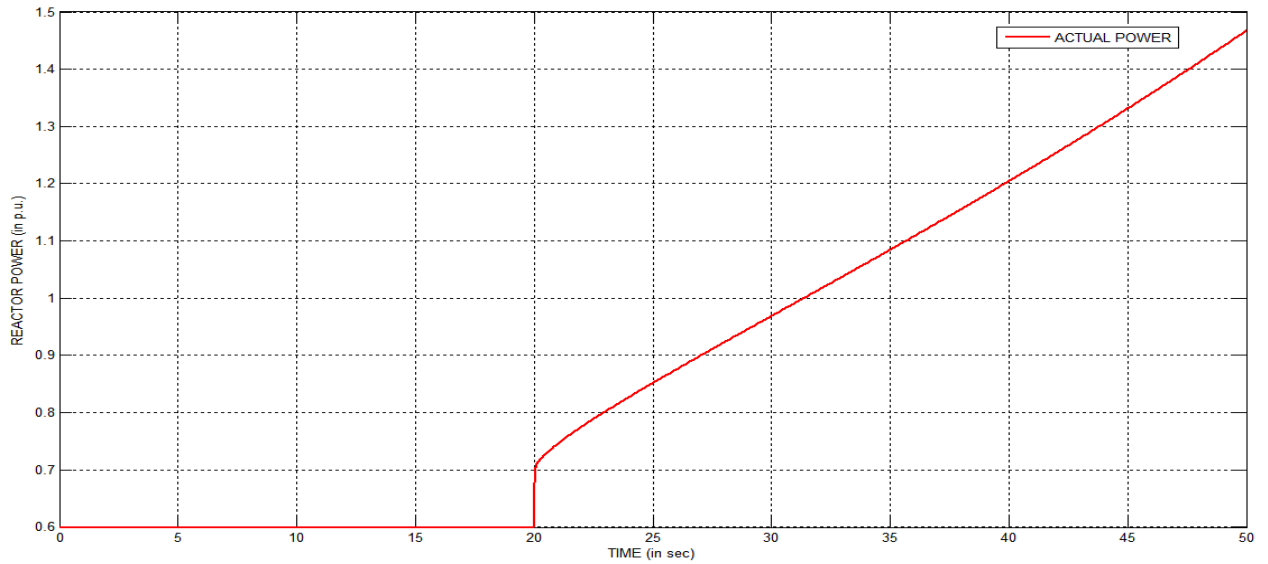


Fig 4.3: Plot of Reactor power of Ex-PKPWR model for 1mK reactivity insertion

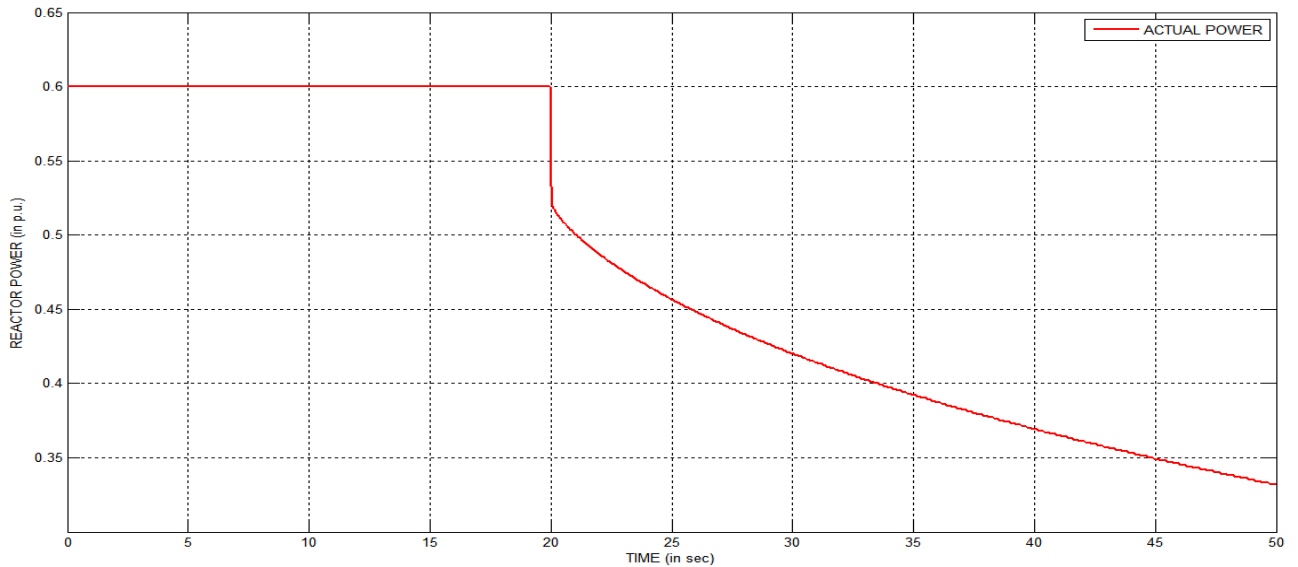


Fig 4.4: Plot of Reactor power of Ex-PKPWR model for -1mK reactivity insertion

4.1.2. *TNPWR ANALYSIS*

This model described the behaviour of the entire primary coolant side. It was controlled only by internal thermal feedback loops for fuel temperature and coolant temperature, without use of the RRS. It was considered that the state space model of the system provides the Power (P_r), fuel temperature (T_f), Average Coolant temperature (T_c), Coolant inlet temperature (T_{cin}) and Coolant outlet temperature (T_{cout}) as outputs and the only input to the system was the feed flow (G_{feed}) as given by the equation (3.53). The feed flow was modelled based on the equations (3.65) and (3.66). It was expected that the thermal feedback would stabilize the system which was not so in case of the PKPWR.

To validate and analyse the behaviour of this model, various scenarios were considered whereby the Reactor deviates from its equilibrium state. The major scenarios that were considered are given below.

- (i) Steady state Operation of TNPWR to understand its equilibrium behaviour.
- (ii) Response of the system to Ramp Changes in demand power.
- (iii) Response of the system to Step Changes in demand power.
- (iv) System response to introduction of unwanted disturbances like noise due to incorrect sensor data or erroneous reactivity variation due to reactivity devices.

4.1.2.1. CASE I: STEADY STATE OPERATION OF TNPWR

During modeling, it was assumed that the reactor is in the power operating regime, i.e. between 20% F.P. and 100%F.P. In this simulation, the system was considered to be operating at a steady state of 60% F.P. ($P_{r0} = 0.6$). The input due to the change in feed flow (G_{feed}) was zero in this case, i.e. the feed flow remained at the same level as the demand power. The simulation was performed for duration of 200 seconds with a step size of 0.01 second. Also, the steady state conditions were specified as given by the equations (3.58)-(3.64) under the section 3.2.2.2. Again, the system behaviour was analysed under two heads, i.e. the behaviour of the Eigen values, controllability and observability and the next being the behaviour of the output variables mentioned above.

- A. The system so formed was found to be state controllable and observable. The Eigen values of this system were found to be,

$$\begin{aligned} eigen(A) = & -30.8931; -4.3056 + 0.9910i; -4.3056 - 0.9910i; \\ & -0.8166 + 1.4896i; -0.8166 - 1.4896i; -0.2147 + 0.2737i; \\ & -0.2147 - 0.2737i; -0.0945, \end{aligned}$$

In the Eigen spectrum as shown above, all the values have negative real parts with the 1st one being on the extreme left of the s-plane and the others forming a cluster near the origin. Also, 3 pairs of complex conjugate Eigen values are present. From such a distribution, it was concluded that system was stable; however the degree of stability was rather low. In order, to further stabilize the system and make the eigen values more negative, the *State feedback model* of the RRS-PWR based on TNPWR similar to the design in section 3.3.2.3 might be used. It should be noted that the eigen values undergo slight variations in values depending on the specified initial power level, but their nature remains the same.

- B. The responses for Power ($P_{r0} = 0.6$) and the corresponding fuel temperature ($T_f = 318.44$), Average Coolant temperature ($T_c = 310.10$ °C), Coolant inlet temperature ($T_{cin} = 309$ °C) and Coolant outlet temperature ($T_{cout} = 311.15$ °C) at this power level showed no variation in their values.

The reactivity plot showed that $\rho = 0$ for the entire period, which reflected that the system was critical and so, there were no inherent deviations in the system when no input was administered. Hence, it is safe to say that the model behaviour was validated for this scenario.

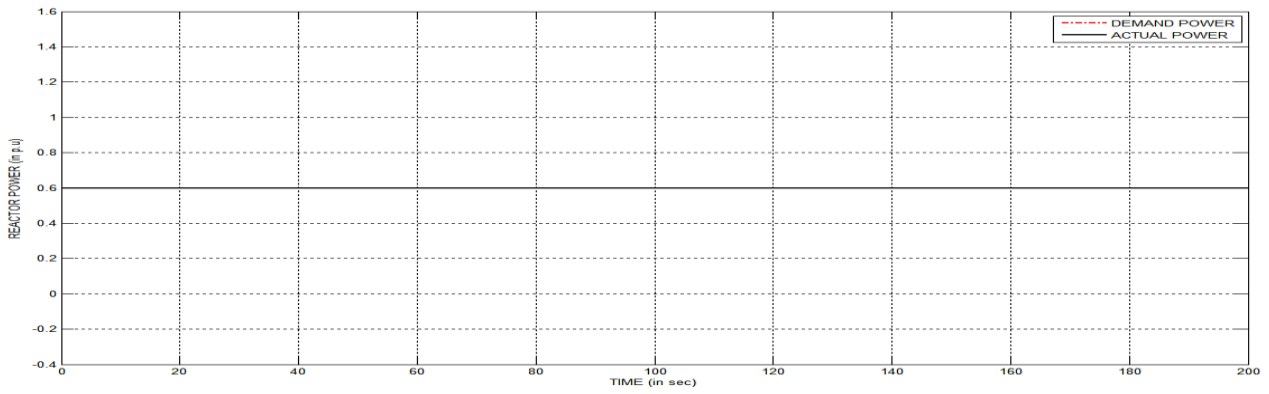


Fig 4.5: Plot of Reactor power of TNPWR model for steady state scenario

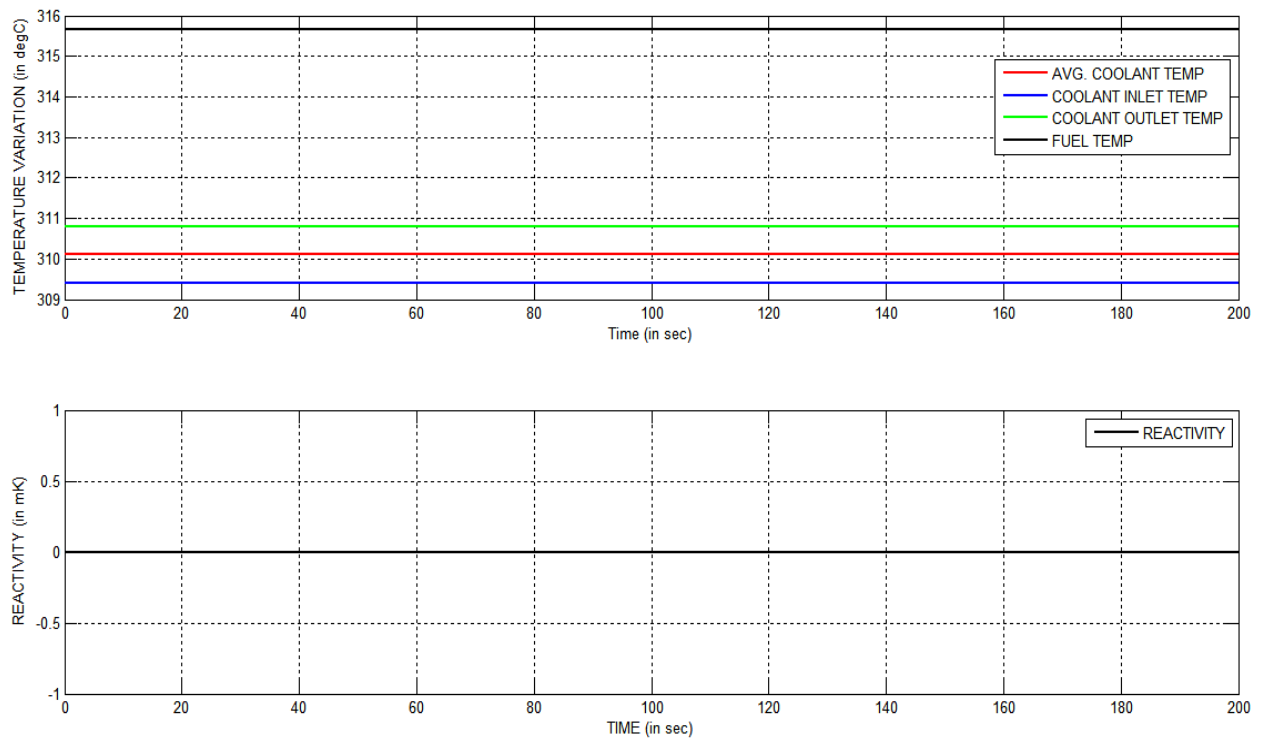


Fig 4.6: Plot of Temperatures and Reactivity of TNPWR model for steady state scenario

4.1.2.2. CASE II: RAMP CHANGES IN DEMAND POWER

In this simulation, the system was again considered to be operating at a steady state of 60% F.P ($P_{r0} = 0.6$) and the ramp variations in demand power take place at ± 1.0 % F.P/sec (ramp rate). These variations were administered to the system at 20 seconds. The simulation was performed for duration of 200 seconds with a step size of 0.01 second. Also, the steady state conditions for the system variables were specified by the equations (3.58)-(3.64) as given under the section 3.2.2.2. Also, the variation of demand power was achieved via the variation of feed flow characteristics as defined by equations (3.65) and (3.66). In each of the cases simulated, the actual power was presented by solid black lines, while dashed red lines showed the change in the demand power. Here, the model was simulated under four different load changing scenarios of ramped nature, mentioned beneath.

- (i) Large rise in demand power from Steady state (60% F.P.) to 100% F.P.
- (ii) Large decline in demand power from Steady state (60% F.P.) to 20% F.P.
- (iii) Small rise in demand power from Steady state (60% F.P.) to 70% F.P.
- (iv) Small decline in demand power from Steady state (60% F.P.) to 50% F.P.

i) Rise in demand power from 60% F.P. to 100% F.P

It is known from section 2.4.3.1 that PWR with its internal thermal feedbacks acts a load follower even without an external RRS. This fact was proved correct by the Power plot of our model as shown in **Fig.4.7**.

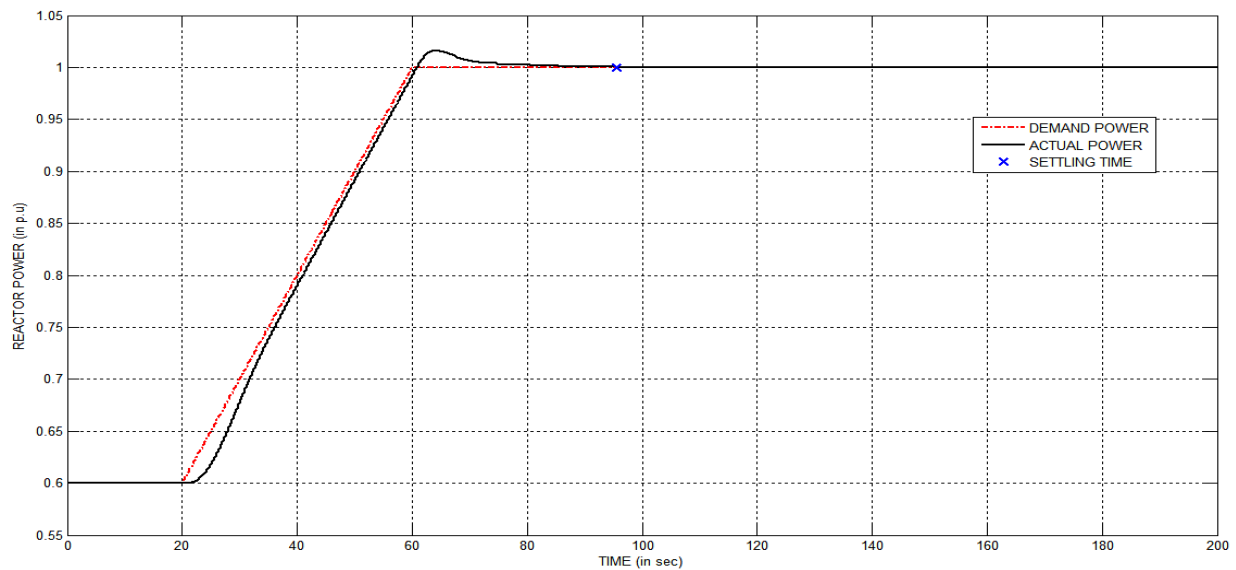


Fig 4.7: Plot of Reactor Power of TNPWR model for ramp rise in power up to 100%F.P.

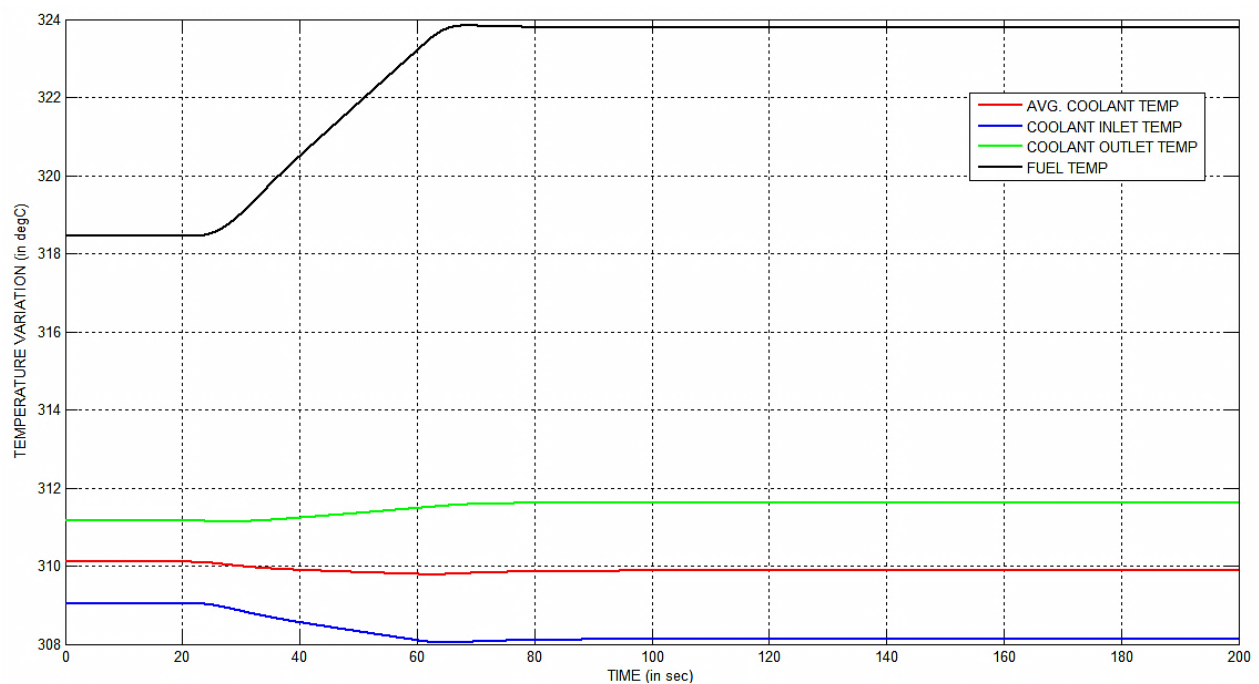


Fig 4.8: Plot of Temperature variation of TNPWR model for ramp rise in power up to 100%F.P.

Also, it is known that, during an increase in load or demand power, the coolant temperature decreases and within that duration the reactor remains supercritical before settling down to the desired power level. The plots for Coolant temperature, Fuel temperature and Reactivity also backed up this fact as given in **Fig.4.8** and **Fig.4.9**.

In the power plot (**Fig.4.7**), as expected the Reactor power followed the demand power. However, closer observation revealed that the reactor power didn't exactly match the demand power until the settling time.

- In the initial stages of the operation, there was a substantial gap between them of about 2%F.P. which can be attributed to the delayed response of the thermal feedbacks due to coolant temperature change as seen in **Fig.4.8**.
- In the transition stage of the power, there existed a continuous steady state error between them owing to the lack of the RRS.
- In the final stages, it was seen that the reactor power overshoot the demand power upto about 101.7% F.P. Here, the thermal feedback came into play and the reactivity decreased as seen in later half of **Fig.4.9**, ultimately bringing the reactor power level down to the demand power at 95.61 sec.

From **Fig.4.8**, it was observed that, the fuel temperature (T_f) rose from its steady state value of 318.44 °C to 323.78 °C during the transition period. The Average coolant temperature (T_c) decreased slowly from 310.10 °C to 309.88 °C. The coolant inlet temperature (T_{cin}) decreased from 309°C to 308.12 °C. The coolant outlet temperature (T_{cout}) increased from 311.15°C to 311.63 °C. Apart from these the **Fig.4.9**, showed that the maximum reactivity (ρ) during the operation went up to around 0.88 mK, from where it decreased steadily to zero at around 120 seconds, thus making the reactor critical again.

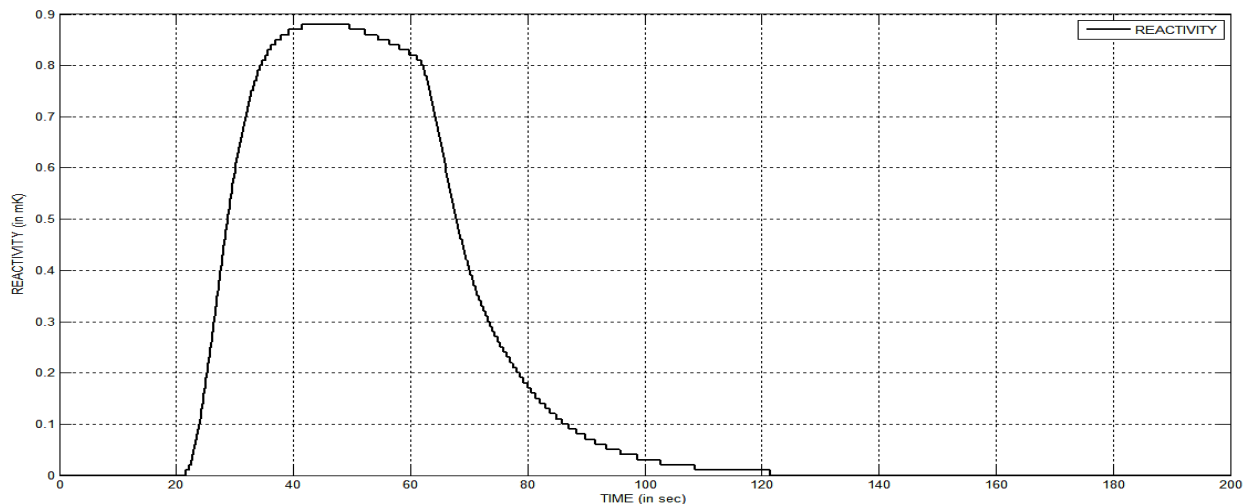


Fig 4.9: Plot of Reactivity variation of TNPWR model for ramp rise in power up to 100%F.P.

ii) Decline in demand power from 60% F.P. to 20% F.P

Similar observations were made, when the demand power declined. However, the nature of the plots were reversed. Here, the coolant temperature increased and within that duration the reactor remained subcritical before settling down to the desired power level.

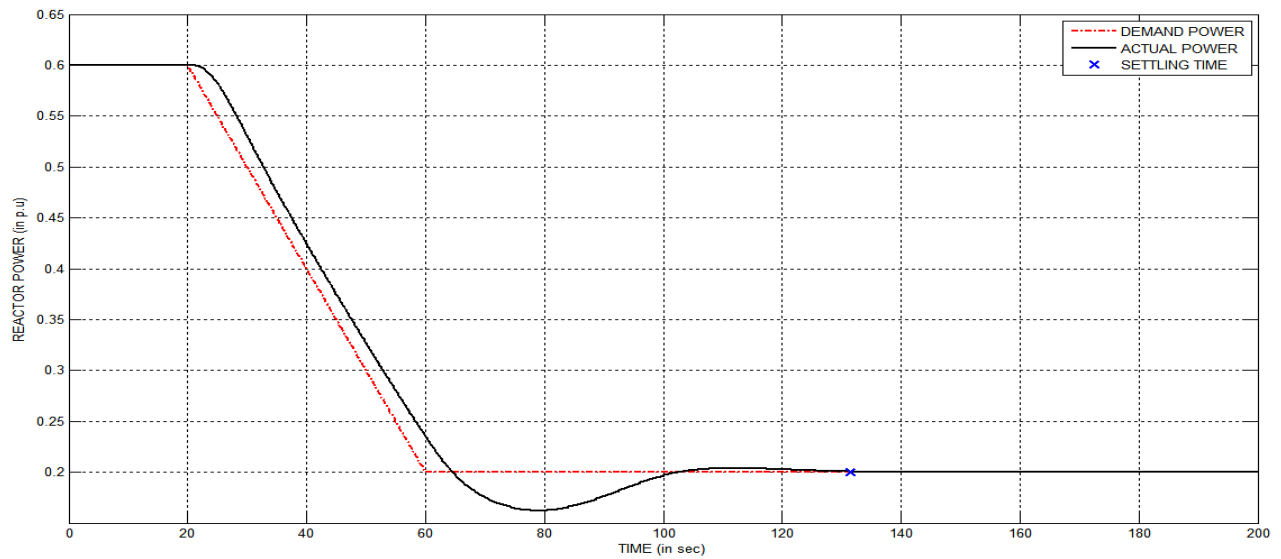


Fig 4.10: Plot of Reactor Power of TNPWR model for ramp decline in power up to 20%F.P.

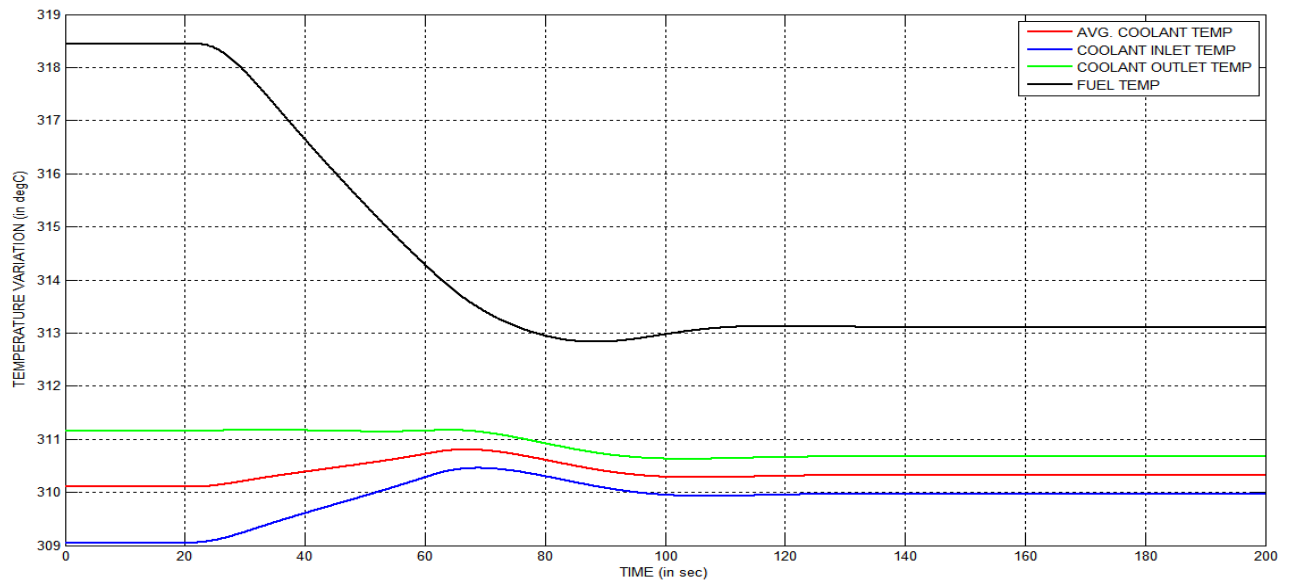


Fig 4.11: Plot of Temperature variation of TNPWR model for ramp decline in power up to 20%F.P.

In this case also, although the reactor power followed the demand power eventually, there were certain initial deviations. From the **Fig.4.10**, the initial gap of 2% F.P. was observed for this case also. There was a substantial continuous steady state error during the transition of power. Also, the reactor power undershot the demand power to about 16.2% F.P. before settling down owing to the internal thermal feedbacks at 131.47 seconds.

In this case, from **Fig.4.11**, it was observed that, the fuel temperature (T_f) first decreased from its steady state value of 318.44 °C to around 312.83 °C and then increased slowly to stabilize itself at 313.10 °C. The Average coolant temperature (T_c), Coolant inlet temperature (T_{cin}) and Coolant outlet temperature (T_{cout}) showed opposite trends by first increasing and then reducing rapidly to settle down at a steady value of 310.32 °C, 309.97 °C and 310.67 °C, respectively which were really close to their initial values.

The reactivity plot in **Fig.4.12**, also showed oscillatory behaviour. First, it grew negative to around -3.64 mK , then increased to 0.3 mK, from where it again became slightly negative before completely settling down at zero around 150 seconds into the transition.

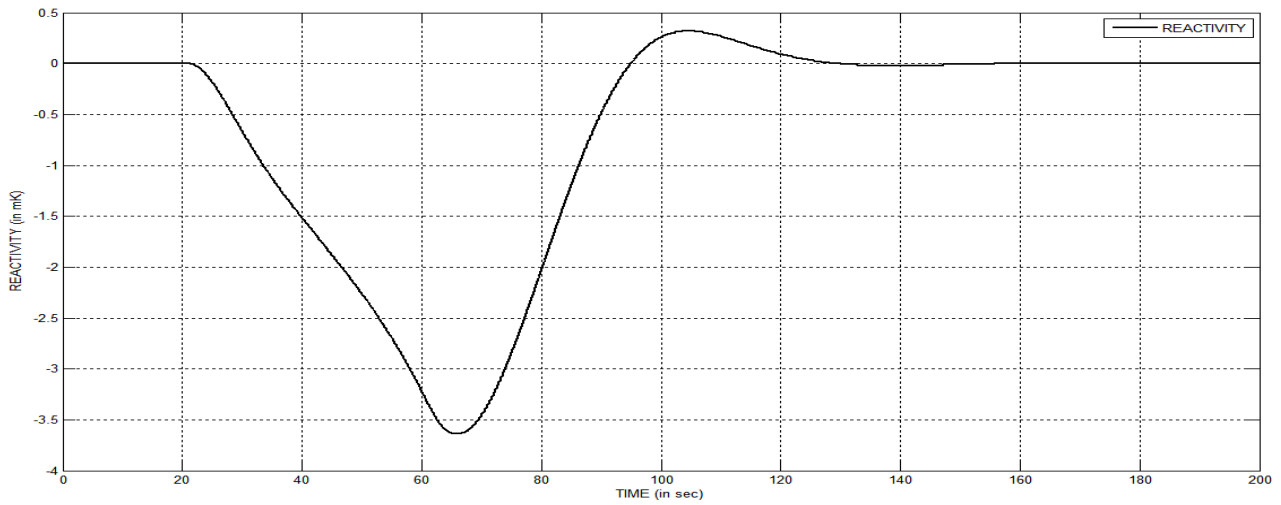


Fig 4.12: Plot of Reactivity variation of TNPWR model for ramp decline in power up to 20%F.P.

iii) Rise in demand power from 60% F.P. to 70% F.P

This case was similar to the case (A). However, upon comparison a few interesting facts came to be noticed. First point was that in the power plot, the transition gap had increased substantially and during the end phase the overshoot magnitude went upto 71.7%F.P. which incidentally was almost the same amount of overshoot as in case (A) being around 1.7% F.P.

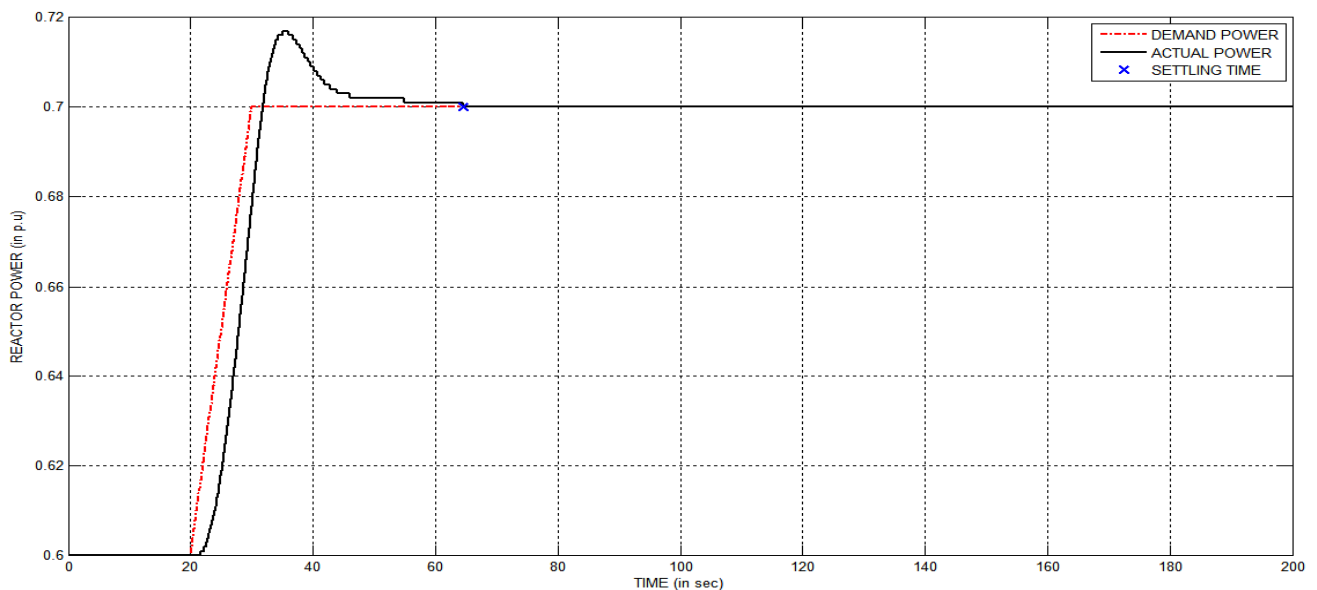


Fig 4.13: Plot of Reactor power of TNPWR model for ramp rise in power up to 70%F.P.

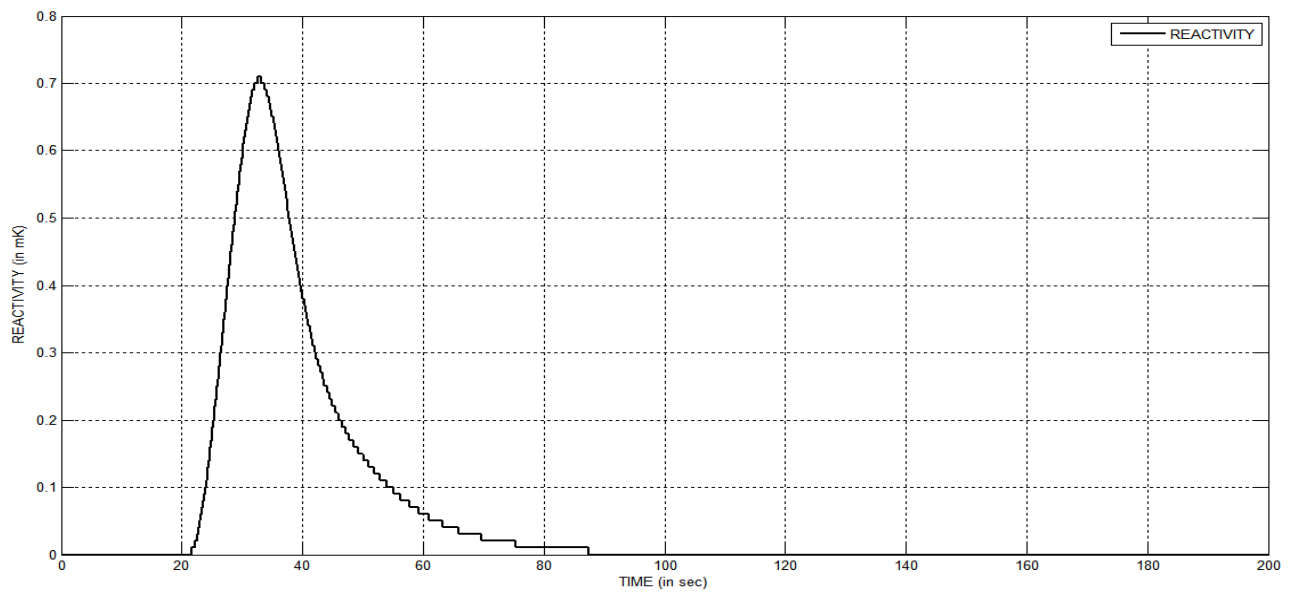


Fig 4.14: Plot of Reactivity variation of TNPWR model for ramp rise in power up to 70%F.P.

Secondly, the nature of the reactivity plot remained the same, however, the peak became much more pronounced (0.71 mK) as compared to the case (A) & settled to zero around 90 seconds much less than for case (A). The fuel and coolant temperatures suffered very little variation and showed similar nature as was observed in case (A).

iv) Decline in demand power from 60% F.P. to 50% F.P

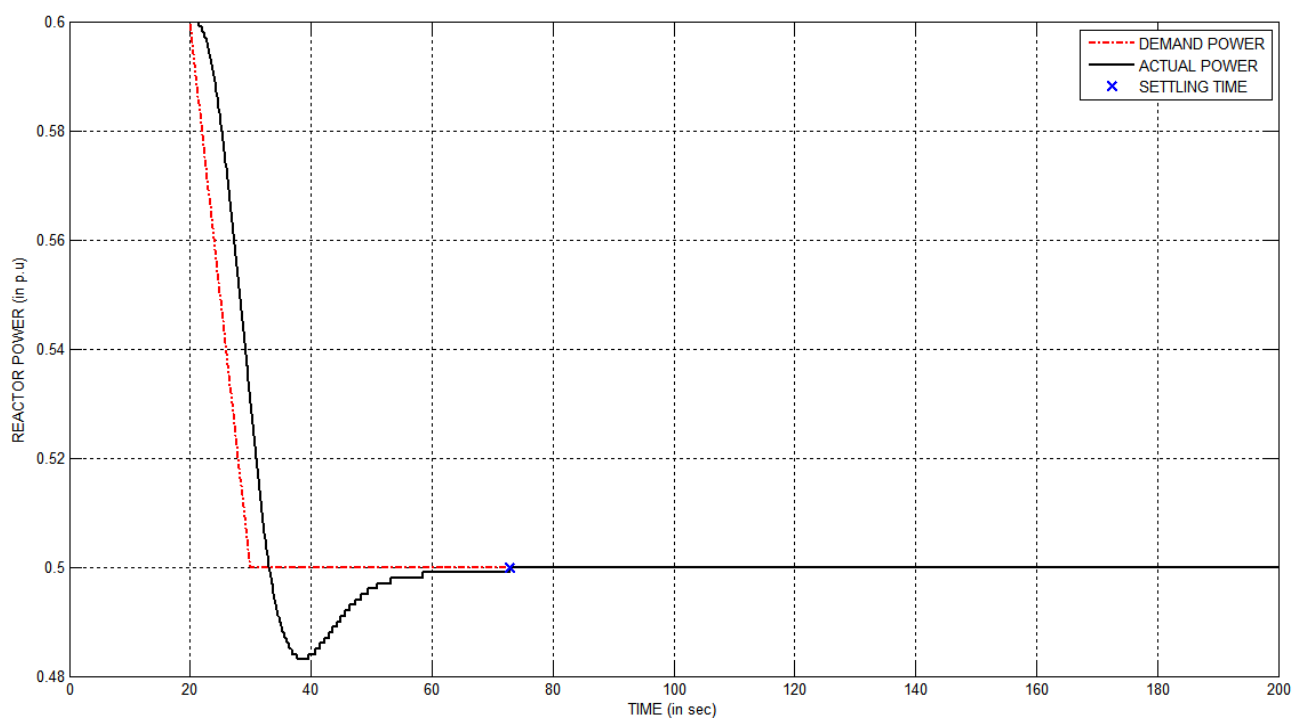


Fig 4.15: Plot of Reactor power of TNPWR model for ramp decline in power up to 50%F.P.

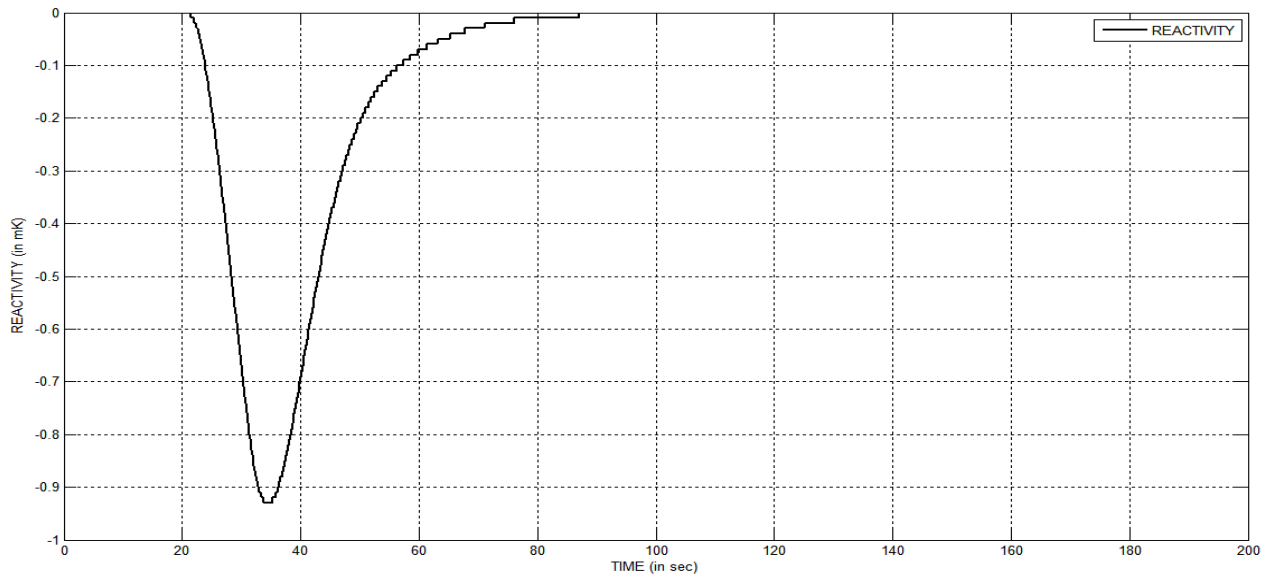


Fig 4.16: Plot of Reactivity variation of TNPWR model for ramp decline in power up to 50%F.P.

In this case, similar to that of case (B), it was observed that, the power plot had similar nature to that in case (B). The difference was the lesser magnitude of the undershoot (48.3% F.P.- exactly 1.7% F.P. less than required value) in this case and the decrease in the settling time (72.84 seconds). Another curious result came from the reactivity plot as shown by the **Fig. 4.16**. Here, the oscillations that were present in **Fig. 4.12** were not observed and the reactivity went to a maximum value of -0.93 mK and then steadily approached zero (criticality), settling down at around 90 seconds.

v) Comparison of the settling times for the above cases

The table below shows a comparison of the settling times for the above discussed scenarios. It is interesting to note that in a power rise scenario the thermal feedbacks help settle the system within 60-100 sec for any sort of change. However for the power decline, this range lies within 70-140 sec.

Table 4.2: Comparison of settling time for various ramp scenarios of TNPWR

TIME (in seconds)	POWER RISE		POWER DECLINE	
	60% TO 100% F.P	60% TO 70% F.P	60% TO 20% F.P	60% TO 50% F.P
Initiation	20	20	20	20
Settling	95.61	64.50	131.47	72.84

4.1.2.3. CASE III: STEP CHANGES IN DEMAND POWER

Similar to the ramp changing scenario, step changes in demand power were analysed based on the same yardstick. In this simulation, the system was considered to be operating at a steady state of 60% F.P ($P_{r0} = 0.6$) and the step change was administered to the system at 20 seconds. The simulation was performed for duration of 200 seconds with a step size of 0.01

second. Also, the steady state conditions for the system variables and the feed flow characteristics remained the same. The actual power was presented by solid black lines, while dashed red lines showed the change in demand power.

Here, the model was simulated under two different load changing scenarios of step nature, mentioned beneath.

- (i) Rise in demand power from Steady state (60% F.P.) to 100% F.P.
- (ii) Decline in demand power from Steady state (60% F.P.) to 20% F.P.

i) Rise in demand power from 60% F.P. to 100% F.P

For such an instant change in power, the thermal feedbacks were relied upon to settle the reactor power to the demand power level. The following figures support this case.

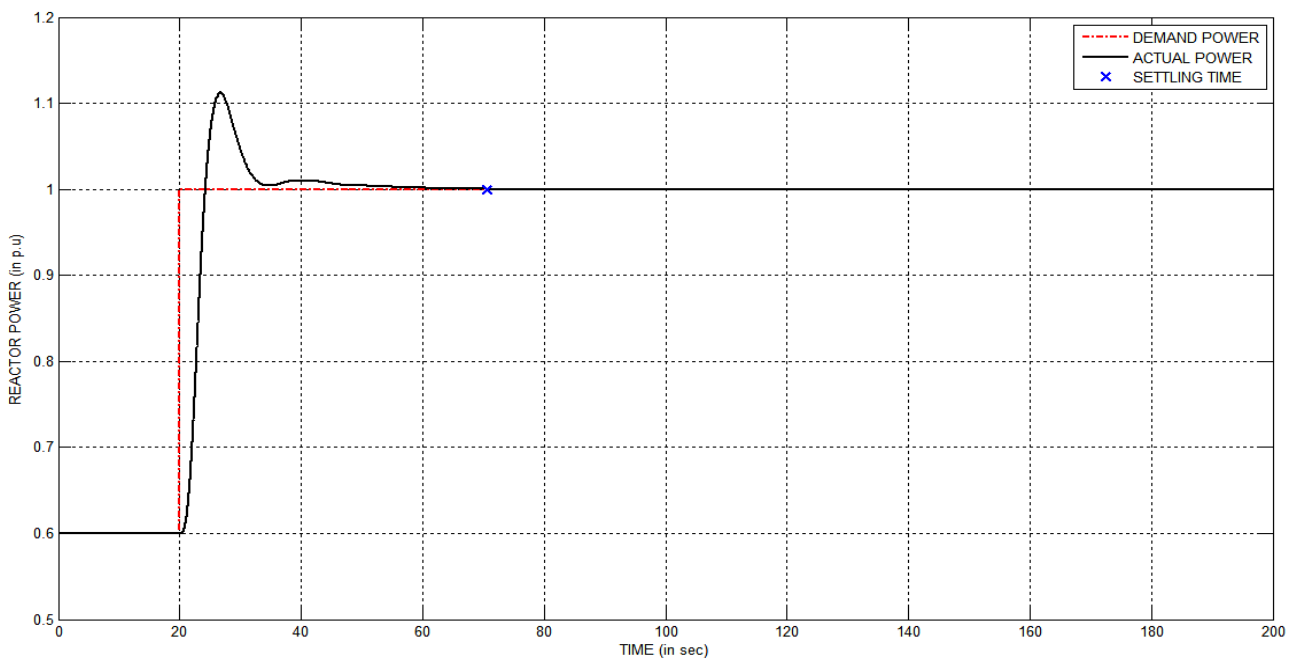


Fig 4.17: Plot of Reactor Power of TNPWR model for step rise in power up to 100%F.P.

From the **Fig. 4.17** showing the Power variation, it was seen that there remained a substantial gap between the demand and reactor powers in the transition phase and at the end the power overshoot up to 111% F.P. before settling down to F.P. at 70.53 seconds, which for the ramp case (A) was about 101.6%F.P. before settling down at 95.61 seconds.

From **Fig. 4.18** the temperature variation showed similar nature to that of the ramp counterpart and the settling values for the fuel temperature (T_f), the Average coolant temperature (T_c), the coolant inlet temperature (T_{cin}) and the coolant outlet temperature (T_{cout}) remained exactly the same as for case (A) of the ramp variations. However, the nature of the reactivity variation approached that for case (C) of the ramp changes and rose up to 2.4 mK before settling down steadily to zero at about 100 seconds as shown in **Fig. 4.18**.

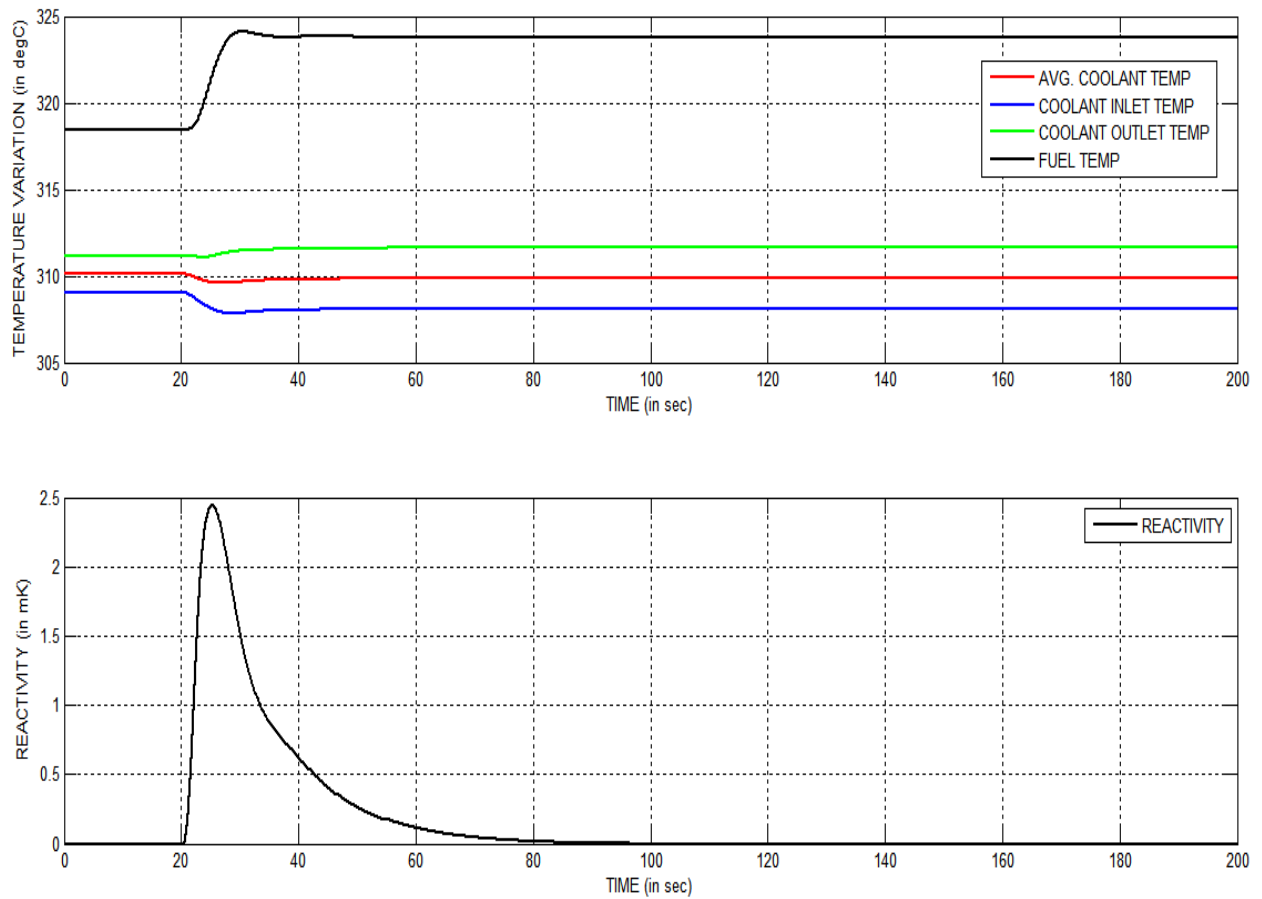


Fig 4.18: Plot of Temperature variation (above) & reactivity variation (below) of TNPWR model for step rise in power up to 100%F.P.

ii) Decline in demand power from 60% F.P. to 20% F.P

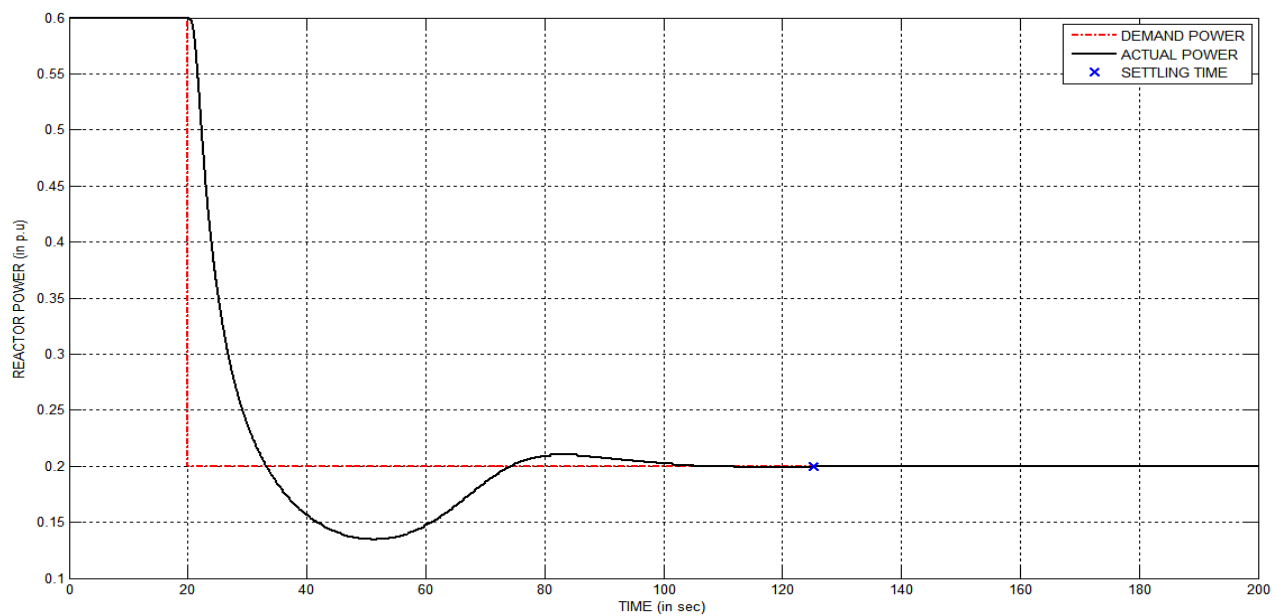


Fig 4.19: Plot of Reactor Power of TNPWR model for step decline in power up to 20%F.P.

For this case, from the Power variation shown in **Fig. 4.19**, it was observed that there was a wide gap between the demand and reactor powers in the transition phase and at the end the power undershot up to 13.4% F.P. before settling down at 125.33 seconds, which for the ramp case (B) was about 16.2%F.P. before settling at 131.47 seconds.

From **Fig. 4.20**, the temperature variation again showed similar nature to that of the ramp counterpart and the settling values for the fuel temperature (T_f), the Average coolant temperature (T_c), the coolant inlet temperature (T_{cin}) and the coolant outlet temperature (T_{cout}) remained exactly the same as for case (B) of the ramp variations. The reactivity variation, on the other hand, showed an oscillatory behaviour starting with a steep decline to about -7.15 mK, almost double the amount as in case (B) of ramp variations, then increasing to 0.72 mK, from where it again became slightly negative before completely settling down at zero around 120 seconds.

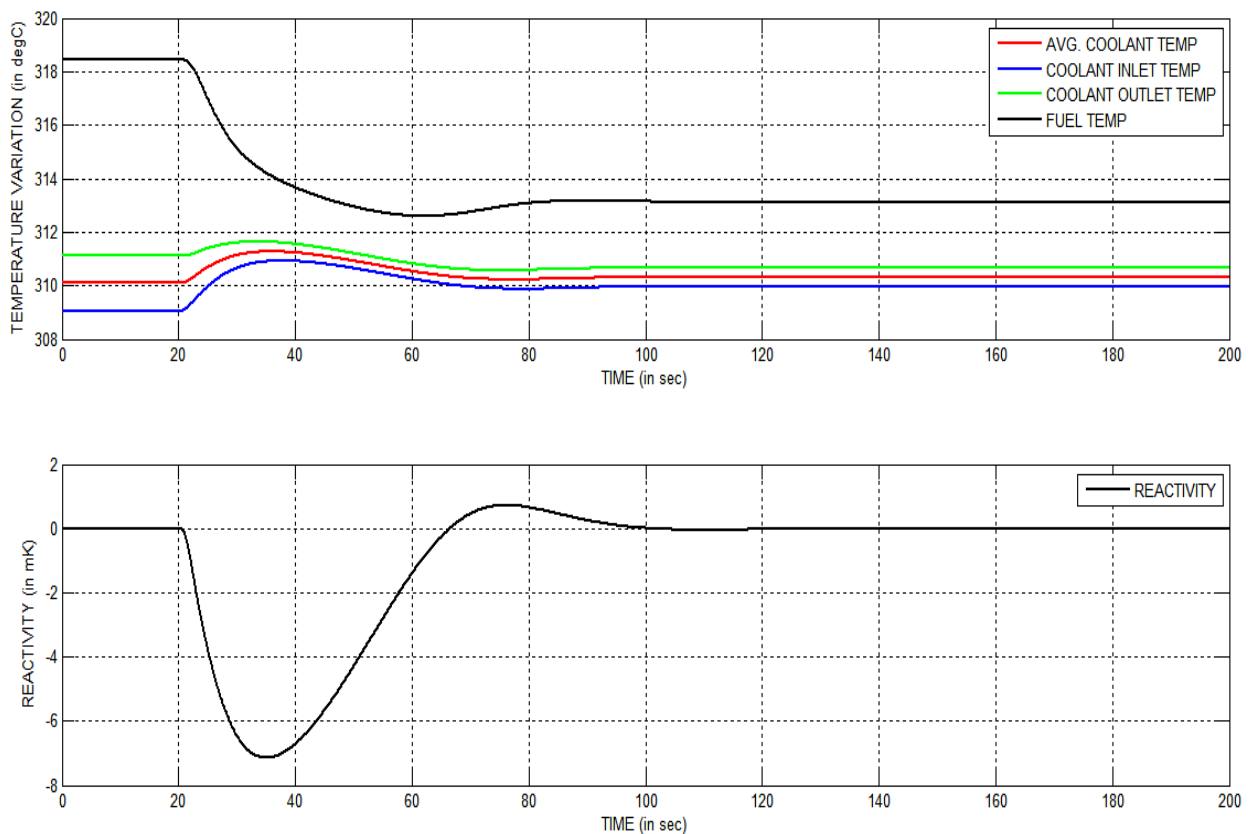


Fig 4.20: Plot of Temperature variation (above) & reactivity variation (below) of TNPWR model for step decline in power up to 20%F.P.

4.1.2.4. CASE IV: SYSTEM RESPONSE TO NOISE

With the system model offering convincing results similar to the established behaviour of PWRs, the model was analysed for any unwanted noise which might trigger positive or negative reactivity insertion. Here, it was assumed that a positive reactivity insertion of 0.5 mK occurs in the system. The choice for such a value was based on the fact that the noise should not be so high as to cause a reactor trip. Here, the basis for the unwanted reactivity insertion was assumed to be the slight upward movement of the CRs caused by the noise. It

was also assumed that after the slight initial movement at a particular instant, the CR became stationary at the new position, thus inserting an instantaneous reactivity. In this simulation, the system was considered to be operating at steady state of 60% F.P ($P_{r0} = 0.6$) and the instantaneous noise was inserted in the system at 20 seconds. Also, the simulation was performed for duration of 200 seconds with a step size of 0.01 second.

Here, three situations were considered, as listed below.

- (i) Instantaneous Noise insertion at steady state power (60%F.P.).
- (ii) Instantaneous Noise insertion at initial stage of ramp rise in demand power from Steady state (60% F.P.) to 100% F.P.
- (iii) Instantaneous Noise insertion at initial stage of step rise in demand power from Steady state (60% F.P.) to 100% F.P.

i) Noise insertion at steady state power of 60% F.P.

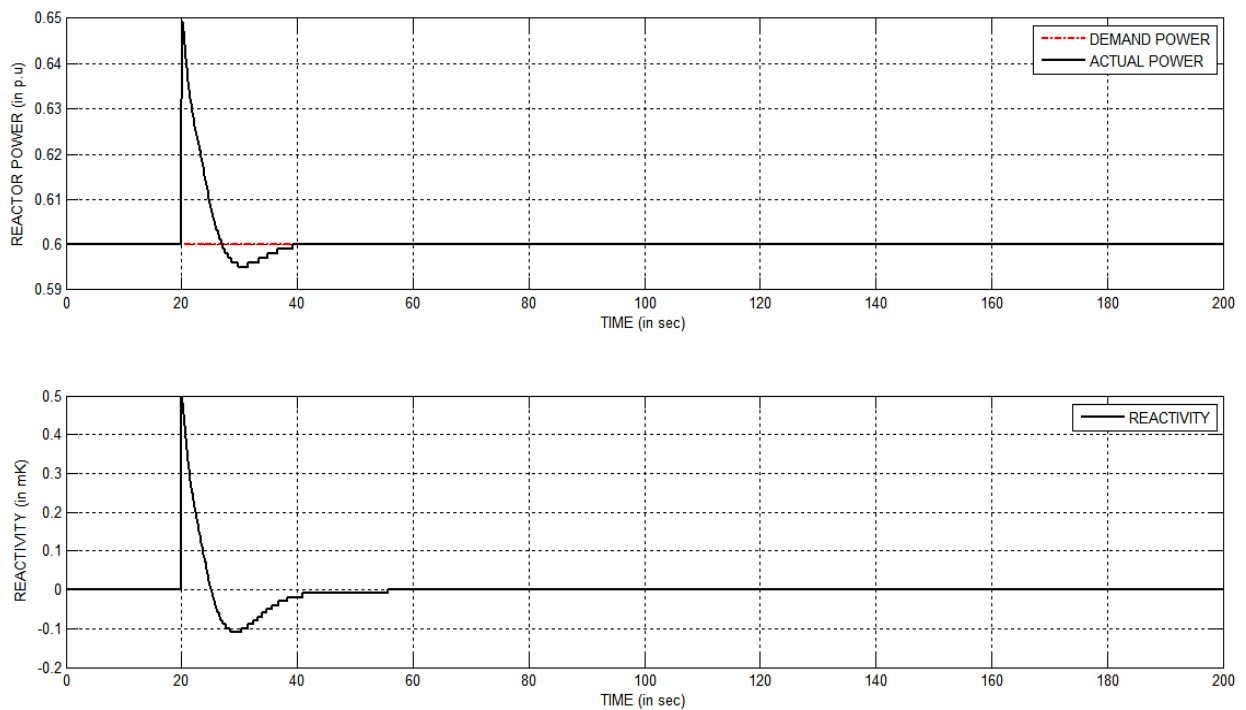


Fig 4.21: Plot of Power variation (above) & reactivity variation (below) of TNPWR model for instantaneous noise insertion of 0.5mK at steady state

From the Power variation in **Fig. 4.21**, it was seen that as the positive reactivity of 0.5 mK was inserted into the system at 20 seconds, the power jumped close to 5% more than the steady state power. But owing to the presence of the thermal feedbacks, the reactor was able to stabilize itself, despite such disturbances and finally it settled down to a steady state level within 40 sec. The main reason behind such behaviour was the slight increase in the Average coolant temperature (T_c) from 310.10°C to 310.167°C thus inducing an equivalent negative reactivity which compensated the noise induced positive reactivity. The reactivity variation in the second figure confirmed this notion whereby after the initial reactivity rise of 0.5 mK, the reactivity dropped to -0.11 mK before settling to zero around 57 seconds.

ii) Noise insertion during ramp power rise

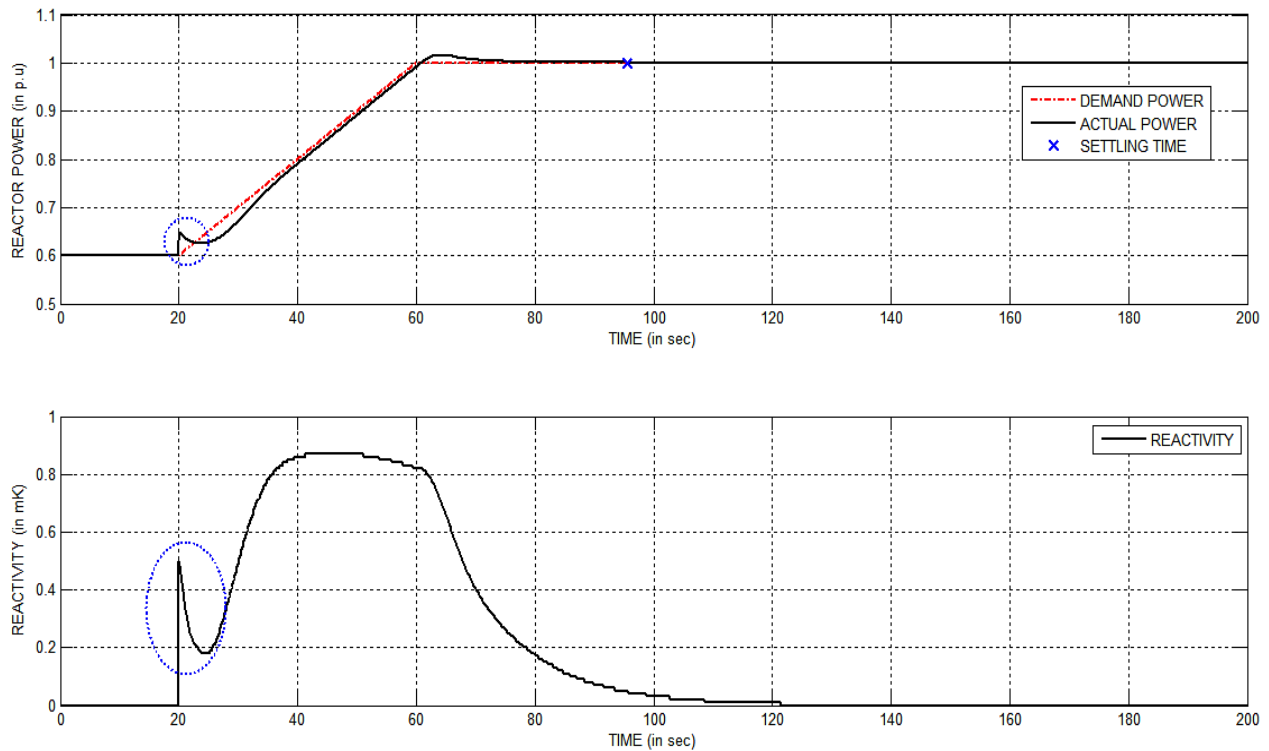


Fig 4.22: Plot of Power variation (above) & reactivity variation (below) of TNPWR model for instantaneous noise insertion of 0.5mK during ramp power rise

In this case, as the power was ramped up from steady state of 60% F.P. to 100% F.P., it was observed that upon insertion of the instantaneous reactivity of 0.5 mK a sudden distortion occurred at the insertion instant of 20 seconds as indicated by the blue circle in the power variation curve in the above figure. However, it didn't distort the system behaviour much after it died down. Also, the reactivity plot confirmed that at the instant of addition, the reactivity had reached 0.5 mK after which owing to the internal feedbacks it came down below 0.2 mK within a few seconds and then started to follow the curve as it did in case of ramp increase in case (A) shown in **Fig. 4.7**. It went up to 0.87 mK and then returned to zero around 120 seconds. In both these plots, the blue circles represent the region of distortion due to the noise induced reactivity.

iii) Noise insertion during step power rise

Similar to the ramp case, as the power was stepped up from steady state of 60% F.P. to 100% F.P., a sudden initial distortion was noticed at the insertion of the instantaneous reactivity. However, its magnitude was extremely low, much less than that for the ramp case. Also, in the reactivity plot the variation showed up just slightly in the initial stages. Apart from these distortions, both these plots exactly followed the behaviour of the case (A) for the step change in power case. In both these plots, the blue circles represent the region of distortion due to the noise induced reactivity. Thus, it was inferred that the model held up nicely for disturbances inducing positive reactivity. It was also found that it held good for disturbances causing insertion of negative reactivity.

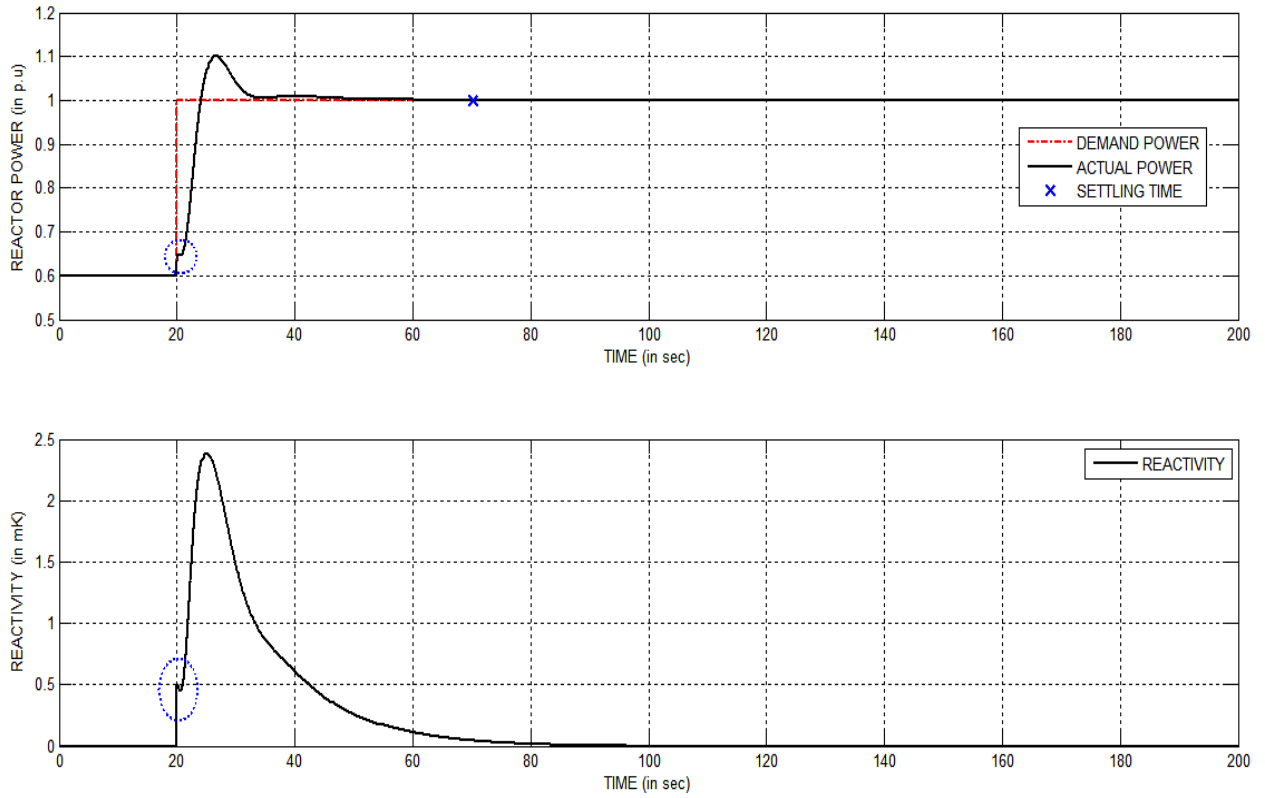


Fig 4.23: Plot of Power variation (above) & reactivity variation (below) of TNPWR model for instantaneous noise insertion of 0.5mK during step power rise

4.1.2.5. TNPWR RESPONSE WITH CONSTANT HOT LEG TEMPERATURE (T_{cout}) PROGRAM

As mentioned in section 3.1.3, there are two kinds of operational programs that are applicable to this model. First was the constant average coolant temperature program, which happens to be the natural program of the PWR. The other kind of program was the Constant hot leg temperature (T_{cout}) program. As mentioned earlier, without the use of external control mechanisms to counter the excess reactivity due to the change of the average coolant temperature (T_c), the power goes on increasing or decreasing depending upon the scenario. Since, no external control mechanisms were used, it was expected that the system power would rise to a level other than the demand power before settling down.

Here, two scenarios were investigated, as listed below to understand the system behaviour in such a case. The plots for them are shown below.

- (i) Ramp rise in demand power from Steady state (60% F.P.) to 100% F.P.
- (ii) Step rise in demand power from Steady state (60% F.P.) to 100% F.P.

The system was again considered to be operating at steady state of 60% F.P ($P_{r0} = 0.6$) and the ramp variations in demand power took place at ± 1.0 % F.P/second (ramp rate), which were administered to the system at 20 seconds. The simulation was performed for duration of 200 seconds with a step size of 0.01 second. Also, the steady state conditions for the system variables and the feed flow characteristics remained the same. The actual power was presented by solid black lines, while dashed red lines showed the change in the demand power.

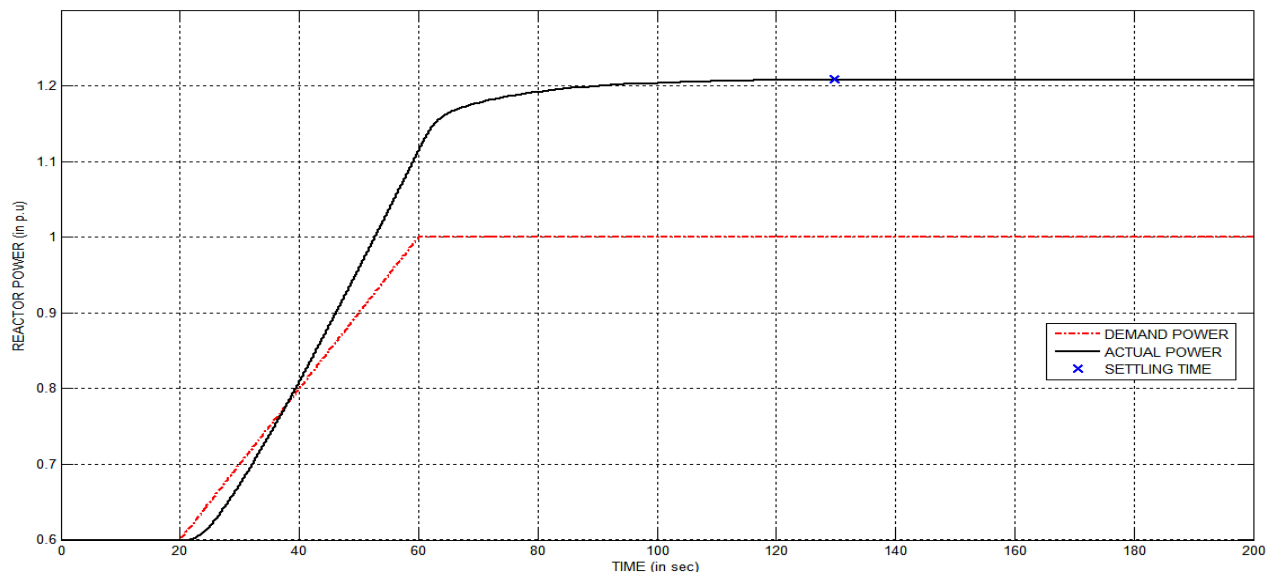


Fig 4.24: Plot of Reactor Power of TNPWR model with Constant Hot leg temperature program for ramp rise in power up to 100%F.P.

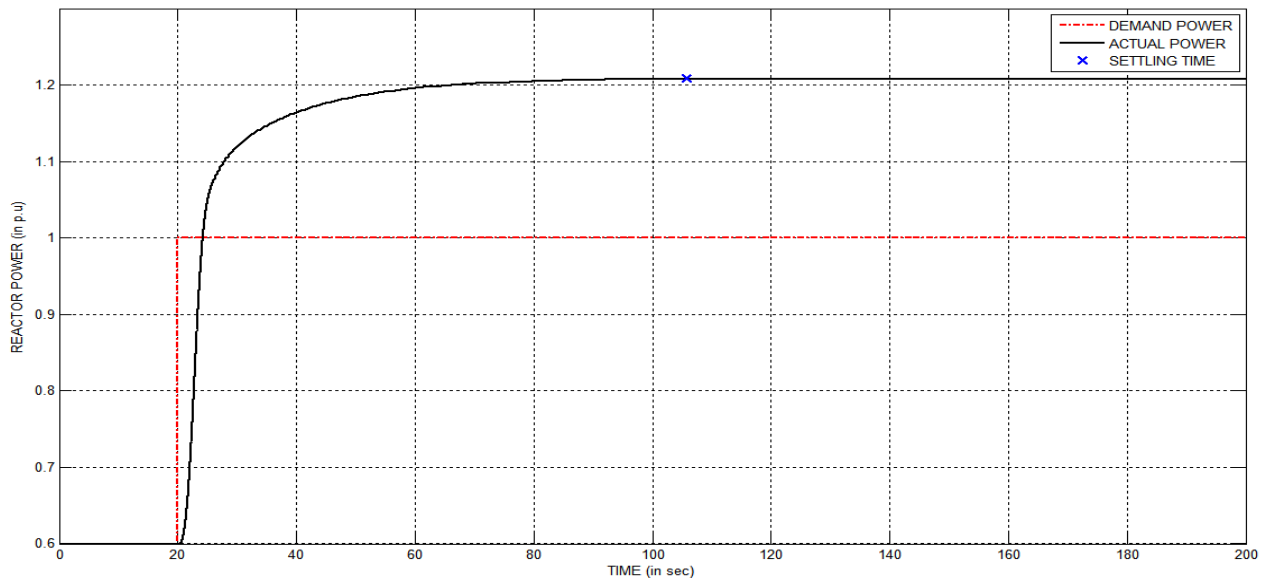


Fig 4.25: Plot of Reactor Power of TNPWR model with Constant Hot leg temperature program for step rise in power up to 100%F.P.

As expected, in both these responses the power overshoot the demand power by almost 20%F.P. and then eventually settled down at 120.8%F.P. The settling time for the ramp response curve was 129.86 seconds whereas for the step response curve was 105.71 seconds. Hence, it was conformed that without an external control mechanism this system program would not turn out to be fruitful. However, the reason that the power settled down eventually could be attributed to the fact that the coolant temperature variation ceased ultimately causing the system to settle down.

Similar results were obtained, just in the reverse order for the decline in power scenario with both ramp and step variation.

4.1.2.6. RO- TNPWR SYSTEM BEHAVIOUR

The reduced dimension TNPWR (RO-TNPWR) is a model using 8 state variables, not using the Coolant outlet temperature or hot leg temperature (T_{cout}) equation. Similar to the other models, the behaviour of this system was also analysed in two parts as below.

- A. In this analysis, the system was considered to be operating at a steady state of 60% F.P. ($P_{r0} = 0.6$). The system so formed was found to be state controllable and observable. The Eigen values of this system are given below.

$$\begin{aligned} eigen(A) = & -30.8931; -2.9214; -1.2215 + 1.7656i; -1.2215 - 1.7656i; \\ & -0.5899; 0.2103; -0.0243; -0.3333 \end{aligned}$$

It was seen that the Eigen spectrum substantially changed in this case as compared to the TNPWR Eigen spectrum, where all the Eigen values had negative real parts and there were three pairs of complex conjugate values. However, in this case apart from the sixth eigen value all others had negative real parts and there was only one pair of complex conjugate eigen values. However, owing to the sixth eigen value being positive, the system was not completely stable. As mentioned earlier, these values are dependent on the specified initial power level and they undergo slight variations in values with the power level specified. But their nature remains the same.

- B. Upon simulation, it was observed that the responses offered by this model were consistent with those obtained from the TNPWR model apart from the slight variations in the overshoot or undershoot magnitude. Also, the reactivity variation and temperature variations remained the same for both step and ramp changes in power level. Hence, they have not been highlighted separately in this study.

4.1.3. TNXPWR ANALYSIS

This model described the behaviour of the entire primary coolant side and in addition to the thermal feedback, also incorporated the poisoning feedback due to xenon. This formed a full scale model of a bare reactor with the entire internal feedback loop. It was considered that the state space model of the system provides the Power (P_r), fuel temperature (T_f), Average Coolant temperature (T_c), Coolant inlet temperature (T_{cin}), Coolant outlet temperature (T_{cout}) and Iodine (I) and Xenon (X) concentrations as outputs and the only input to the system was the feed flow (G_{feed}) as given by the equation (3.82). The feed flow was modelled based on the equations (3.65) and (3.66).

As discussed earlier, due to xenon poisoning being a slow acting process and the considered reactor being small, it was expected that the reactivity addition due to xenon wouldn't have much of an effect on the changes in demand power which could be considered as relatively fast changes. For analysis, it was considered that the system was operating at a steady state of 60% F.P ($P_{r0} = 0.6$). The simulation was performed for duration of 200 seconds with a step size of 0.01 second and the steady state conditions as given under the section 3.2.3.2.

- A. The system so formed was found to be state controllable and observable. The Eigen values of this system are shown below. The Eigen value spectrum in this case had 11 values one for each equation. There were 3 pairs of complex conjugate values and all had negative real parts. However, it should be noted that the 10th and 11th eigen values were not exactly zero but a negative number of order e-017, which was considered zero. These made the system rather marginally stable. Also the eigen values underwent slight variations with the initial power level as mentioned earlier.

$$\begin{aligned} \text{eigen}(A) = & -30.8931; -4.3056 + 0.9910i; -4.3056 - 0.9910i; \\ & -0.8166 + 1.4896i; -0.8166 - 1.4896i; -0.2147 + 0.2737i \\ & -0.2147 - 0.2737i; -0.0945; -0.0000; -0.0000; -0.3333 \end{aligned}$$

In order, to stabilize the system and make the 10th and 11th eigen values negative, the *State feedback model* of the RRS-PWR as designed in section 3.3.2.3 might be used.

- B. In order to investigate the behaviour of this model, two cases were considered.
- (i) Ramp rise in demand power from Steady state (60% F.P.) to 100% F.P.
 - (ii) Step rise in demand power from Steady state (60% F.P.) to 100% F.P.

i) Ramp Rise in demand power from 60% F.P. to 100% F.P.

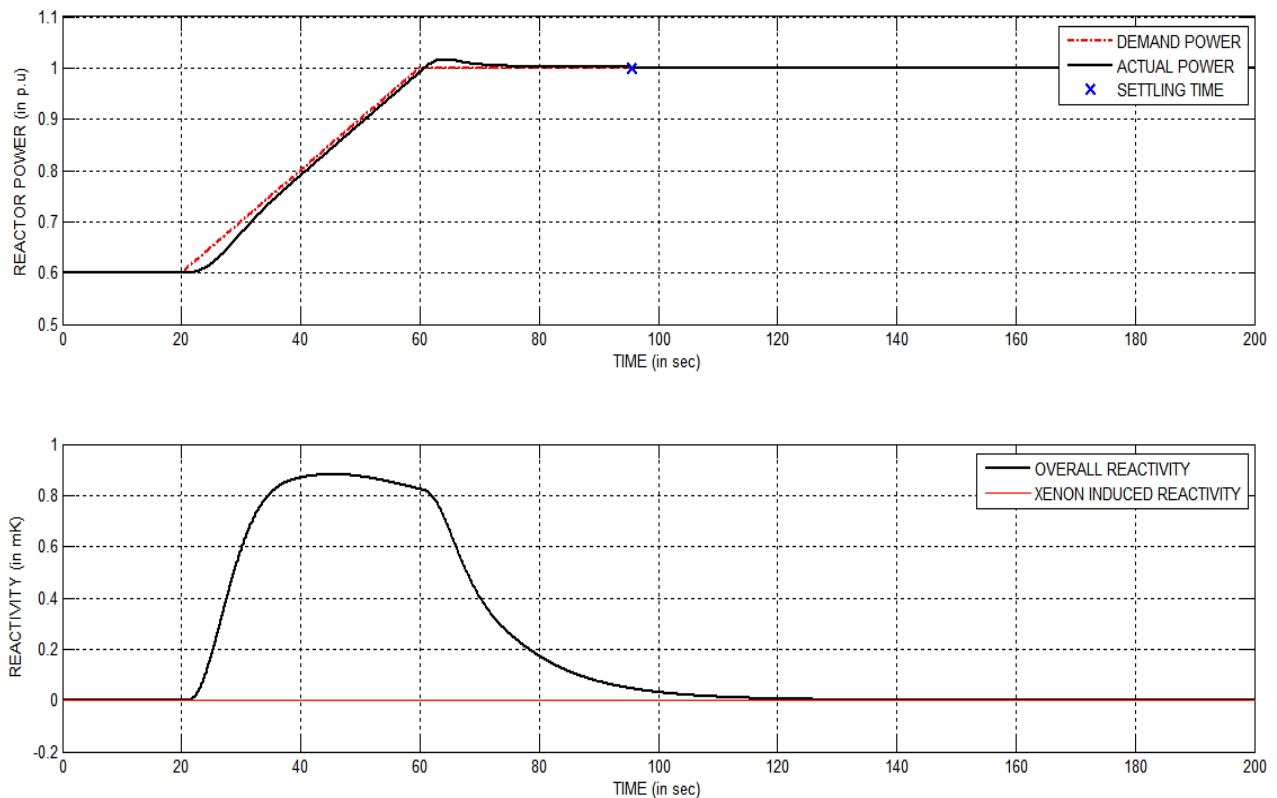


Fig 4.26: Plot of Reactor power (above) & reactivity variation (below) of TNXPWR model for ramp rise in power up to 100%F.P.

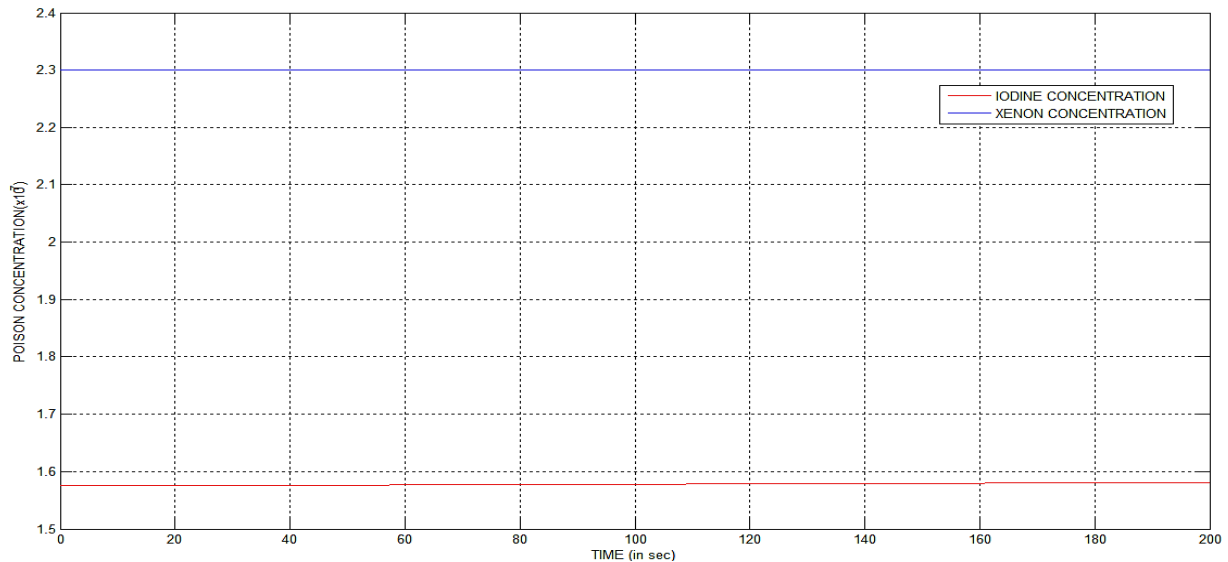


Fig 4.27: Plot of Xenon and Iodine concentrations of TNXPWR model for ramp rise in power up to 100%F.P.

Few curious observations were made from the above results which are given as follows.

- The power plot in **Fig.4.26** was exactly the same as for TNPWR as shown in **Fig. 4.7**, settling at 95.61 seconds. However, upon very close observations, was seen that the power oscillated around the demand power, however, the magnitude of these oscillations were rather negligible. This behaviour could be attributed to the xenon effect and the absence of any RRS to tackle the scenario.
- The reactivity plot showed very slight variations as compared to the **Fig.4.9**, settling down at around 130 seconds. The xenon induced reactivity contribution to the overall reactivity seemed to be zero, however closer observation revealed that poison reactivity in the range, $\rho_x = -0.18\mu K$ existed, which was the reason for the slight power oscillations. However, these effects were so minute that they practically didn't have much effect on the system, hence were considered negligible.
- Lastly, from the xenon and iodine concentration plot shown in **Fig.4.27**, it was seen that the xenon concentration remained almost the same and iodine concentration varied slightly. Closer observation revealed that the xenon concentration did vary, however, the magnitude of variation was so low, that it didn't cause any practical effect during load changing scenarios.

ii) Step Rise in demand power from 60% F.P. to 100% F.P.

In this case, the power plot behaved similar to **Fig.4.17**, settling down at about 70.53 seconds. The reactivity plot also followed the **Fig.4.18** and the poison reactivity stayed in the range, $\rho_x = -0.18\mu K$, which caused power oscillations of very low magnitude, which was classified as negligible. Also, the variation in xenon and iodine concentrations was found to be almost constant.

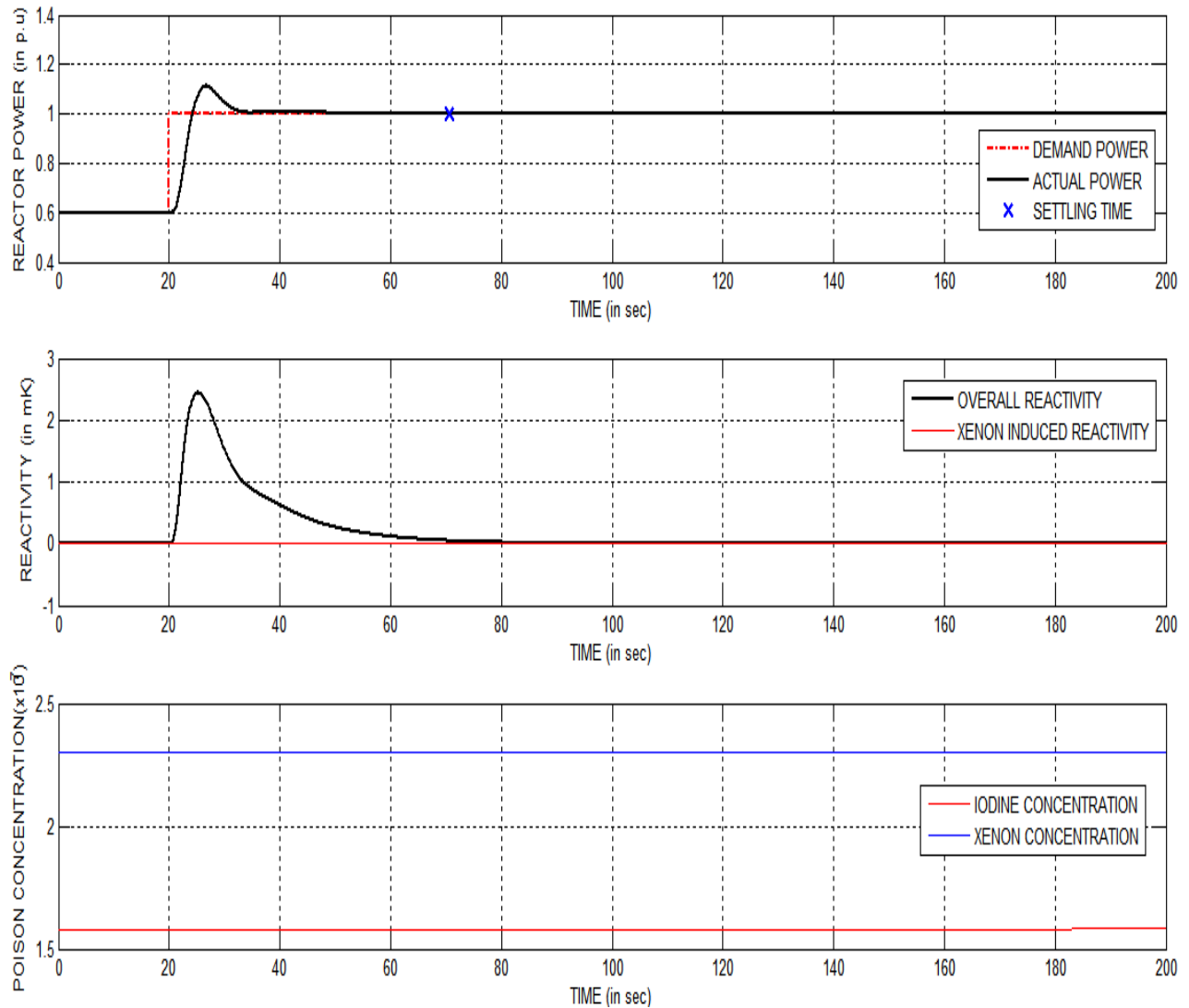


Fig 4.28: Plot of Reactor power (above), Reactivity variation (middle) & Xenon and Iodine concentrations (below) of TNXPWR model for step rise in power up to 100%F.P.

The performance analysis can be concluded with the fact that the PWR model designed in this study worked fairly well under various circumstances that were examined, behaving as expected. With the incorporation of the RRS designed in the section 3.3, the system responses could be further improved with the elimination of overshoots (or undershoots), reduction of the 2%F.P. gap during transition phase and providing a better settling time for the system.

4.2. BLOCK CONTROL MODEL OF PWR

Apart from the analytical model of the PWR system, a Block Control model was also developed for the primary loop of PWR plant during the course of this study. The model thus formed was analogous to our coded model and it was expected that the results would show resemblance with the results already obtained for the earlier model, thus establishing that the fundamental behaviour of the PWR-NPP was successfully captured [1],[9],[25],[28],[30]. This section describes the formation of this model and the results obtained from simulation.

4.2.1. FORMATION OF THE MODEL

The formation of the Block Control model of the PWR was based on the fundamental point kinetic equation (normalised) for six groups of delayed neutrons and the thermodynamic equations of the reactor as mentioned earlier. The basic equations for the model construction are given by (3.27), (3.43), (3.46), (3.47), (3.48), (3.49), (3.51) and (3.52). The reactor was considered to be of 100MWth capacity & the general assumptions under section 3.1.2 were assumed to be valid in this case also. As considered earlier in assumption (e) of the Reactor thermal systems, it was assumed that $F_r = 1.0$. This made the constant $k_4 = 0$ in equation (3.46) for average coolant temperature variation.

The formation of the model was divided in to three modules. Firstly, the Neutronic module for the core, then the thermal module for the core and lastly, the Primary coolant loop module. Later, these three were coupled together along with the control signal to form a complete PWR model.

1. Core Neutronic Module

The Neutronic module was based on the normalised point kinetic equations for six groups of delayed neutrons given by (3.27). The Block module had a two – fold structure. The outer system represented only the input and output signals, while the Subsystem depicted the structure responsible for processing these signals. **Fig. 4.29** and **Fig.4.30** shows the Outer system and the subsystem, respectively [9].

As shown in **Fig. 4.29**, the only input to this module was the reactivity (ρ) and the output signals were the neutron density (n) and the delayed neutron precursor (C) concentration. From the module structure, it is clear that there is a resemblance with the Ex-PKPWR model.

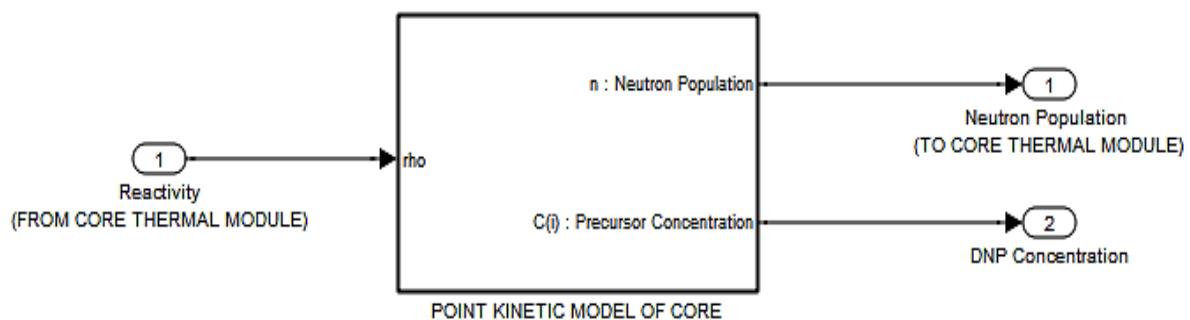


Fig 4.29: Block model of the Neutronic module of the PWR core

The subsystem as shown in the **Fig. 4.30** had two interconnected loops. The upper one represented the neutron balance equation and the bottom one represented the DNP concentration equation, both based on (3.27). The neutron balance equation was composed of three terms, which were computed independently and added together in the Summation block. A path coming from the lower loop carried the matrix, $[(\beta_i/l) * C_i]$, where i denoted the number of delayed groups (6 in this case). It was sent to the 1st positive port of the sum block.

The second positive input to the sum block originated at the output neutron population. Then it got multiplied with the gain block, $1/l$ and the input reactivity (ρ) to form the signal, $[(n/l) * \rho]$. The input to the negative port of the sum block came from the multiplication of the gain block β/l with the output neutron population (n). The output of the sum block was passed on to an integrator block initialized to a value $N0 = n_{r0}$. The neutron population, $n(t)$ was obtained as the output of the integrator block.

Similarly, the DNP concentration equation was composed of two terms and was added together in the Summation block. The input to the positive port came from element wise multiplication of the gain block for λ_i and the neutron population (n), made available from the upper loop. The input to the negative port came from element wise multiplication of the gain block for λ_i and the DNP concentration, $C_i(t)$ available at the output of this loop. The integrator block in this case was initialized to a value $C0 = \vec{C}_{r0}$.

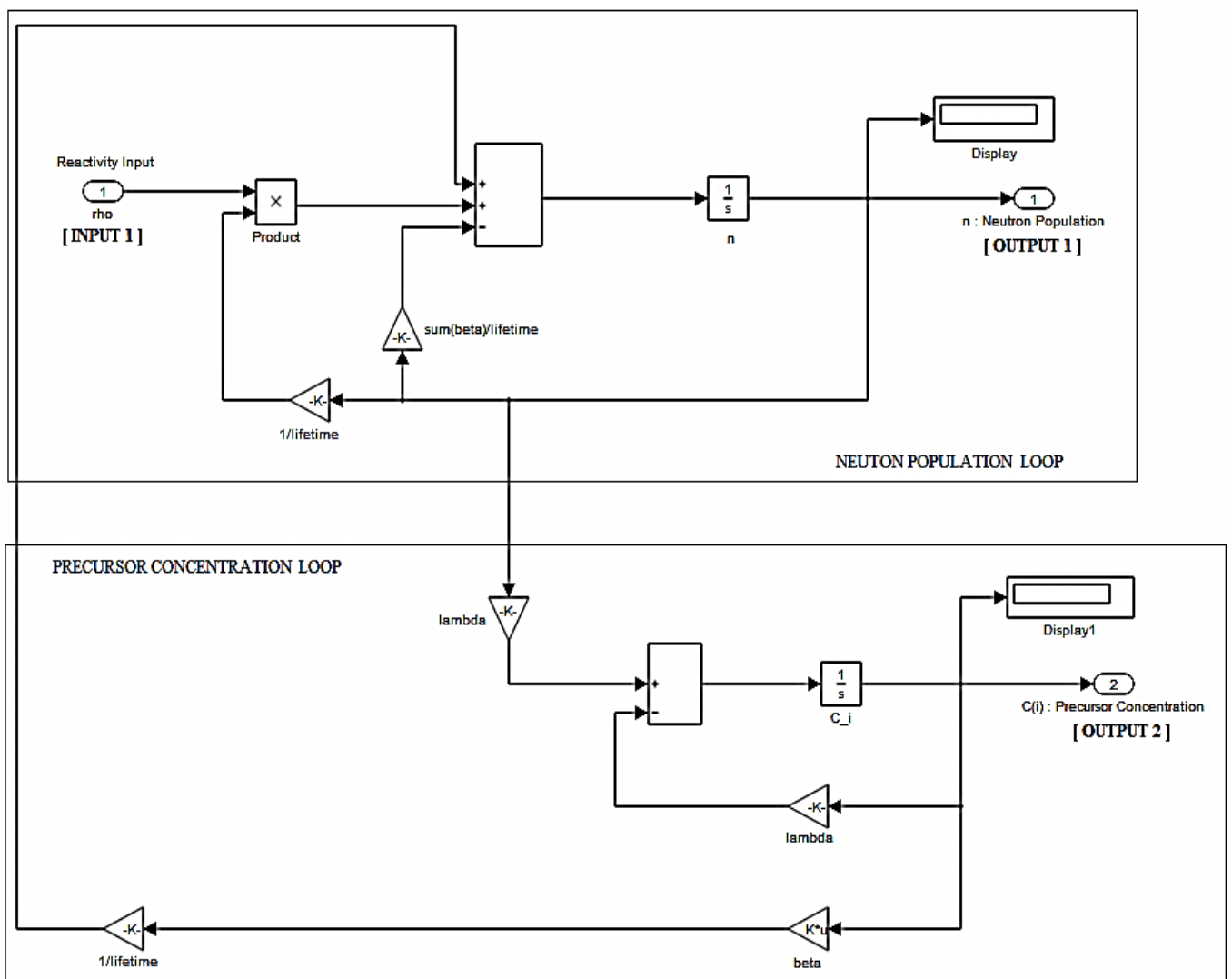


Fig 4.30: Subsystem of the Neutronic module of the PWR core

2. Core Thermal Module

The Core thermal module was based on the fuel temperature (T_f) and average coolant temperature (T_c) equations given by (3.43) and (3.46). Similar to the Neutronic module, this Block module also had a two – fold structure, where the outer system represented only the input and output signals and the Subsystem depicted the internal structure. **Fig. 4.31** and **Fig.4.32** shows the Outer thermal system and the subsystem, respectively.

As shown in **Fig. 4.31**, this module had two inputs, namely, the neutron density (n) and the coolant inlet temperature (T_{cin} or $T_{i,c}$). Out of these two inputs, the neutron density was obtained from the Neutronic module and the coolant inlet temperature from the third module developed for the Primary Coolant loop later. There were two sets of outputs emerging from this module. The first set comprised of the Fuel and Average Coolant temperatures (T_f and T_c). The second set consisted of the fuel and coolant reactivities (ρ_f and ρ_c) based on the equations (3.33) and (3.34).

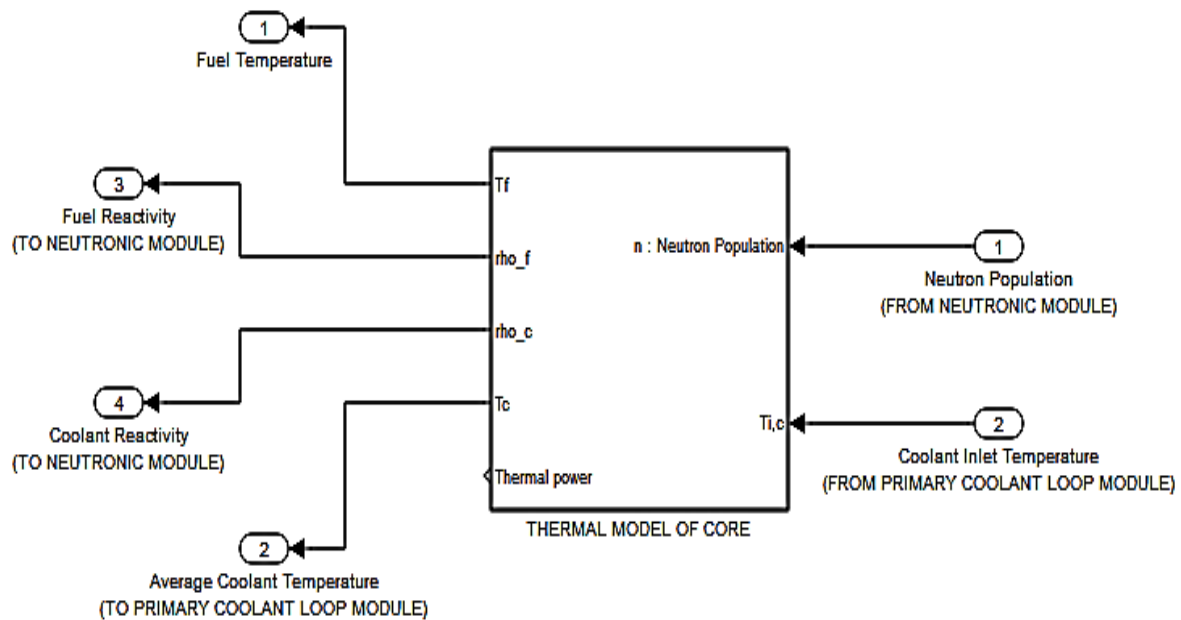


Fig 4.31: Block model of the Thermal module of the PWR core

In this case also, the subsystem consisted of two interconnected loops as shown in **Fig. 4.32**. The upper one represented the Fuel temperature equation and the bottom one represented the Average Coolant temperature equation based on (3.43) and (3.46).

The Fuel temperature loop was composed of two terms, one being a modified version of the input neutron population (INPUT 1) and the other being a modified version of the difference equation ($T_f - T_c$). As shown in **Fig. 4.32**, at the right hand side of the upper loop, the neutron population got multiplied by the gain, $P_0 F_r$ and this signal was sent to the positive port of the main summation block. The input to the negative port of the main summation block came from a set of two successive blocks. Firstly, the secondary summation block had

the Fuel temperature as the input to its positive port and the Average coolant temperature as the input to its negative port, obtained directly from the output of the Bottom loop. So, the output of this summation block was $(T_f - T_c)$. This signal got modified when multiplied by the gain, $h_f A_f$ to give $h_f A_f (T_f - T_c)$ as the output signal. This was fed to the negative port of the main summation block. Then the output from the main summation block got multiplied by a gain, $1/MC_f$ before moving to the integrator block. The integrator block in this case was initialized to a value $Tf0 = T_{f0}$. The Fuel temperature, T_f was obtained as the output (OUTPUT 1.1) of the integrator block. This output was further used as the input to the positive port of a tertiary summation block. The output from this block was $(T_f - T_{f0})$ which got multiplied by the gain α_f to give the fuel reactivity (ρ_f) as the output (OUTPUT 2.1) according to the equation (3.33).

Similarly, the Average Coolant temperature loop at the bottom consisted of two terms. The input to the negative port of the main summation block came via a set of two blocks. The first one was a secondary summation block with Coolant inlet temperature (T_{cin} or $T_{i,c}$) as its negative port input (INPUT 2) and the Average coolant temperature (T_c) being the positive port input. The output from this summation block got multiplied by a gain, $(2 * GC_c)$. This signal was fed to the negative port of the main summation block. The input to the positive port of the main summation block was obtained by following similar steps used in the case of the negative port input to the main summation block of the fuel temperature loop. The value of this signal was $h_f A_f (T_f - T_c)$. Then the output from the main summation block got multiplied by a gain, $1/MC_c$ before moving to the integrator block. The integrator block in this case was initialized to a value $Tc0 = T_{c0}$. The Fuel temperature, T_c was obtained as the output (OUTPUT 1.2) of the integrator block. This output was further used as the input to the positive port of a tertiary summation block. The output from this block was $(T_c - T_{c0})$ which got multiplied by the gain α_c to give the coolant reactivity (ρ_c) as the output (OUTPUT 2.2) according to the equation (3.34).

3. Primary Coolant Loop Module

The Coolant loop module was based on the remaining equations available for the TNPWR model, viz. the Coolant inlet temperature (T_{cin} or $T_{i,c}$), the Coolant outlet temperature (T_{cout} or $T_{o,c}$), the Steam generator inlet temperature (T_{SGin}), the Average steam generator temperature (T_{SG}) and the steam generator power (sgp) as given by the equations (3.47), (3.48), (3.49), (3.51) and (3.52). Again, a two – fold structure was used for the Block Model. **Fig. 4.33** and **Fig.4.34** shows the Outer primary loop system and the subsystem, respectively.

As shown in **Fig. 4.33**, this module had two inputs, viz. the Average Coolant temperature (T_c) and the Feed Flow (G_{feed}), which were obtained from the Core thermal module and the control block, respectively. The two major outputs emerging from the system were the Coolant inlet temperature (T_{cin} or $T_{i,c}$) and the Coolant outlet temperature (T_{cout} or $T_{o,c}$). The coolant inlet temperature became one of the inputs to the Core thermal module.

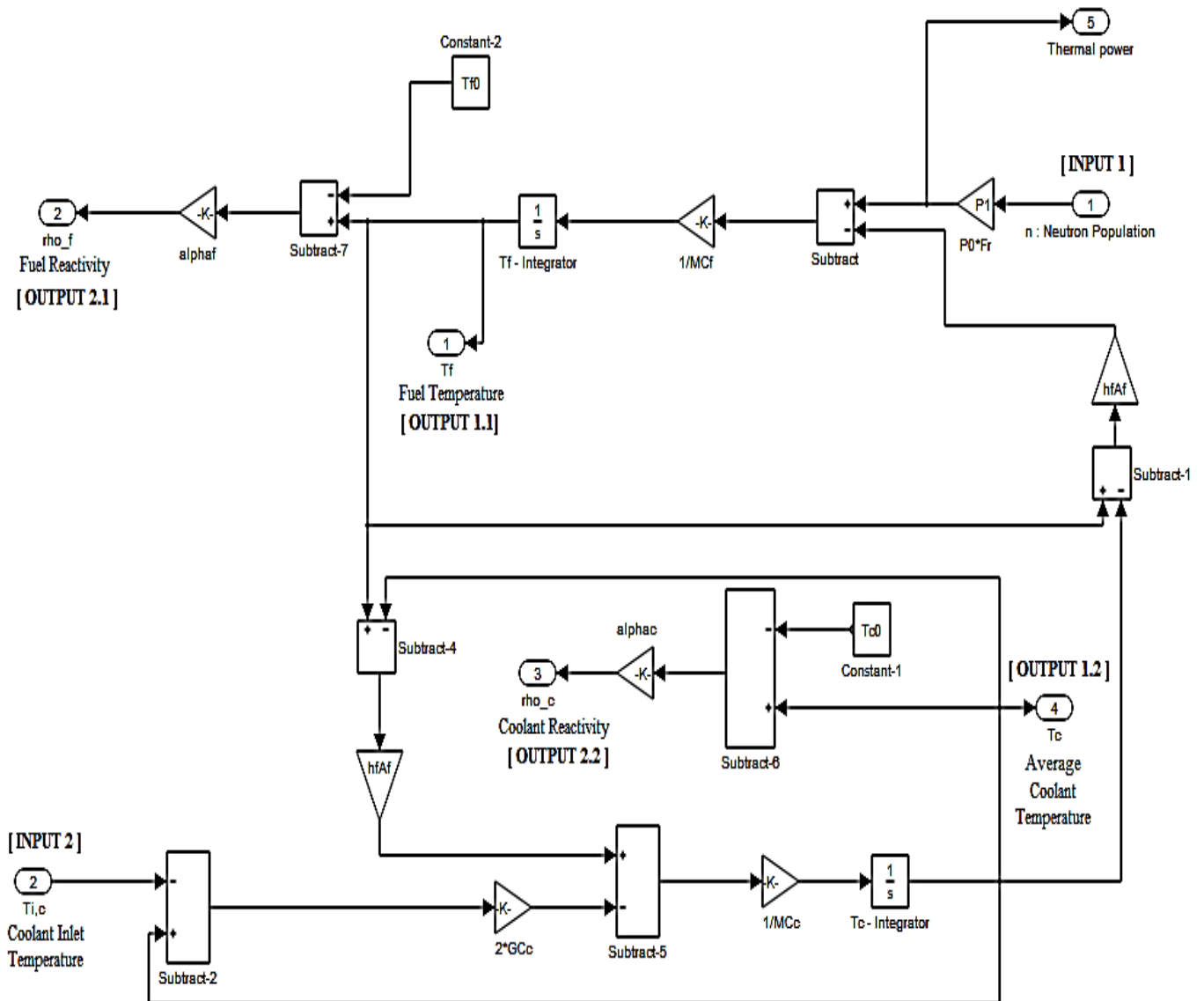


Fig 4.32: Subsystem of the Thermal module of the PWR core

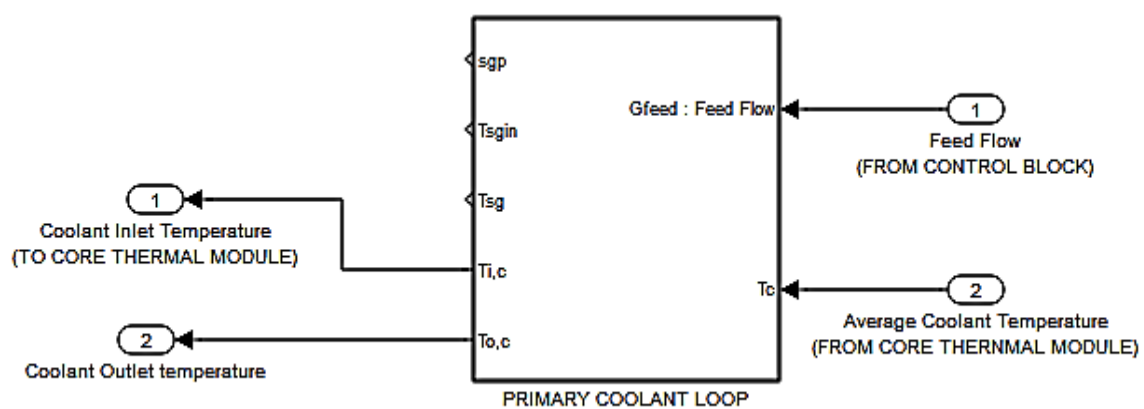


Fig 4.33: Block model of the Primary Coolant loop module of the PWR

In this case, the subsystem consisted of five interconnected loops as shown in **Fig. 4.34**. The upper one was for the steam generator power equation; the next one represented the Steam generator inlet temperature equation, the third one was for the Average steam generator temperature equation, the fourth one represented the Coolant inlet temperature and the last one was for the Coolant outlet temperature. These were based on the set of equations given by (3.47), (3.48), (3.49), (3.51) and (3.52).

The upper steam generator power loop was composed of two terms. One was the Feed flow input (INPUT 2) and the other was the steam generator power itself, thus forming a closed loop. The output signal from the main summation block ($G_{feed} - sgp$) went through a gain block of value (1/3) and then moved onto the integrator, which was initialized at sgp_0 .

The subsequent loop was for the steam generator inlet temperature. The inputs to this loop were the Coolant outlet temperature (positive port) and the steam generator inlet temperature (negative port) itself. The output signal from the main summation block ($T_{o,c} - T_{SGin}$) went through a gain block of value ($1/\tau_{hld}$) before moving onto the integrator, which was initialized at T_{SGin0} .

The next loop was concerned with the average steam generator temperature. The input to the negative port of the main summation block of this loop came from the steam generator power loop as sgp . It was then multiplied by two subsequent gain blocks of value $P_0 F_r$ and $1/MC_{sgw}$ before moving to the port. The input to the positive port came through a secondary summation block and a gain block. The secondary summation block had steam generator inlet temperature (T_{SGin}) from the previous loop as the positive port input and the average steam generator (T_{SG}) from this loop as the negative port input. The value of the gain block for the output of this summation block was ($2/\tau_{SG}$). The integrator was initialized at T_{SG0} .

The fourth loop was for the Coolant inlet temperature ($T_{i,c}$) and it was composed of three terms. The first one was the steam generator inlet temperature (T_{SGin}) which formed the input signal to the first negative port of the main summation block. The second one was the average coolant temperature (T_{SG}) which went to the positive middle port of the main summation block after passing through a gain block of value 2. The third term was the Coolant inlet temperature itself which went to the second negative port of the main summation block. The output from the main summation block again went through a gain of value ($1/\tau_{cld}$) before moving on to the integrator, which was initialized at $T_{i,c0}$.

The last loop was used to process the Coolant outlet temperature ($T_{o,c}$). This was also composed of three terms, with the first one being the Coolant inlet temperature ($T_{i,c}$) from the previous loop. It formed the input signal to the first negative port of the main summation block. The second term was the Average coolant temperature (T_c) coming from the Core Thermal module (INPUT 1), which formed the input signal to the positive port of the main summation block. However, before being sent to the positive port it was multiplied by a gain block of value 2. The third term was the Coolant outlet temperature itself which went to the second negative port of the main summation block. The output from the main summation block again went through a gain of value ($1/\tau_{ct}$) before moving on to the integrator, which was initialized at $T_{o,c0}$.

4. Complete PWR Primary Loop Model

The complete primary loop model of the PWR was formed by coupling all the three modules explained above and by providing the control signal from the control block, as shown in **Fig. 4.35**. The interfacing between the three modules was done with the help of common signals and summation blocks as given below.

- The Neutron Population signal was the first output of the Core Neutronic module and was the first input of the Core thermal module (INPUT 1).
- The Average Coolant Temperature (T_c) signal was the second output (OUTPUT 1.2) of the Core thermal module. At the same time, it was an input to the Primary Coolant loop module (INPUT 1).
- The Coolant inlet temperature ($T_{i,c}$) signal was the first output (OUTPUT 1) of the Primary Coolant loop module and subsequently was the second input of the Core thermal module (INPUT 2).
- The fuel and coolant reactivities were the second set of output signals (OUTPUT 2.1 and OUTPUT 2.2) from the Core thermal module. The Core Neutronic module had reactivity (ρ) as its input.

The first three signals were used to link the corresponding modules simply by forming a path between them. However, the last signal was linked to the Core Neutronic module through a summation block, whose output was considered as the reactivity (ρ) input to the module. An additional block was formed here (E_R) which represented any external reactivity addition to the system. This block was then linked to a port of the summation block.

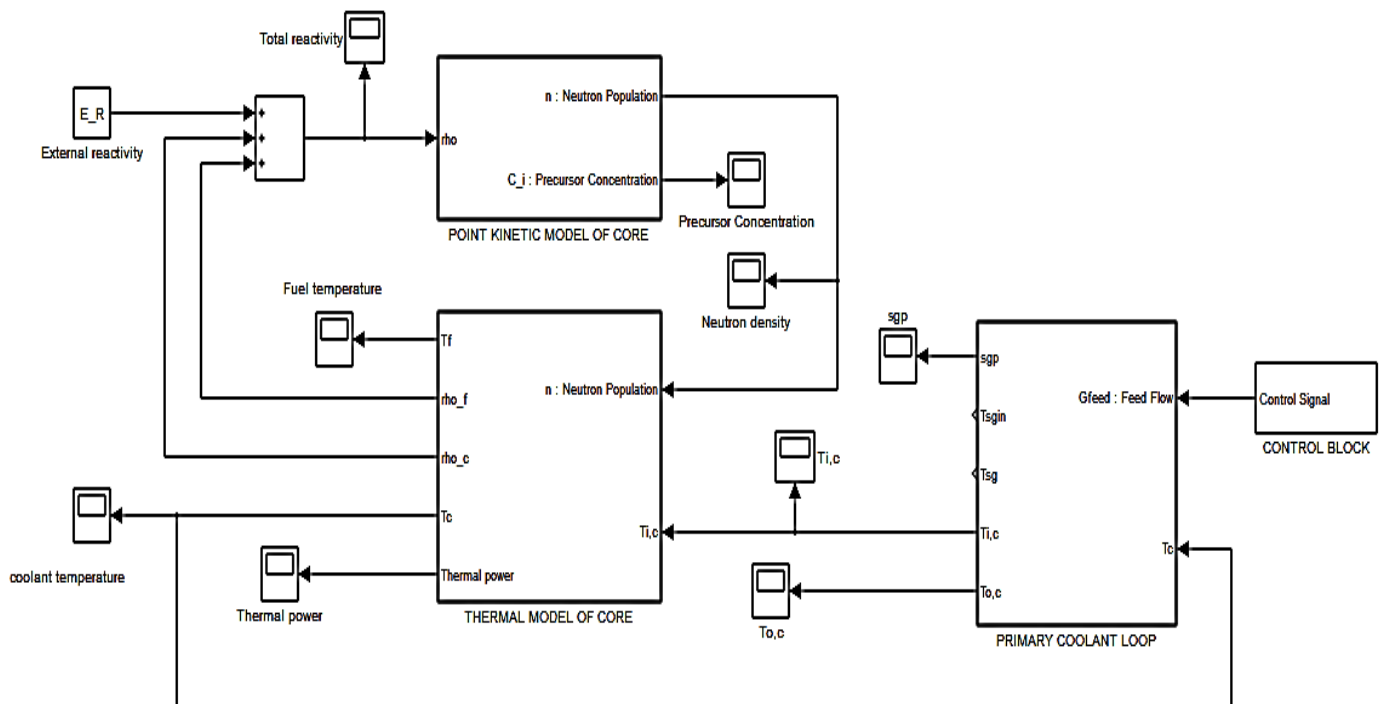


Fig 4.35: Complete Block Control model of the Primary loop of PWR-NPP

The control block in the above model was used to generate the signal for the feed flow (G_{feed}), which was the only external input to this system. The values necessary for initializing the integrators and the corresponding values of the gain blocks were assumed to be the same as those used for the performance analysis of the coded model. These values are provided in **Appendix B**.

4.2.2. RESULTS OBTAINED FROM SIMULATION

The Block model developed above was simulated under similar circumstances, which were used to analyse the models developed previously. The initial values of the integrators and the gain blocks were stored on a script file and from there the simulation was initiated. The simulation was carried out in two phases: Firstly on the Neutronic module alone and then on the Complete model of the PWR.

1. Core Neutronic Module

Here, it was considered that the system was operating at steady state power level of 60 % F.P ($P_{r0} = 0.6$). The simulation was performed for duration of 50 seconds with a step size of 0.01 second. Also, the same initial conditions and steady state values as given under section 3.2.1.4 were specified.

Since, the only input to the Core neutronic module was reactivity (ρ); a step input in reactivity was inserted in the system at 20 seconds. The value of the reactivity insertion was 1mK for the first simulation and -1mK for the second. The results obtained from these simulations of the neutronic module (for 6 delayed groups), matched those obtained for the Ex-PKPWR model as shown in **Fig. 4.3** and **Fig. 4.4** for insertion of 1mK and -1mK reactivity. The values obtained from the Block model were same as those tabulated in **Table 4.1**. The respective plots are shown below in **Fig. 4.36** and **Fig. 4.37**.

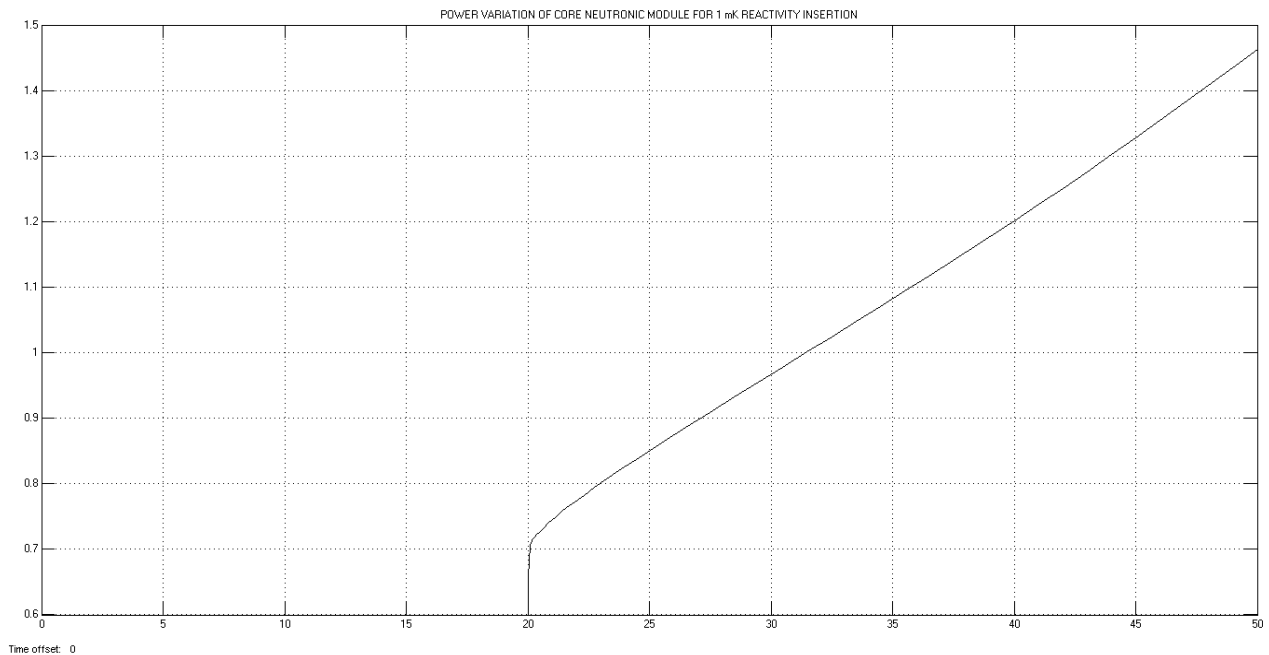


Fig 4.36: Plot of Power variation for 1 mK reactivity insertion obtained from the Neutronic module of the Block Control model

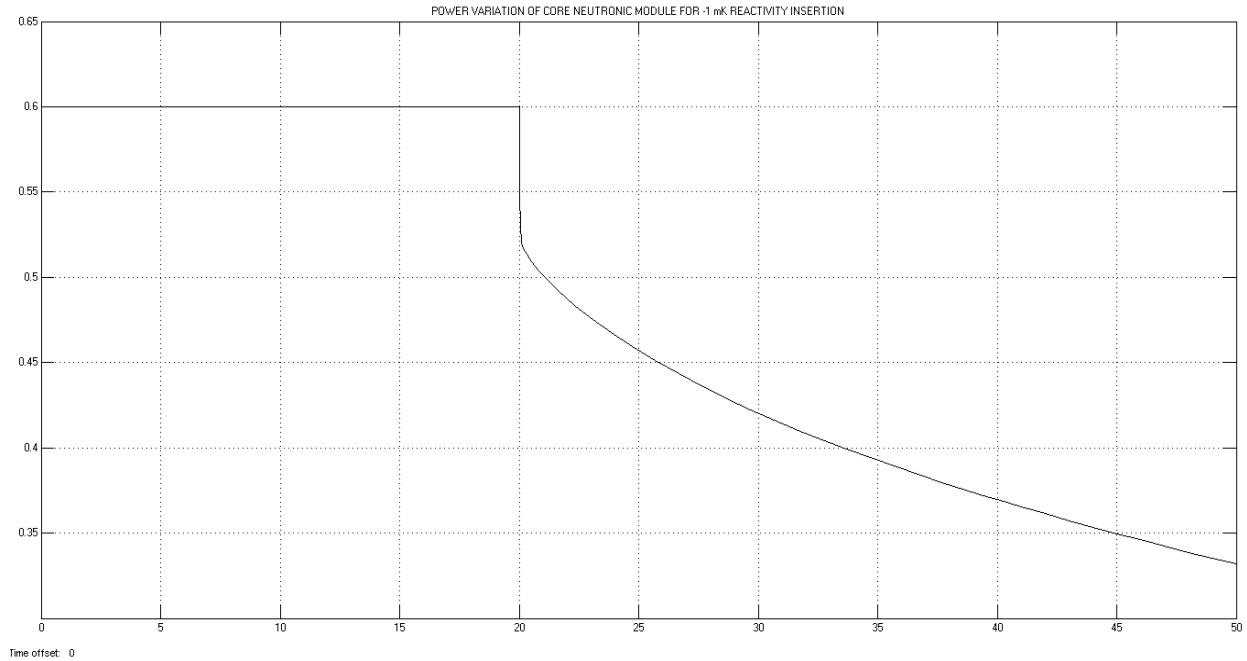


Fig 4.37: Plot of Power variation for -1 mK reactivity insertion obtained from the Neutronic module of the Block Control model

2. Complete PWR Primary Loop model

The complete PWR model as shown in **Fig. 4.35** had only the feed flow (G_{feed}) as an external input to the system. The control block was used to generate the control signals for the feed flow. The simulation was performed for two types of control signals, step and ramp. Since this model had a similar structure as that of TNPWR, so the steady state conditions were specified as given under the section 3.2.2.2 by the equations (3.58)-(3.64). As done previously, the simulation was performed for duration of 200 seconds with a step size of 0.01 second. Also, the input (the change in demand power via the feed flow) was administered to the system at 20 seconds.

i) Step change in power

For this case, the system was again considered to be operating at steady state power level of 60 % F.P ($P_{r0} = 0.6$). The demand power level was assumed to be 100 % F.P ($P_r = 1.0$) for power rise and 20 % F.P ($P_r = 0.2$) for power decline. For both the cases, the responses were found to be similar to those obtained for the TNPWR step change scenario. However, few minute differences were noted, as given below.

- In the step rise in power scenario, the Neutron density (or power) variation as shown in **Fig.4.38** settled down to the demand power after undergoing oscillations around that power level. In case of TNPWR, these oscillations were very low after the initial overshoot magnitude as shown in **Fig.4.17**. Also, the Reactivity plot showed oscillations after the initial rise around 40 seconds as given in **Fig.4.39**. These oscillations were not present in the reactivity plot for the TNPWR case as shown in **Fig.4.18**. Here, the reactivity died down exponentially after the initial rise.

- In the case of step decline in power, the opposite trend was noticed. The pronounced oscillations in power and reactivity that were present in the TNPWR case as shown in **Fig.4.19** and **Fig.4.20** were not present in this case. The **Fig.4.40** and **Fig.4.41** show the corresponding power and reactivity variations.
- Another important point was observed that the reactivity plots never exactly become zero at the end of the simulation. This could be attributed to the difference in simulation techniques.

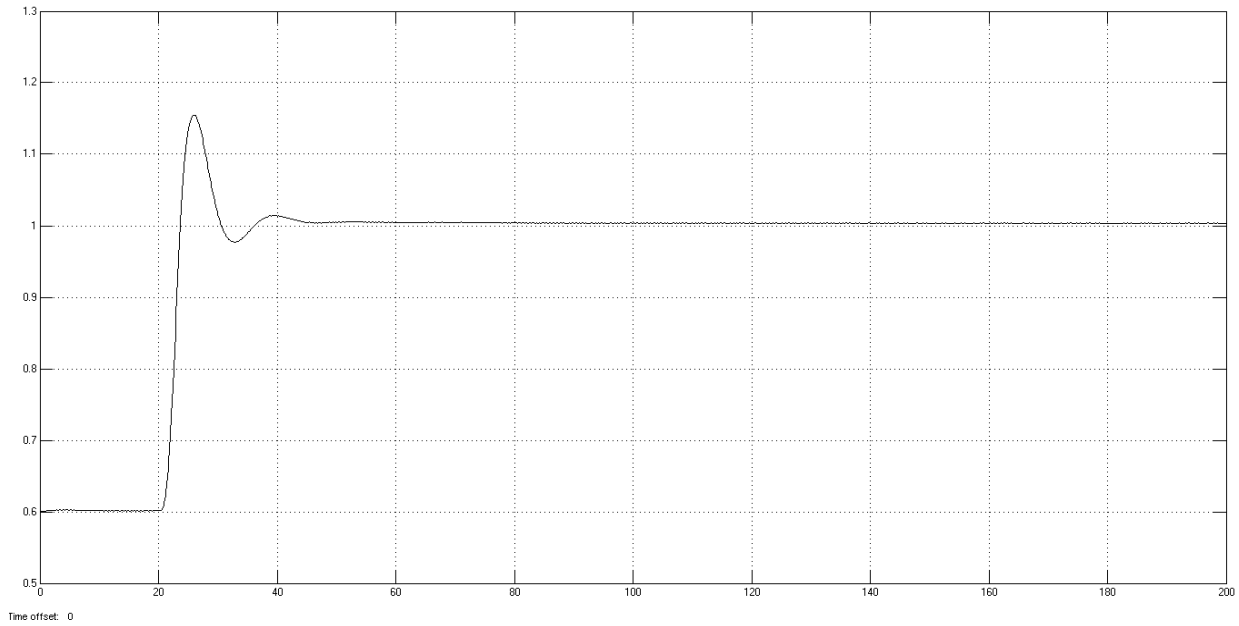


Fig 4.38: Plot of Neutron density variation for step rise in power scenario

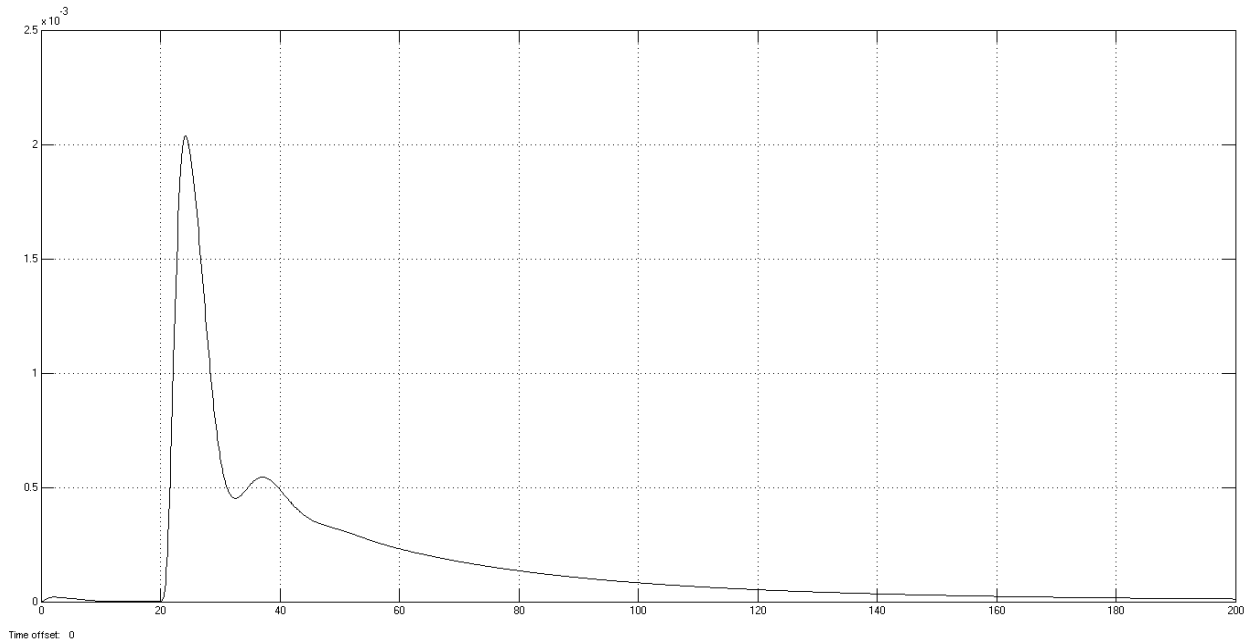


Fig 4.39: Plot of Total reactivity variation for step rise in power scenario

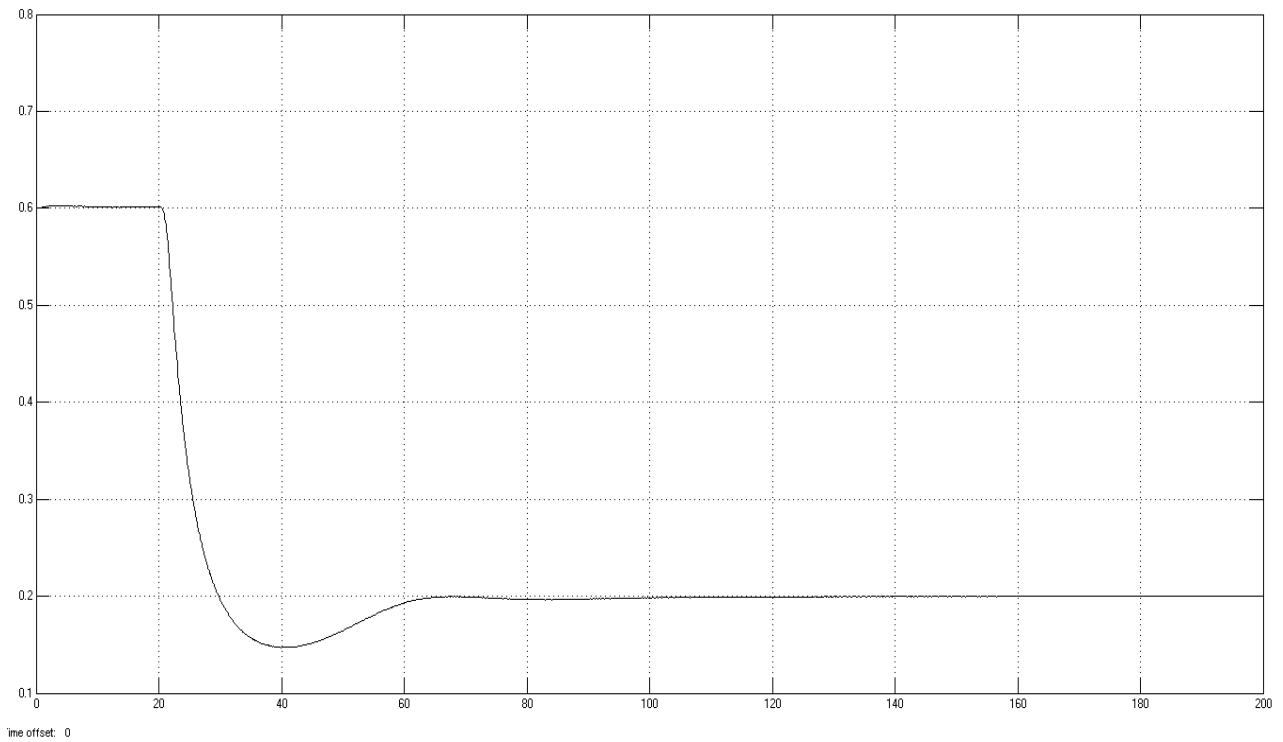


Fig 4.40: Plot of Neutron density variation for step decline in power scenario

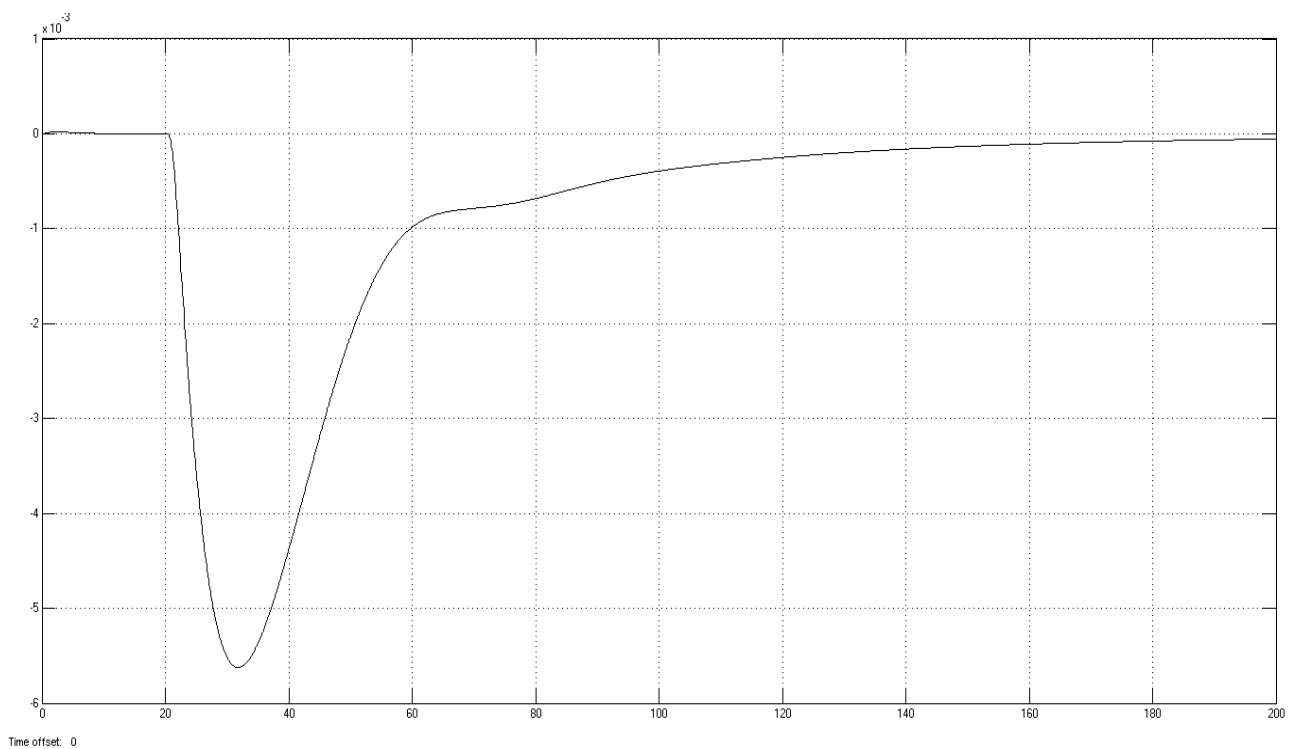


Fig 4.41: Plot of Total reactivity variation for step decline in power scenario

ii) Ramp change in power

The same circumstances that were used in the simulation of the step power change were considered in this case also. The ramp increase in power was considered to occur at a rate of $\pm 1.0\%$ F.P/second (ramp rate).

Although, the nature of the plots remained the same, few minute differences were observed in this case also, as given below.

- In the ramp rise in power scenario, the Neutron density (or power) variation as shown in **Fig.4.42** behaved similar to that was observed in case of TNPWR as shown in **Fig.4.7**. Although, the Reactivity plot (**Fig.4.43**) showed slight oscillations during the steady state phase (before 20 seconds) and during the transition phase, the reactivity reached up to 0.55mK around 40 seconds and then slowly rose to the maximum value of 0.62mK at around 60 seconds. After that it declined exponentially, however, it didn't become exactly zero. In case of TNPWR, the reactivity plot (**Fig.4.9**) was exactly at zero during the steady state and reached the maximum value of 0.88mK around 40 seconds. Then, it slowly declined to 0.8mK around 60 seconds, after which it steadily declined to zero. These differences were attributed to the variation in the simulation techniques.
- In case of ramp decline in power, the Neutron density (or power) variation as shown in **Fig.4.44** behaved similar to that was observed in case of TNPWR as shown in **Fig.4.10**. The reactivity plot as shown in **Fig.4.45** declined up to -2.88mK and then started to rise exponentially towards zero as the simulation progressed. However, it was never able to reach zero during the simulation interval. In comparison, the reactivity plot in case of TNPWR (**Fig.4.12**) showed oscillatory behaviour. First, it grew negative to around -3.64 mK , then increased to 0.3 mK, from where it again became slightly negative before completely settling down at zero.

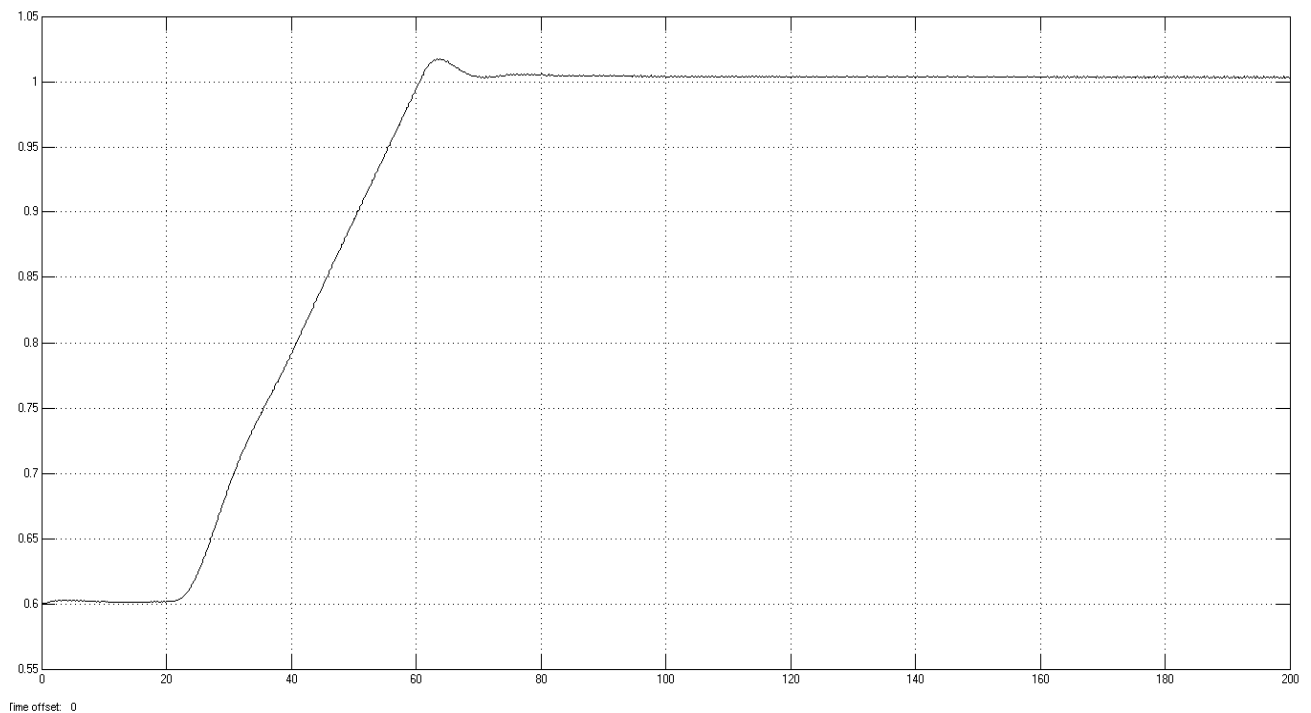


Fig 4.42: Plot of Neutron density variation for ramp rise in power scenario

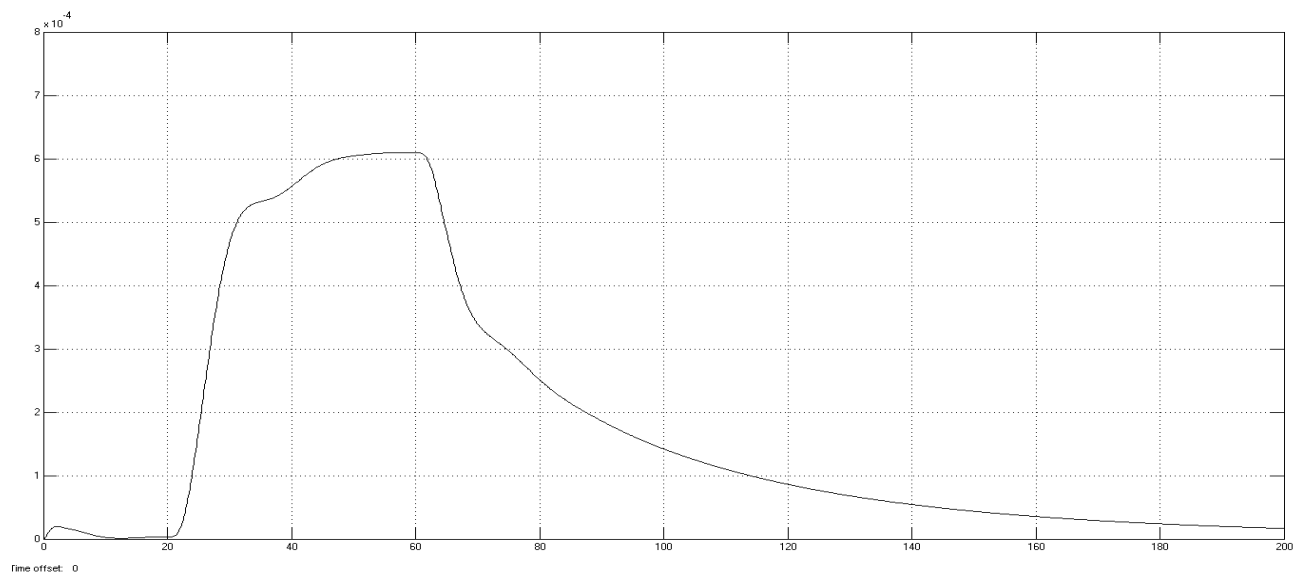


Fig 4.43: Plot of Total reactivity variation for ramp rise in power scenario

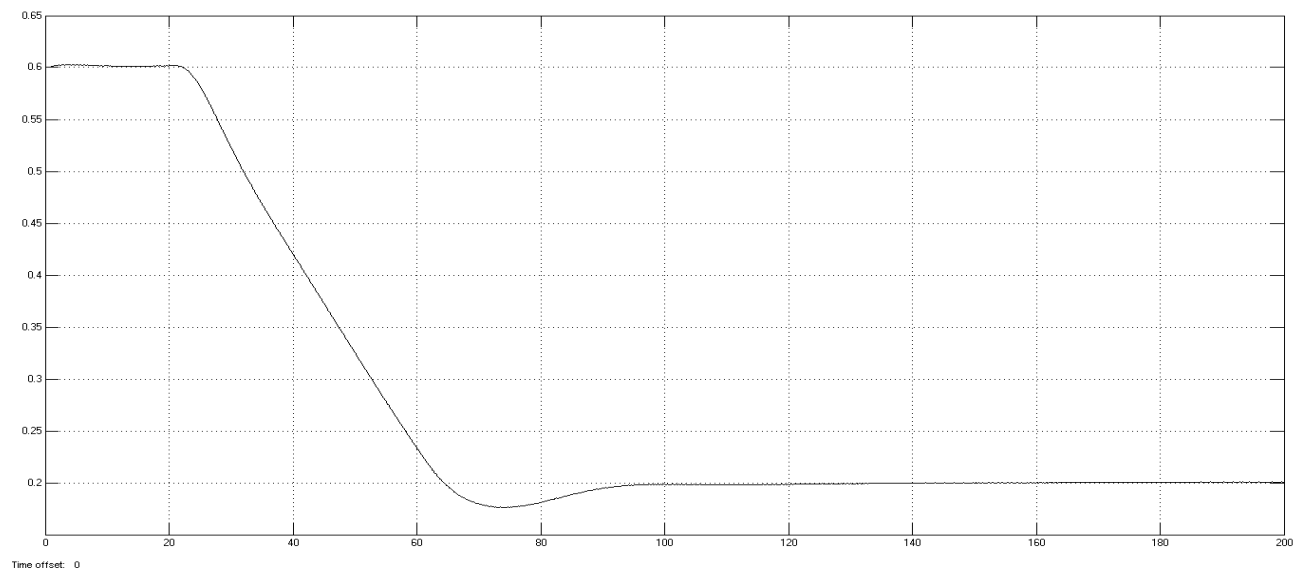


Fig 4.44: Plot of Neutron density variation for ramp decline in power scenario

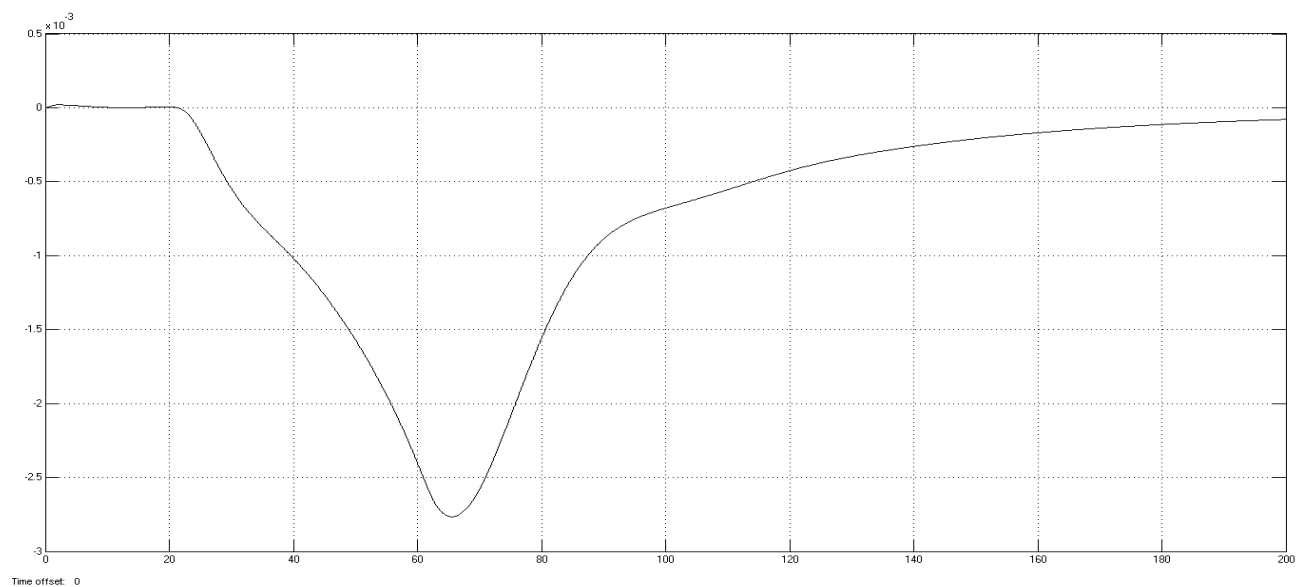


Fig 4.45: Plot of Total reactivity variation for ramp decline in power scenario

4.3. STABILITY ANALYSIS OF PWR MODELS

Nuclear reactors show non-linearity which can be justified by the fact that the point kinetics equations are itself non-linear in nature. During formation of the PWR models, this non-linear part of the point kinetic equation (the term containing $\delta n \delta \rho$) was neglected while linearizing. Since this non-linearity plays a major role in practical situations, it is necessary to analyse its effect on the system stability.

To achieve this task, the Lyapunov stability analysis given under section 2.5 for reactor system stability was used. In the Lyapunov stability theory, if it can be proved that the linear part of the system is asymptotically stable, and then a finite region of stability can be specified for the entire non-linear system (refer **Appendix D.2**). For this, it was necessary to express the non-linear reactor system as a combination of linear and non-linear part. However, during linearization of the PWR models this part had already been covered. So by analysing the stability of these linearized PWR models with the Lyapunov stability theory, the stability conditions for the non-linearity of aforesaid models was also analysed. In this study, the linear matrix inversion commands available in the software package were used.

1. Stability of TNPWR model

During the performance analysis, it was seen that the Eigen spectrum for TNPWR had negative real parts. The 1st value was on the extreme left of the s-plane and the others formed a cluster near the origin. This implied that the system was stable. Also, the performance of this model was analogous to its expected behaviour. Despite the disturbances given to the system, it was able to stabilize itself around an equilibrium point without external mechanisms.

Now, for the Lyapunov analysis, a study approach was used whereby the $\mathbf{P}$ matrix was specified as positive definite. In order for the system to be asymptotically stable, it was necessary to prove that the $\mathbf{Q}$ matrix was positive definite or ϕ was negative definite, according to the Lyapunov equation [37],

$$-\mathbf{Q} = \phi = (\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A}) \quad (4.2)$$

It was considered that the P matrix was of the order of the system matrix ($\mathbf{A}$), in this case 9x9 square matrix as shown in equation (3.57). Also, it was considered that the system was initially operating at 60% F.P. ($P_{r0} = 0.6$). Now, for the TNPWR system to be asymptotically stable the following conditions were to be met.

- › The symmetric matrix P must be positive definite, for which the Eigen values of the P matrix must be positive.
- › The ϕ matrix must be negative definite, for which the Eigen values of the ϕ matrix must be negative.

Using the linear matrix inversion, the Eigen values of the P matrix were found to be,

$$\begin{aligned} eigen(P) = & 4.9347; 229.9181; 462.8777; 958.2498; 2151.6156; \\ & 2394.3406; 10725.9730; 56440.9194; 117708.5561 \end{aligned}$$

And the Eigen values of the ϕ matrix as

$$\begin{aligned} \text{eigen}(\phi) = & -29733.1392; -27097.1023; -20741.8353; \\ & -12583.9592; -12306.8862; -1744.5925; \\ & -877.4936; -83.1464; -0.9117 \end{aligned}$$

These Eigen values satisfied the necessary conditions and hence it was proved that the TNPWR model was asymptotically stable.

2. Stability of TNXPWR model

During the performance analysis, it was seen that the Eigen spectrum for TNXPWR had proper negative real parts for all values except, the 10th and the 11th eigen values. The real parts of these values were very close to zero, which made the system marginally stable.

Similar analysis was performed on the TNXPWR model as in the previous case and the Eigen values of the P matrix and ϕ matrix were calculated. Here, the system matrix was considered as described by the equation (3.84) and the system was considered to be operating at 60% F.P. ($P_{r0} = 0.6$). Using the linear matrix inversion commands, the Eigen values of the P matrix and the ϕ matrix were found to be,

$$\text{eigen}(P) = 1.0e + 006 * (0.0000; 0.0000; 0.0150; 0.0170; 0.0570; 0.1197; 0.2189; 0.4714; 0.7688; 2.4246; 3.3010)$$

$$\begin{aligned} \text{eigen}(\phi) = & 1.0e + 006 * (-1.2079; -1.0343; -0.9785; -0.6330; \\ & -0.4987; -0.3702; -0.2578; -0.0121; \\ & -0.0031; -0.0000; -0.0000) \end{aligned}$$

These Eigen values proved the fact that the TNXPWR model was also asymptotically stable.

3. Stability of RO-TNPWR model

The analysis of the RO-TNPWR showed that the 6th Eigen value was positive, which made the entire system unstable. Similar analysis was performed on this model also and the Eigen values of the P matrix and ϕ matrix were calculated. Here, the system matrix was considered to be defined as per the equation described in (3.69) and the system was considered to be operating at 60% F.P. ($P_{r0} = 0.6$). Using the linear matrix inversion commands, the Eigen values of the P matrix and the ϕ matrix were found to be,

$$\text{eigen}(P) = 1.0e + 005 * (-0.0000; 0.0101; 0.0153; 0.1267; 0.2899; 0.4550; 1.0255; 1.5373)$$

$$\begin{aligned} \text{eigen}(\phi) = & 1.0e + 005 * (-1.4278; -0.7748; -0.7472; -0.5145; -0.3177 \\ & -0.1890; -0.0012; 0.000) \end{aligned}$$

Here, it was seen that the 1st Eigen value of the P matrix was negative and the last Eigen value of the ϕ matrix was positive. This indicated that the system was unstable in the sense of Lyapunov.

From the stability analysis of the PWR models, the following facts were inferred.

- The TNPWR model which has only the thermal feedbacks associated with the bare reactor is asymptotically stable as proved by the Lyapunov analysis
- The TNXPWR model which includes poisoning feedback in addition to the thermal feedback is also asymptotically stable; however, the degree of stability can be improved by incorporating the RRS into the model.
- The RO-TNPWR model is unstable not only in the sense of Lyapunov but also based on the Eigen values of the system matrix itself. Hence, in order to successfully implement this model and improve its stability, it is necessary to use the RRS along with it.

4.4. CHAPTER SUMMARY

In this chapter, the results of the simulations conducted on the PWR models are discussed in detail. The simulations are performed based on the order of development of the models, starting from the Bare Reactor model. Ramp and step rise in power scenarios are considered for the simulation. Also, the effect of disturbances in the form of small amount of reactivity addition is investigated.

The results of the simulations show that the behaviour of the PWR models resembles the behaviour of practical PWR-NPP. The results show that the Bare Reactor model can't stabilize itself due to lack of any feedback mechanism. However, as expected, it is seen that PWR models with internal feedbacks can stabilize themselves under any perturbation scenario, without the use of external control system. Although, it is observed that the addition of poisoning feedback didn't change the settling time or reactivity by much as compared with the thermal feedbacks. So, it is concluded that even without the poisoning feedback the PWR model can behave similar to that of actual plant. Apart from these results, the eigen values of the PWR models are noted. They show that except the reduced order thermal feedback model, all other systems are stable, although the degree of stability varies from one model to another.

Succeeding this discussion, the formation of a Block Control model has been explained. The notion behind the formation of this model is to validate the responses of the models formed previously, by comparing them with the results obtained from the Block model. It is seen that apart from a few differences, the nature of the responses remain more or less the same. These differences between the responses have been clearly indicated. Further, based on the Lyapunov direct method as mentioned in the literature survey, the stability analysis of the feedback models has been carried out. The results show that apart from the reduced order thermal feedback model, the other two models viz. one with only thermal feedback and the other with both thermal and poisoning feedback are stable in the sense of Lyapunov.

CONCLUSION

In the recent years, apart from our PHWR programme, we have made advances in the study of the PWRs. In that context, this dissertation was aimed at studying the behaviour of the PWR by formulating an analytical model and a Block Control model. Then their performances were to be evaluated under various expected scenarios. Also, the stability issues of the analytical models were to be studied in brief.

5.1. SUMMARY OF THE PRESENT STUDY

During the course of this study, an analytical model of the primary loop of the PWR was developed using first principles for the power regime operation of the reactor. The model was designed, implemented and investigated in three different phases, on a commercially available software and simulation package.

1. Firstly, the model of the PWR primary loop was progressively developed from the basic point kinetic model (PKPWR) to the closed loop models with the aid of State Space modelling approach. The closed loop models were formed by stepwise addition of the internal feedbacks (TNPWR & TNXPWR) along with the equations describing the behaviour of the primary loop. The behaviour of these models was investigated with the help of various inputs (step and ramp) which simulated the power level variations in a typical PWR-NPP. Apart from these regular inputs, the effect of disturbances was also investigated during such power level variations. The responses offered by these models for the array of inputs and disturbances, were found to be consistent with the known behaviour of a PWR. In addition to the State Space modeling of the bare reactor, a non-linear reactor regulating system was designed for the reactor's power operational regime, employing the state feedback approach. This could be implemented with the aforesaid models to improve their responses for the simulated variations and disturbances. Also, two different operational programs, viz. the constant average coolant temperature program and constant hot leg temperature program, were investigated for the closed loop system (TNPWR). The responses obtained for these two programs suggested that the constant average coolant temperature program was the natural program of a PWR since the plant was able to stabilize itself even without the RRS, as opposed to the other program. This fact was validated from the principles available in literature [33].
2. In the second phase, a Block Control model of the PWR was developed based on the same mathematical principles and equations. In the developmental stage, three different modules were created each pertaining to a different part of the primary loop of the PWR. These modules were then joined together to form the complete model of the PWR-NPP. In order to investigate the performance of the developed Block model, a

control block was created. Also, an external reactivity block was added to it, which could be used to investigate the effect of disturbances (noise) in the system.

The model behaviour was again investigated with the help of various inputs to the control block (step and ramp) which simulated the power level variations in a typical PWR-NPP. The responses offered by the neutronic module of the reactor for input reactivities (1 mK and -1mK) were a complete match to that offered by the PKPWR models. Also, the responses of the complete reactor system bore very close resemblances to those offered by the TNPWR model. This established the fact that the analytical model of the reactor so developed gave same responses even for different simulation procedures, thus making it reliable.

3. Lastly, the stability analysis of the system models formed in the first phase was done using the Lyapunov stability criteria because of the inherent non-linearity in the system. The Lyapunov criterion was applied to the linearised models of the systems, whose stability in the sense of Lyapunov, indicates that the entire system is stable in that region [37]. Based on the above principles the investigation revealed that the TNPWR model was asymptotically stable. The TNXPWR model although stable, had a low degree of stability. Also, the RO-TNPWR model was found to be unstable in the sense of Lyapunov. The application of the RRS model to these systems might improve their stability.

These results established the fact that the analytical models so developed behaved analogous to the known behaviour of PWRs. Also, the models so formed were mostly stable and their stability could be improved upon by employing simple state feedback strategies.

5.2. SCOPE FOR FUTURE STUDIES

In case of further research along these lines, few recommendations can be made so as to improve upon the present study.

The models developed here could be made more accurate and robust by reducing the number of assumptions made here. The system behaviour can be checked for scenarios where 100% heat transfer at all the exchange points, doesn't take place ($F_r < 1.0$). Also, the reactor operation can be considered for start-up and shutdown regions together with the power regime and the model updated based on these conditions. Along with these improvements in the model, the system behaviour could be studied when the RRS is incorporated within the model structure.

Apart from the first principle approach, another way to model the PWR would be to collect several sets of input-output data from PWR simulation software [11], [41]. Then these data could be utilized based on System identification principles [19] to develop several models of the system. In order to select the best performing model among them the best fit approach could be utilized. After that, a control mechanism representing the RRS could be developed using control strategies like MPC [22], [23] or robust control [42] for the best performing model. Then, the performance of the controller could be studied as compared to the existent controllers.

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APPENDIX A

DERIVATIONS

A.1. Linearization of the Neutron Density (or Power) Equation

The normalised neutron density (or Power) variation equation is given by,

$$\frac{d}{dt}P_r = \frac{(\rho(t) - \beta)}{l}P_r + \frac{\beta}{l}C_r \quad (\text{A.1})$$

It is known that, the fuel, coolant and the xenon poisoning effects contribute to the overall reactivity of the system. So, it can be said that,

$$\rho(t) = \rho_f(t) + \rho_c(t) + \rho_x(t) \quad (\text{A.2})$$

Substituting $\rho(t)$ in the (A.1) yields,

$$\frac{d}{dt}P_r = \frac{(\rho_f(t) + \rho_c(t) + \rho_x(t) - \beta)}{l}P_r + \frac{\beta}{l}C_r \quad (\text{A.3})$$

Now considering perturbation from steady state, it can be written that,

$$\begin{aligned} P_r(t) &= P_{r0} + \delta P_r \\ C_r(t) &= C_{r0} + \delta C_r \\ \rho_f(t) &= \rho_{f0} + \delta \rho_f \\ \rho_c(t) &= \rho_{c0} + \delta \rho_c \\ \rho_x(t) &= \rho_{x0} + \delta \rho_x \end{aligned} \quad (\text{A.4})$$

In steady state,

$$P_{r0} = C_{r0} \text{ (DNP concentration variation equation) and,} \quad (\text{A.5})$$

$$\rho(0) = \rho_f(0) + \rho_c(0) + \rho_x(0) = 0 \quad (\text{A.6})$$

Substituting (A.4) in the (A.3) and rearranging gives,

$$\begin{aligned} \frac{d}{dt}P_{r0} + \frac{d}{dt}\delta P_r &= \frac{(\rho_{f0} + \rho_{c0} + \rho_{x0})}{l}(P_{r0} + \delta P_r) - \frac{\beta}{l}(P_{r0} + \delta P_r) + \frac{\beta}{l}(C_{r0} + \delta C_r) + \\ &\quad \frac{(\delta \rho_f + \delta \rho_c + \delta \rho_x)}{l}(P_{r0} + \delta P_r) \end{aligned} \quad (\text{A.7})$$

Now, the time variation of steady state values is zero, i.e.

$$\frac{d}{dt}P_{r0} = 0$$

$$\Rightarrow \frac{(\rho_{f0} + \rho_{c0} + \rho_{x0})}{l} P_{r0} - \frac{\beta}{l} P_{r0} + \frac{\beta}{l} C_{r0} = 0 \quad (\text{A.8})$$

First expanding (A.7) and then by substituting equation (A.8) in (A.7), the following equation can be formed.

$$\begin{aligned} \frac{d}{dt} \delta P_r = & \frac{(\rho_{f0} + \rho_{c0} + \rho_{x0})}{l} \delta P_r + \frac{(\delta \rho_f + \delta \rho_c + \delta \rho_x)}{l} P_{r0} + \frac{(\delta \rho_f + \delta \rho_c + \delta \rho_x)}{l} \delta P_r \\ & - \frac{\beta}{l} \delta P_r + \frac{\beta}{l} \delta C_r \end{aligned} \quad (\text{A.9})$$

The first term on the RHS becomes zero, since $\rho_f(0) + \rho_c(0) + \rho_x(0) = 0$ from (A.6).

The third term on the RHS is a non-linear term which gives the terms $\delta \rho_f \delta P_r, \delta \rho_c \delta P_r, \delta \rho_x \delta P_r$, thus introducing non-linearity in the equations. So as to linearize the point kinetics equations, this term is neglected by making the assumption that value of the perturbation is small. So, the normalised and linearized point kinetic equation can be formed by rearranging (A.9).

$$\boxed{\frac{d}{dt} \delta P_r = -\frac{\beta}{l} \delta P_r + \frac{\beta}{l} \delta C_r + \frac{P_{r0}}{l} \delta \rho_f + \frac{P_{r0}}{l} \delta \rho_c + \frac{P_{r0}}{l} \delta \rho_x} \quad (\text{A.10})$$

A.2. Linearization of the Xenon Concentration Equation (Normalised)

The normalised xenon concentration equation is given by,

$$\frac{dX}{dt} = (\gamma_X \Sigma_f P_0) P_r + \lambda_I I - \lambda_X X - \sigma_X P_0 P_r X \quad (\text{A.11})$$

Now considering perturbation from steady state, it can be written that,

$$\begin{aligned} P_r(t) &= P_{r0} + \delta P_r \\ I(t) &= I_0 + \delta I \\ X(t) &= X_0 + \delta X \end{aligned} \quad (\text{A.12})$$

Substituting (A.12) in the (A.11) and rearranging gives,

$$\begin{aligned} \frac{d}{dt} X_0 + \frac{d}{dt} \delta X = & (\gamma_X \Sigma_f P_0) (P_{r0} + \delta P_r) + \lambda_I (I_0 + \delta I) - \lambda_X (X_0 + \delta X) - \\ & \sigma_X P_0 (P_{r0} + \delta P_r) (X_0 + \delta X) \end{aligned} \quad (\text{A.13})$$

Now, the time variation of steady state values is zero, i.e.

$$\begin{aligned} \frac{d}{dt} X_0 &= 0 \\ \Rightarrow & (\gamma_X \Sigma_f P_0) P_{r0} + \lambda_I I_0 - \lambda_X X_0 - \sigma_X P_0 P_{r0} X_0 = 0 \end{aligned} \quad (\text{A.14})$$

First expanding (A.13) and then by substituting equation (A.14) in (A.13), the following equation can be formed.

$$\frac{d}{dt}\delta X = (\gamma_X \Sigma_f P_0)\delta P_r + \lambda_I \delta I - \lambda_X \delta X - \sigma_X P_0 P_{r0} \delta X - \sigma_X P_0 X_0 \delta P_r - \sigma_X P_0 \delta P_r \delta X \quad (\text{A.15})$$

The last term on the RHS of equation (A.15) is a non-linear term, since it contains term $\delta P_r \delta X$. So as to linearize the equation, this term is neglected by making the assumption that value of the perturbation is small. So, the linearized xenon concentration equation can be written as,

$$\frac{d}{dt}\delta X = (\gamma_X \Sigma_f P_0 - \sigma_X P_0 X_0)\delta P_r + \lambda_I \delta I - (\lambda_X + \sigma_X P_0 P_{r0})\delta X \quad (\text{A.16})$$

If , $k_X = \gamma_X \Sigma_f P_0$ is considered, then the equation (A.16), can be rewritten as,

$$\frac{d}{dt}\delta X = (k_X - \sigma_X P_0 X_0)\delta P_r + \lambda_I \delta I - (\lambda_X + \sigma_X P_0 P_{r0})\delta X \quad (\text{A.17})$$

This is the linearized form of Xenon concentration equation that was used in this study.

APPENDIX B

DATA USED FOR SIMULATIONS

The values of the various parameters used for simulation of the developed PWR were obtained from relevant literature [4],[7],[16],[27],[33],[42] available on the topic. The values of few significant parameters are provided below.

Neutronic Parameters

Prompt Neutron Life Time (l): 2.1×10^{-4} seconds.

Delayed Neutron Fraction of Precursor Groups (β_i): 0.000215; 0.00142; 0.00127; 0.00257; 0.00075; 0.00027

Decay Constant of the Precursor Groups (λ_i): 0.0124; 0.0305; 0.111; 0.301; 1.1; 3.0 (sec^{-1}).

Decay Constant of Iodine (λ_I): 2.9306×10^{-5} seconds.

Decay Constant of Xenon (λ_X): 2.1066×10^{-5} seconds.

Microscopic absorption cross section of Xenon (σ_X): 2.65×10^6 barns.

Fractional yield of Iodine (γ_I): 0.0610 (6.1%).

Fractional yield of Xenon (γ_X): 0.003 (0.3%).

Fuel Temperature Coefficient (α_f): $-2.97 \times 10^{-5} \Delta k/k/^\circ\text{C}$.

Coolant Temperature Coefficient (α_c): $-7.2 \times 10^{-4} \Delta k/k/^\circ\text{C}$.

Thermodynamic Parameters

Total mass of Primary Coolant (m_c): 5.868×10^3 kg.

Total mass of Fuel (m_f): 8.517×10^4 kg.

Specific Heat of Primary Coolant (C_c): 5594.2 J/kg. $^\circ\text{C}$.

Specific Heat of fuel (C_f): 236.77 J/kg. $^\circ\text{C}$.

Fuel to Coolant Heat Transfer Area (A_f): 3867.7 m^2 .

Neutron Population to Reactor Thermal Power Conversion Factor (F_r): 1.0

Fuel to coolant heat transfer coefficient (h_f): 1859.03 $\text{W}/\text{m}^2.^\circ\text{C}$

Primary coolant mass flow rate (G_c): 4.8634×10^3 kg/seconds.

Average Fuel Temperature (T_f): 445.27 $^\circ\text{C}$.

Coolant Outlet (hot leg) Temperature (T_{cout}): 328.2 $^\circ\text{C}$.

Coolant Inlet (cold leg) Temperature (T_{cin}): 292 $^\circ\text{C}$.

APPENDIX C

FEW RELATED TOPICS

C.1. Brief Overview of the Indian Nuclear Programme

The importance of nuclear energy, as a sustainable energy resource for our country, was recognised at the very inception of our atomic energy programme conceived more than four decades ago. The main architect behind this ingenious formulation was *Dr. Homi Jehangir Bhabha*. A three-stage nuclear power programme based on a closed nuclear fuel cycle, was chalked out by him, bearing in mind the sole objective of utilizing the domestic thorium & uranium reserves in an optimal manner so as to cater for the country's long – term energy requirements [12]. The most worthwhile feature of the Indian policy is to achieve closed fuel cycle utilization. This principle now forms the bedrock of our nuclear policy. The three stages envisage:

1. Natural uranium fuelled Pressurised Heavy Water Reactors (PHWRs).
2. Fast Breeder Reactors (FBRs) utilising plutonium based fuel.
3. Advanced nuclear power systems for utilisation of thorium.

Dr. Bhabha summarized the rationale for the three-stage approach as follows:

“The total reserves of thorium in India amount to over 500,000 tons in extractable form, while the known reserves of uranium are less than a tenth of this. The aim of long range atomic power programme in India must therefore be to base the nuclear power generation as soon as possible on thorium.” [12].

1. First stage of the programme –Development of PHWR technology

In the first stage, natural uranium dioxide based fuel is used in heavy water (D₂O) moderated and cooled Pressurised Heavy Water Reactors (PHWRs), which generates Plutonium – 239 as the by- product [12],[14].

PHWRs was a natural choice for implementing the first stage because it had the most efficient reactor design in terms of uranium utilization, and the existing Indian infrastructure in the 1960s allowed for quick adoption of the PHWR technology. India correctly calculated that it would be easier to create heavy water production facilities (required for PHWRs) than uranium enrichment facilities (required for LWRs).

To kick start the nuclear power generation programme, the first two plants were of the Boiling water reactor type (BWRs) at Tarapur, Maharashtra and were based on imported technology [12], [14]. Subsequent plants were of PHWR type and through the R&D efforts have achieved complete self-reliance in this technology.

2. Second stage of the programme –Development of FBR technology

In the second stage, Fast breeder reactors (FBRs) would use a mixed oxide (MOX) fuel made from **Plutonium-239**, contained in the spent fuel from the PHWR & would be recovered by reprocessing spent fuel and natural uranium [12],[14]. In FBRs, plutonium-239 undergoes fission to produce energy, while the uranium-238 present in the mixed oxide fuel transmutes to additional plutonium-239. Thus, the Stage II FBRs are designed to “breed” more fuel than they consume. Once the inventory of plutonium-239 is built up thorium can be introduced as a blanket material in the reactor and transmuted to **U-233** for use in the third stage.

The surplus plutonium bred in each FBR can be used to set up more such reactors, and thus grow the Indian civil nuclear power capacity till the point where the third stage reactors using thorium as fuel can be brought online. Using FBRs an energy output of 350 GWe can technically be sustained using indigenously available uranium resources. In pursuit of second stage, the design of our first fast breeder, called *Prototype Fast Breeder Reactor* (PFBR), was done by IGCAR [12].

3. Third stage of the programme –Development of Thorium technology

The third stage of the programme comprises of dedicated breeder reactors based on **Uranium-233 and Thorium** are planned to be constructed. These reactors would serve as the mainstay of the final, thorium utilization stage, of the Indian nuclear programme. The currently known Indian thorium reserves amount to 200,000 GWe-yr of the electrical energy [34] and can easily meet the energy requirements during the next century and beyond.

C.2. Upcoming Technologies

It was realised that there would be a long delay before direct thorium utilisation can begin with the conventional three stage programme. Keeping this in mind, the nuclear programme is now focussing at reactor designs that will lead to the timely development of the thorium based technologies necessary for the third stage of the programme. The main highlights of this new approach have been the development of the indigenous AHWRs that allow direct use of thorium [36] and the addition of PWRs in the form of VVERs & EPRs to the nuclear programme [14]. Further, to conserve fuel, technology for the fabrication of mixed oxide (MOX) fuel has been developed [12].

1. Light Water Reactors

Considering the growing energy demands and the necessity to increase the energy potential, a second line of Light water reactors (LWRs) is being added to this current indigenous programme of Pressurized Heavy Water Reactors (PHWRs). As a first step, two Light Water Reactors of 1000 MWe VVER units are being installed at Koodankulam (KNPP) in collaboration with the Russian Federation [13]. These Reactors belong to the family of the Pressurized Water Reactors (PWRs).

The salient features of this reactor are Horizontal steam generators, hexahedral fuel assemblies & High-capacity pressurisers providing a large reactor coolant inventory with a negative void coefficient [13]. It happens to have a passive heat removal system. VVER

uses low enriched (2.4 – 4.4% ^{235}U) uranium dioxide (UO_2) or equivalent pressed into pellets and assembled into fuel rods [12]. Later, EPRs, a class of generation III+ LWRs are being planned at Jaitapur, Maharashtra which would be the largest NPP in India according to capacity [12], [14].

2. Advanced Heavy Water Reactor

Considering the large thorium reserves in India, the future systems, in the third stage of Indian nuclear power program, will be based on thorium– ^{233}U fuel cycle [34]. While the initiation of the third stage will take place in the future, there is a need for the timely development of thorium-based technologies for the entire thorium fuel cycle. The Advanced Heavy Water Reactor (AHWR) is being developed to fulfil this need. The AHWR is a vertical; pressure tube type, heavy water (D_2O) moderated and boiling light water (H_2O) cooled natural circulation reactor designed to generate 300MWe (920MWth) and 500m³/day of desalinated water [35],[36]. The AHWR is fuelled with (Th– ^{233}U) O_2 pins and (Th–Pu) O_2 pins. The fuel is designed to maximize generation of energy from thorium, to maintain self-sufficiency in ^{233}U .

APPENDIX D

RELEVANT THEORIES

D.1. Six factor formula

The *Six factor formula* represents the effective multiplication factor as shown in the equation below.

$$k_{eff} = \varepsilon P_f p P_t f \eta \quad (\text{D.1})$$

Each of these factors serves a special significance in the context of the neutron population, which is discussed below.

i. Fast Fission Factor (ε)

The first process that the neutrons of one generation may undergo is fast fission. Fast fission is fission caused by neutrons that are in the fast energy range. The mathematical expression of *fast fission factor* (ε) is shown below.

$$\varepsilon = \frac{\text{Number of fast neutrons produced by all fissions}}{\text{Number of fast neutrons produced by thermal fissions}} \quad (\text{D.2})$$

The value of ε is essentially 1.00 for a homogenous reactor where the fuel atoms are surrounded by moderator atoms. However, in a heterogeneous reactor like PWRs, the closely packed arrangement of the core elements results in a value of about 1.03. The value of ε is not significantly affected by variables such as temperature, pressure, enrichment, or neutron poison concentrations.

ii. Fast Non-Leakage Factor (P_f)

In a realistic reactor of finite size, some of the fast neutrons leak out of the boundaries of the reactor core before they begin the slowing down process. This is taken care of by the Fast Non-Leakage Factor, whose mathematical expression is provided beneath [11].

$$P_f = \frac{\text{Number of fast neutrons that do not leak from the reactor}}{\text{Number of fast neutrons produced by all fissions}} \quad (\text{D.3})$$

iii. Resonance Escape Probability (p)

When the fast neutrons are slowing down through the resonance region of uranium-238, which extends from about 6 eV to 200 eV [7], there is a chance that some neutrons will be captured. The probability that a neutron will not be absorbed by a resonance peak is called the resonance escape probability, as expressed below.

$$p = \frac{\text{Number of neutrons that reach thermal energy}}{\text{Number of fast neutrons that start to slow down}} \quad (\text{D.4})$$

The value of the resonance escape probability is determined largely by the fuel-moderator arrangement and the amount of enrichment of uranium-235. As in PWR, raising the temperature of the fuel will raise the resonance absorption in uranium-238 due to the Doppler Effect (or broadening) [2]. The increase in resonance absorption lowers the resonance escape probability.

iv. Thermal Non-Leakage Factor (P_t)

Neutrons can also leak out of a finite reactor core after they reach thermal energies. This leakage is accounted for by the Thermal non-leakage factor [11].

$$P_t = \frac{\text{Number of thermal neutrons that don't leak from the reactor}}{\text{Number of neutrons that reach thermal energies}} \quad (\text{D.5})$$

v. Thermal utilization factor (f)

Once thermalized, the neutrons continue to diffuse throughout the reactor and are subject to absorption by other materials in the reactor as well as the fuel. The thermal utilization factor describes how effectively thermal neutrons are absorbed by the fuel, or how well they are utilized within the reactor.

$$f = \frac{\text{Number of thermal neutrons absorbed in the fuel}}{\text{Number of thermal neutrons absorbed in all reactor materials}} \quad (\text{D.6})$$

The thermal utilization factor will always be less than one because some of the thermal neutrons absorbed within the reactor will be absorbed by atoms of non-fuel materials. In heterogeneous water-moderated reactors (like PWRs), it is affected by moderator temperature, the amount of enrichment of uranium-235 and the poison concentration [2].

vi. Reproduction factor (η)

Most of the neutrons absorbed in the fuel cause fission, but some do not. The *reproduction factor* (η) showcases this ratio mathematically as,

$$\eta = \frac{\text{Number of fast neutrons produced by thermal fission}}{\text{Number of thermal neutrons absorbed in the fuel}} \quad (\text{D.7})$$

The rate of production of fast neutrons by thermal fission can be determined by the product of the fission reaction rate ($\Sigma_f^U \phi^U$) and the average number of neutrons produced per fission (ν). The rate of absorption of thermal neutrons by the fuel is $\Sigma_a^U \phi^U$. In the case where the fuel contains several fissionable materials, it is necessary to account for each material. In the case of a PWR core containing both uranium-235 and uranium-238, the reproduction factor would be calculated as shown below.

$$\eta = \frac{N^{U-235} \sigma_f^{U-235} \nu^{U-235}}{N^{U-235} \sigma_a^{U-235} + N^{U-238} \sigma_a^{U-238}} \quad (\text{D.8})$$

D.2. Lyapunov Direct method

Lyapunov stability theory, given by the Russian mathematician Alexander Mikhailovich Lyapunov in his work entitled '*The general problem of motion stability*', [37] provides a general approach for studying the stability of nonlinear control systems. This theory adopts two different methods to deal with the problem of *motion stability*, i.e. whether a system remains close to its original motion trajectory once it has been perturbed away from it. The first method is the *linearization method* which investigates the local stability of nonlinear systems around an equilibrium point. It depends on the idea that a nonlinear system should behave equivalently to its linear approximation for small ranges. Whereas, the second method or the so called *direct method* is not restricted to local stability. It emphasizes about the extent of stability and investigates the stability 'in the large'. The direct method of Lyapunov stability analysis has become a prime tool for the analysis of stability of dynamical systems, mostly nonlinear systems. The reason behind it is the intuitive approach it provides for the solution of the stability problem, without actually finding the state trajectories by solving the differential equations.

1. Basic Philosophy

The underlying philosophy of the direct method is based on the general observation that : *When any mechanical or electrical system dissipates its total energy continuously, despite the issue of degree of nonlinearity in that system, it must eventually settle down to an equilibrium point* [37]. This basic idea can be used to convey the stability of any system by investigating the variation of a *single scalar function of energy* and in essence is a more general expression of a Hamiltonian.

Upon comparing, the ideas of stability and the total mechanical energy of a system which is nothing but a summation of its kinetic and potential energy, a relation can be observed amongst them as explained beneath.

- Zero mechanical energy corresponds to an equilibrium point.
- The convergence of mechanical energy to zero implies Asymptotic stability.
- The undamped growth of mechanical energy implies Instability.

Hence, this method attempts to explain the stability of the dynamical systems by forming '*energy-like*' functions. It is used when faced with a set of nonlinear differential equations [37]. This energy function can be seen as a measure of *distance* of the state from the origin, in the state space and the variation of this distance provides the bounds of stability regions.

2. PROPERTIES OF ENERGY FUNCTION

The energy function, generally expressed as $V(x)$, has two main properties [37],[31] :

- The first property is that the function should be strictly positive. This property is formalised as 'Positive Definite Functions'.

- The second property is associated with the dynamics of the system and tells that the function should be monotonically decreasing. This property is formally known as ‘Lyapunov Functions’.

a. Positive Definite Functions

For autonomous systems, a scalar function $V(x)$ continuous for $|x| \leq A$, $A > 0$ is said to be positive definite if $V(0) = 0$ and if $V(x) > 0$ for $x \neq 0$. For non-autonomous systems, a scalar function $V(x, t)$ continuous for (x, t) in some region R is said to be positive definite if there exists a positive function $V(x)$, such that $|V(x, t)| \leq \epsilon$ for all (x, t) in R , and if $V(0, t) = 0$.

b. Lyapunov Functions

Now, it is assumed that $V(x, t)$ possesses continuous first partial derivatives with respect to t and x in the region R described at the beginning of this section and suppose that $x = x(t)$. This positive definite function $V(x, t)$ which admits an infinitely small upper bound and for which $\frac{dV}{dt}$ is negative semi-definite is called a *Lyapunov function* [17]. The criteria for Lyapunov function candidate are given below for autonomous & non- autonomous systems.

i. For autonomous systems:

The Lyapunov function, $V(x)$, is defined as a positive definite scalar function with the following properties:

- $V(x)$ is continuous with its first partial derivatives in a certain open region about the origin.
- $V(0) = 0$ only.
- Outside the origin and always inside the open region, $V(x)$ is non-negative and vanishes only at the origin. The origin is an isolated minimum of $V(x)$.
- For $\dot{V}(x) \leq 0$ in the open region
 - If $\dot{V}(x) = 0$, the stability is neutral.
 - If $\dot{V}(x) < 0$, the stability is asymptotic.

ii. For non-autonomous systems:

The Lyapunov function, $V(x, t)$, is defined as a positive definite scalar function with the following properties:

- $V(x, t)$ is defined in the open region for all $t \geq 0$.
- $V(0, t) = 0$ for all $t \geq 0$.
- $\dot{V}(x, t) \leq 0$ in the open region

Where,

$$\dot{V}(x, t) = \frac{d}{dx} V(x, t) = \frac{\partial V}{\partial x} \dot{x} = \frac{\partial V}{\partial x} f(x) \quad (\text{D.9})$$

Summarizing these conditions, the three main steps to verify if a scalar function is a Lyapunov stable function are:

1. $V(x, t) > 0$.
2. $V(0, t) = 0$.
3. $\dot{V}(x, t) \leq 0$.

3. ANALYSIS OF GENERAL SYSTEMS

In this section, the Lyapunov analysis for both linear as well as for non-linear systems has been discussed.

I. Lyapunov Analysis of Linear Systems

Lyapunov functions can be systematically formulated to describe stable linear systems. Since nonlinear systems may include linear components, it is important that the Lyapunov analysis appropriately examines their stability.

Formulation

Let us consider a linear autonomous system of the form, $\dot{x} = Ax$ (D.10)

From the Lyapunov theorem, Lyapunov function candidate, $V(x)$ is assumed to be a class of scalar functions called the Quadratic form and it is represented as,

$$V(x) = x^T Px \quad (D.11)$$

Where, P is symmetric positive definite matrix. The definiteness of the quadratic form depends upon the matrix P . the usual means of determining of a quadratic form is through the application of *Sylvester's theorem* [37], which states that the necessary and sufficient conditions that this form is positive definite are that all the successive minors of P are positive. If any of these minors become zero, then the function is only positive semi-definite. This form becomes negative definite or semi-definite if $-P$ becomes positive definite or semi-definite.

Let, $V(x) > 0$ and for the system to be stable, $\dot{V}(x) < 0$ needs to be proved.

Hence, differentiating the positive-definite function, $V(x)$ along the system trajectory gives,

$$\dot{V}(x) = \dot{x}^T Px + x^T P \dot{x} \quad (D.12)$$

Now, from the equation (D.10), $\dot{x}^T = x^T A^T$ (D.13)

Substituting (D.13) and (D.10) in equation (D.12) gives,

$$\begin{aligned} \dot{V}(x) &= (x^T A^T) Px + x^T P Ax \\ \Rightarrow \dot{V}(x) &= x^T (A^T P + P A) x \\ \Rightarrow \dot{V}(x) &= -x^T Q x \end{aligned} \quad (D.14)$$

Where, Q is also a symmetric positive definite matrix and, $Q = -(A^T P + P A)$ is known as the *Lyapunov equation* [37].

Study Approaches

In order to analyse the LTI systems, either of the two approaches given below might be taken up [37].

- Consider a symmetric positive definite matrix, $\mathbf{P}$ and then solve the Lyapunov equation to obtain $\mathbf{Q}$. Then, check whether $\mathbf{Q}$ is positive definite or not.
- An easier approach would be to derive the positive definite matrix, $\mathbf{P}$ for a given positive definite matrix $\mathbf{Q}$. So, firstly consider a symmetric positive definite matrix, $\mathbf{Q}$ and then solve the Lyapunov equation to obtain $\mathbf{P}$. Then, check whether $\mathbf{P}$ is positive definite or not. If $\mathbf{P}$ is a positive-definite matrix then $V(x)$ is a Lyapunov function of the linear system and so, global asymptotic stability is guaranteed. This technique generally leads to conclusive results for stable linear systems.

II. Lyapunov Analysis of Non - Linear Systems

Since, the stability of non-linear systems is not global; the emphasis should be on finding the largest region surrounding the origin in which stability can be shown. A non-linear autonomous system is described as,

$$\dot{x} = f(x) \quad (\text{D.15})$$

The above system can be expressed as a combination of linear and non-linear part as shown.

$$\dot{x} = f(x) = \mathbf{A}x + g(x) \quad (\text{D.16})$$

Where, $g(x)$ is the non-linear part of $f(x)$.

Now, again considering the quadratic scalar function as in the linear case, $V(x) = x^T \mathbf{P} x$ where, $\mathbf{P}$ is symmetric positive definite matrix.

Again, let, $V(x) > 0$, i.e. positive definite and for the system to be stable $\dot{V}(x) < 0$ needs to be proved. Hence, differentiating the positive-definite function, $V(x)$ along the system trajectory gives,

$$\dot{V}(x) = \dot{x}^T \mathbf{P} x + x^T \mathbf{P} \dot{x} \quad (\text{D.17})$$

$$\text{Now, from the equation (D.15), } \dot{x}^T = (\mathbf{A}x + g(x))^T \quad (\text{D.18})$$

Substituting (D.15) and (D.18) in equation (D.17) gives,

$$\begin{aligned} \dot{V}(x) &= (\mathbf{A}x + g(x))^T \mathbf{P} x + x^T \mathbf{P} (\mathbf{A}x + g(x)) \\ \Rightarrow \dot{V}(x) &= x^T (\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A}) x + 2x^T \mathbf{P} g(x) \\ \Rightarrow \dot{V}(x) &= -x^T \mathbf{Q} x + 2x^T \mathbf{P} g(x) \end{aligned} \quad (\text{D.19})$$

Where, $\mathbf{Q}$ is also a symmetric positive definite matrix

From the equation, it may be concluded that if the linear part of the dynamics is asymptotically stable, then a finite region of stability is assured for the non-linear system [15]. Hence, the stability of the linear part of the non-linear system as expressed by (D.14) is to be determined. This approach was taken for the analysis of the PWR models in this study.

D.3. Moderator Effectiveness

The effectiveness of a material as a Moderator can be determined in terms of the ratios mentioned beneath.

1. Moderating Ratio

The moderating ratio is merely the ratio of slowing down power to the macroscopic absorption cross section. The higher the moderating ratio, the more effectively the material performs as a moderator. Mathematically it can be expressed as,

$$MR = \frac{\zeta \Sigma_s}{\Sigma_a} \quad (\text{D.20})$$

Here, $\zeta \Sigma_s$ = macroscopic slowing down power (MSDP).

ζ = logarithmic energy decrement per collision.

Σ_s = macroscopic scattering cross section for neutrons

Σ_a = macroscopic absorption cross section for neutrons

2. Moderator to Fuel Ratio

The *Moderator-to-fuel ratio* (N^m/N^u) plays a significant part in determining the effectiveness of moderators under transient scenarios.

i. *Over Moderation*

When the reactor designer increases the amount of moderator in the core (i.e. N^m/N^u increases), neutron leakage decreases. Neutron absorption in the moderator (Σ_a^m) increases & causes a decrease in the *thermal utilization factor* (f).

ii. *Under Moderation*

When the reactor designer has insufficient amount of moderator in the core (i.e. N^m/N^u decreases), it causes an increase in slowing down time and results in a greater loss of neutrons by *resonance absorption*. This also causes an increase in neutron leakage.

Since, the moderator-to-fuel ratio affects the thermal utilization factor (f) and the resonance escape probability (p), it also affects the effective multiplication factor (k_{eff}).